

Homochiral ideas

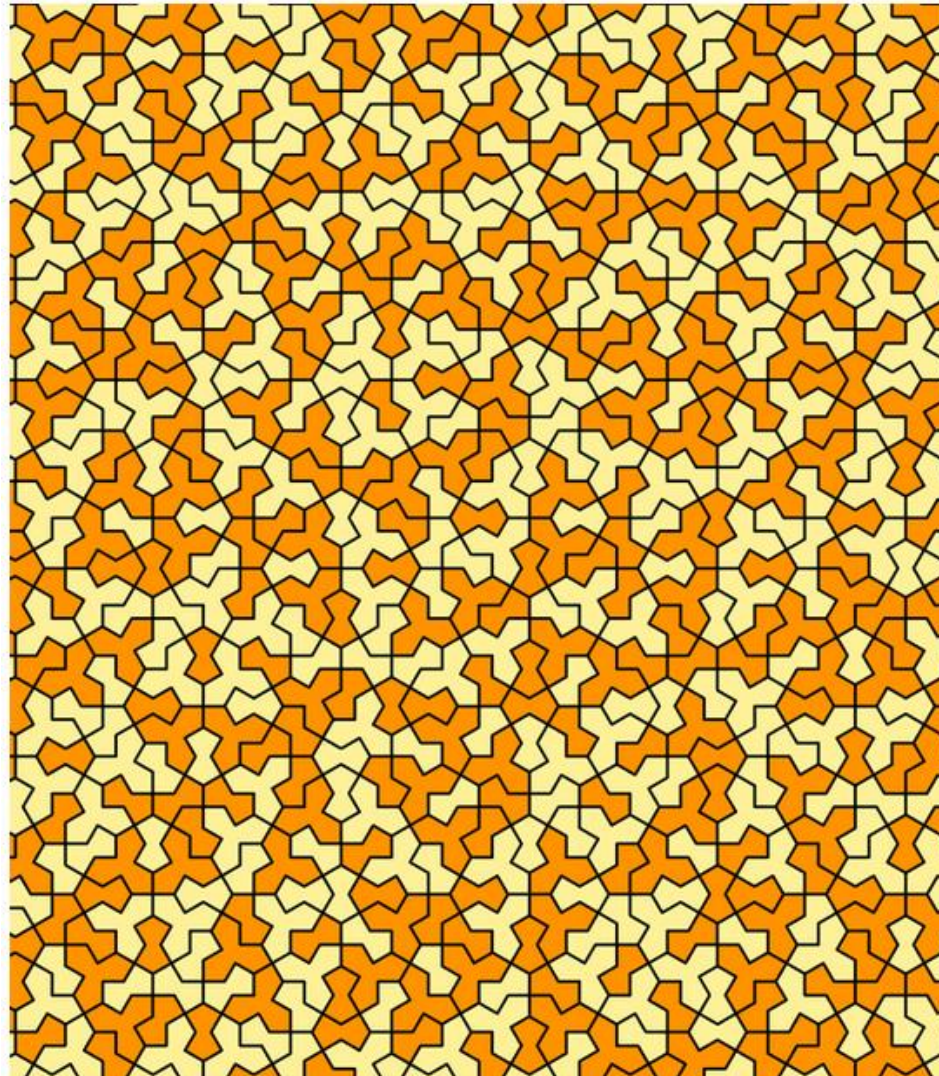
Marianne Imperor-Clerc

April 2026

Wheel tiling

<https://tilings.math.uni-bielefeld.de/substitution/wheel-tiling/>

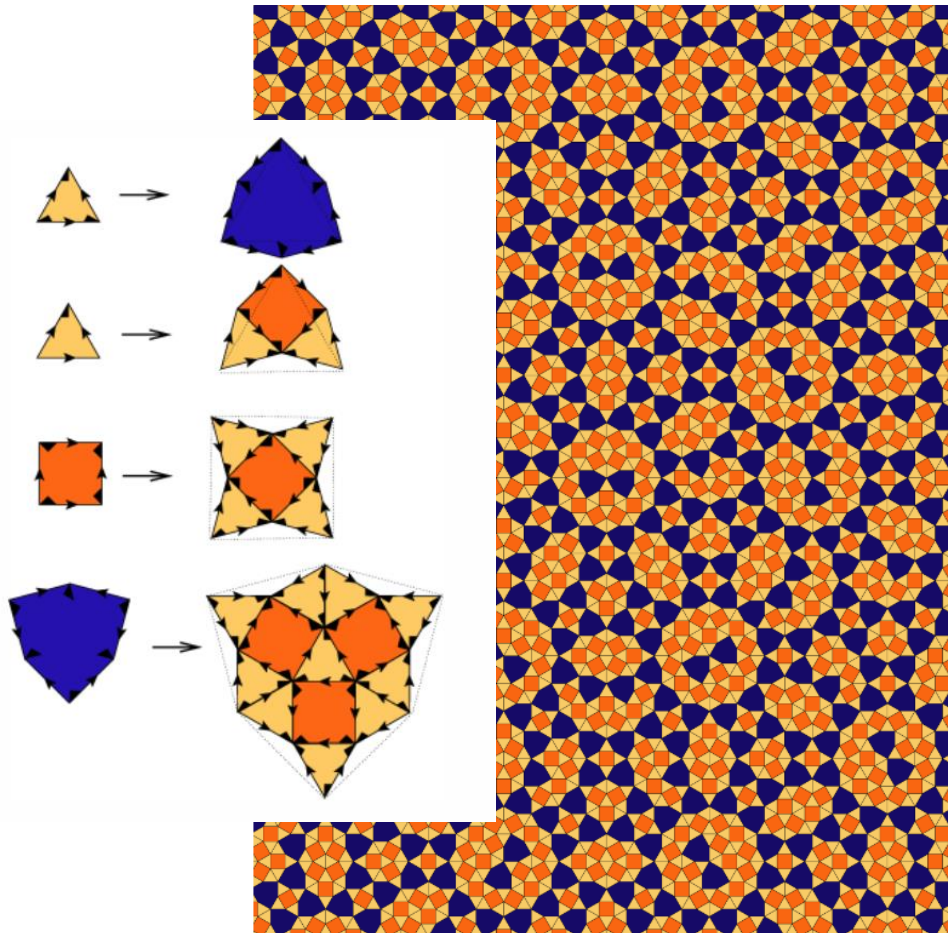
Nissen, Hans-Ude



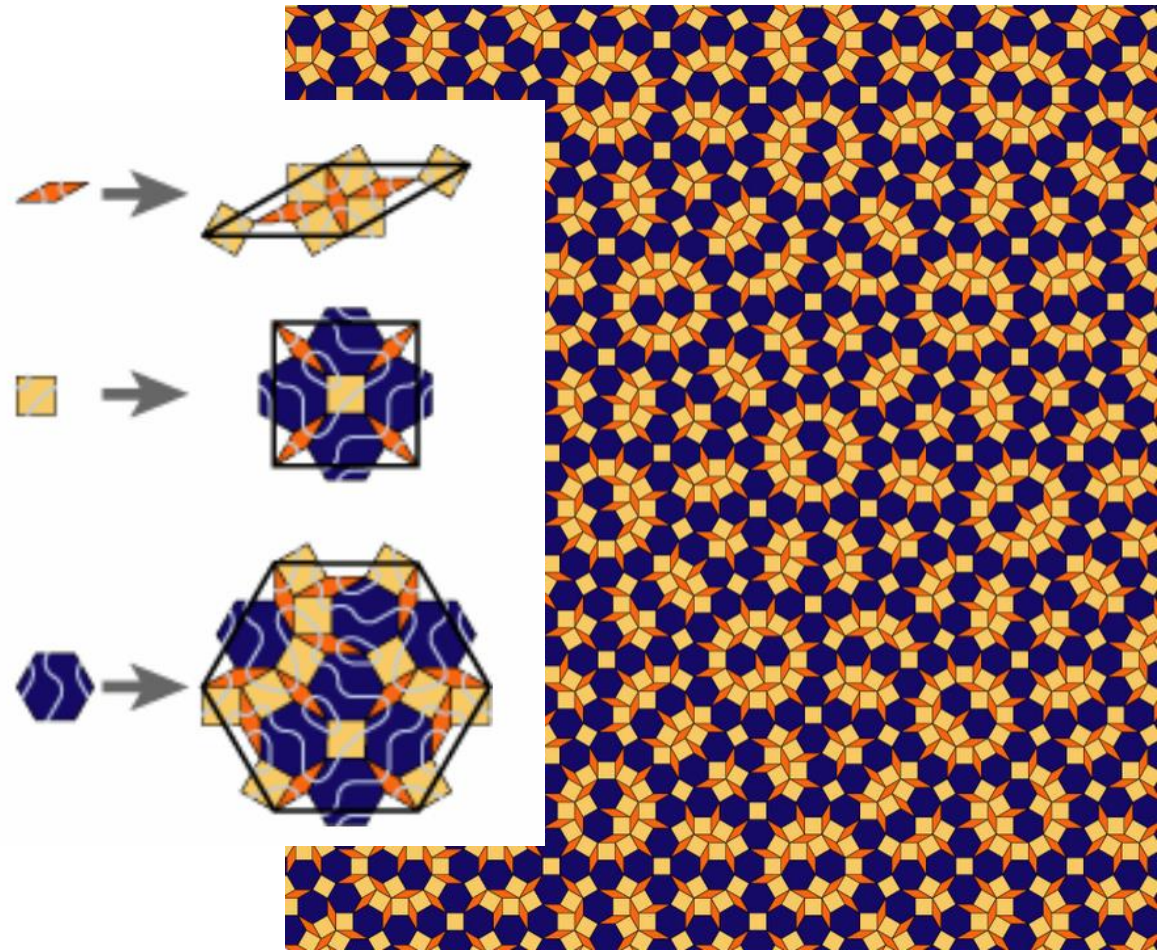
Wheel tiling -> shield and Socolar tilings

There is a very simple rule to transform the wheel tiling into the [shield](https://tilings.math.uni-bielefeld.de/substitution/shield/) tiling: Replace each edge in the tiling by an edge orthogonal to it, of equal length, such that the old and new edge intersect in their midpoints. Applying this rule to the wheel tiling yields the shield tiling and vice versa.

<https://tilings.math.uni-bielefeld.de/substitution/shield/>



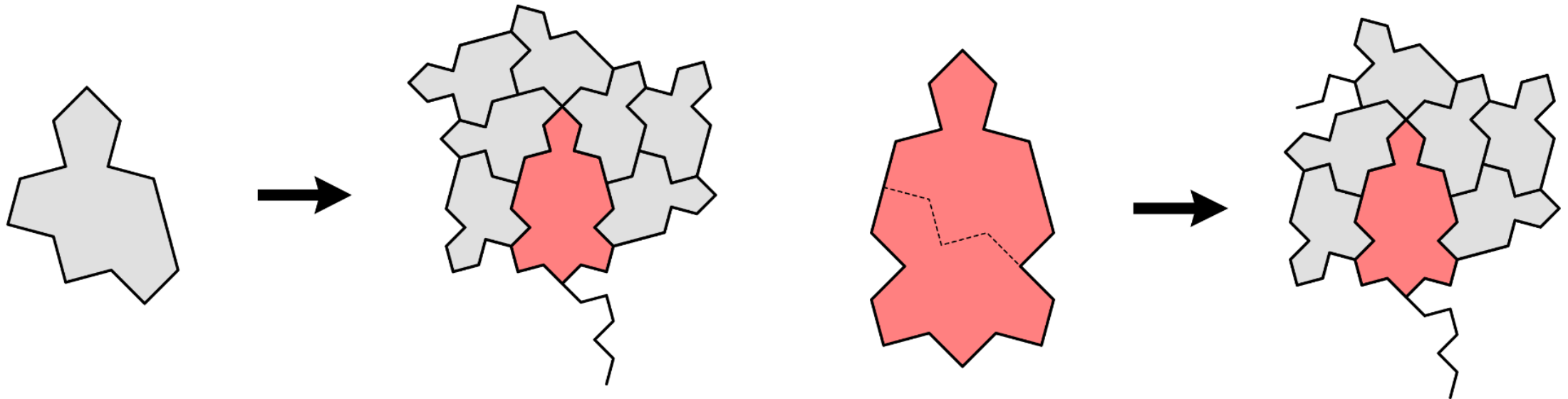
<https://tilings.math.uni-bielefeld.de/substitution/socolar/>



H7/H8 by Simon Tathan

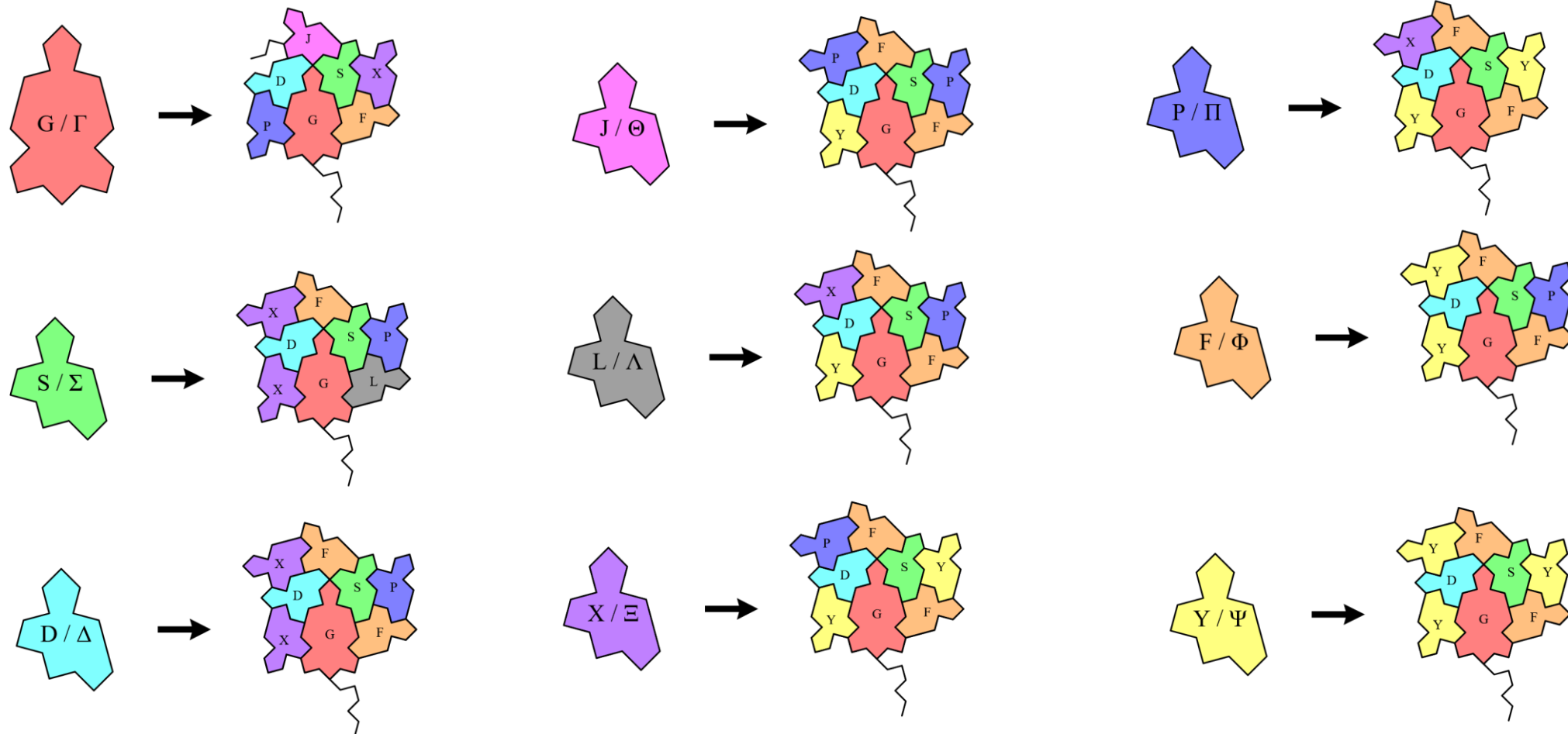
<https://www.chiark.greenend.org.uk/~sgtatham/quasiblog/aperiodic-refine/>

<https://arxiv.org/abs/2512.16595>



The nine homochiral substitutions by Simon Tathan

<https://www.chiark.greenend.org.uk/~sgtatham/quasiblog/aperiodic-refine/>



Homochiral python code on github

- <https://github.com/marimperorclerc/homochiral>

python code for drawing clusters with $\text{Tile}(a,b)$ in svg graphics

References

Code is based on the drawsvg package:

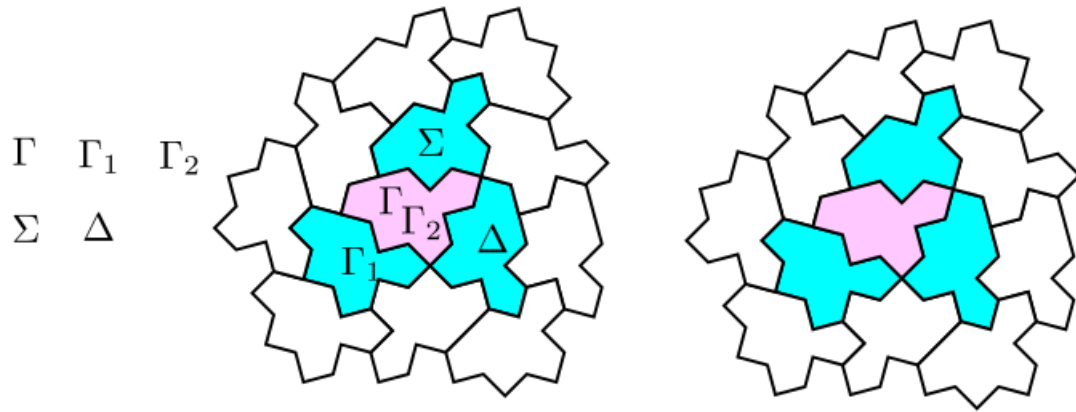
<https://pypi.org/project/drawsvg/>

And the python code with alternate chirality for inflation for spectre tiling:

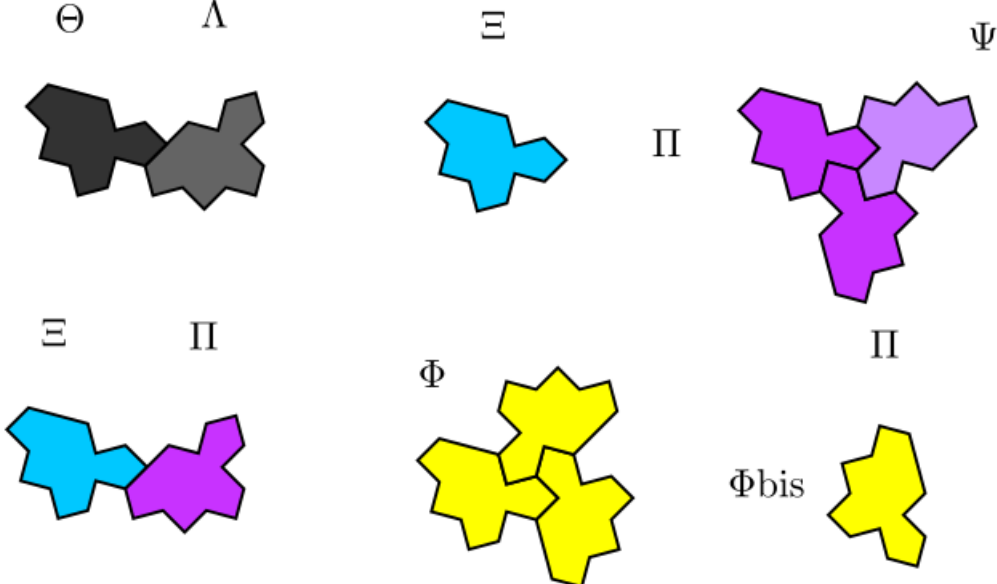
<https://github.com/shrx/spectre/blob/master/spectre.py>

- Adapted here (spectre-colored-tiles.ipynb) with different color choices
- See some .svg files in \svg-spectre folder

Identification of the greek letters for tiles



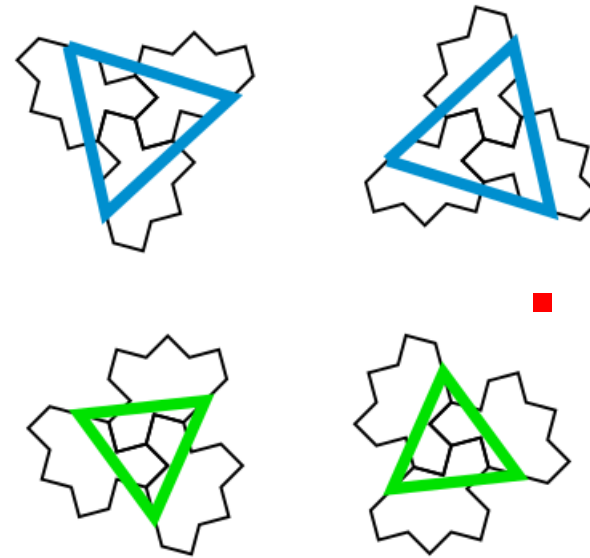
- $\backslash\theta$ and $\backslash\lambda$ always in couple



- All $\backslash\chi$ tiles are linked to a $\backslash\pi$ tile and consequently to a large star.

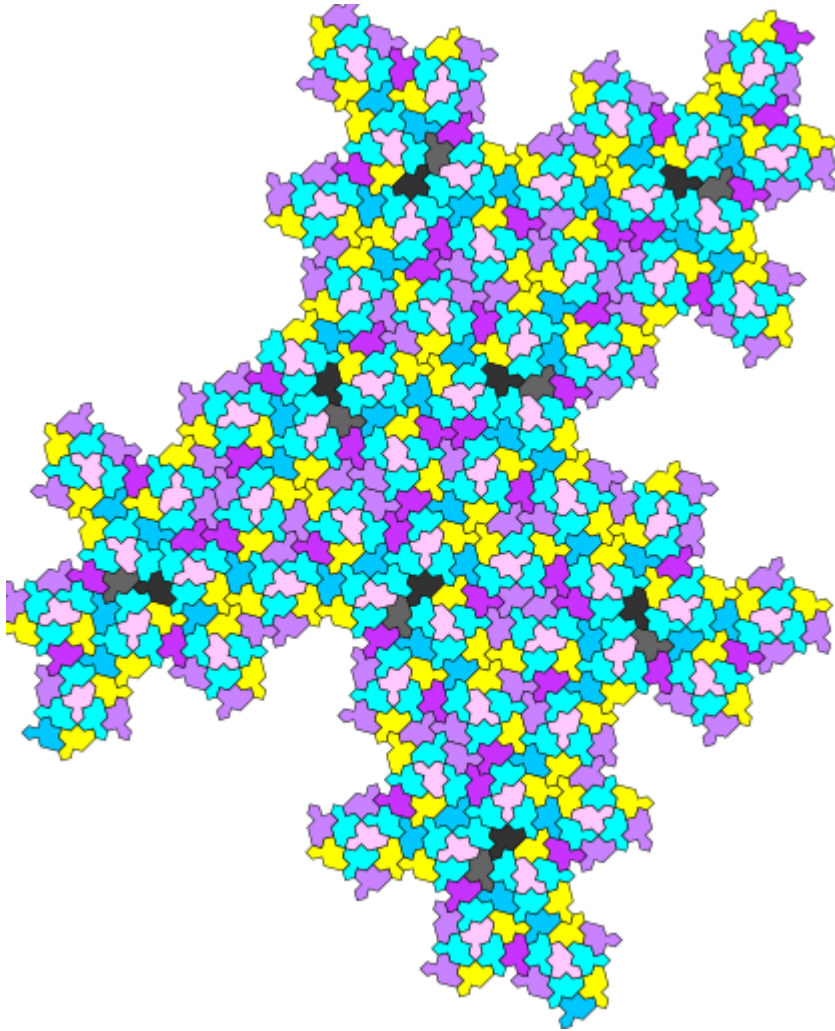
- Consider a cluster of four tiles instead of 3 for horn in the anatomy ? (by adding the $\backslash\Delta$ tile)

- Large stars correspond to all $\backslash\pi$ and $\backslash\psi$ tiles



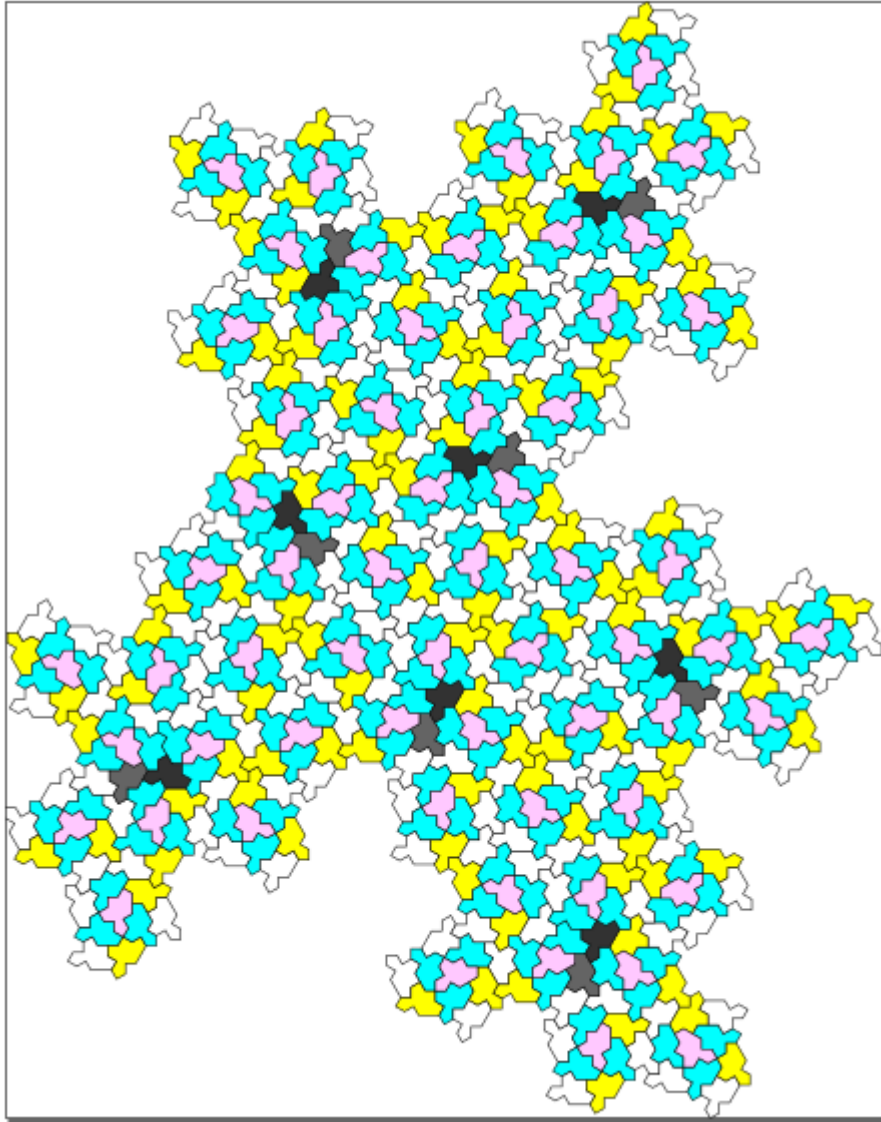
- $\backslash\phi$ tiles give all small stars plus isolated ones. Look at the position of the isolated ones (called here $\backslash\phi_{bis}$) ?

Third inflation with color code from greek letters



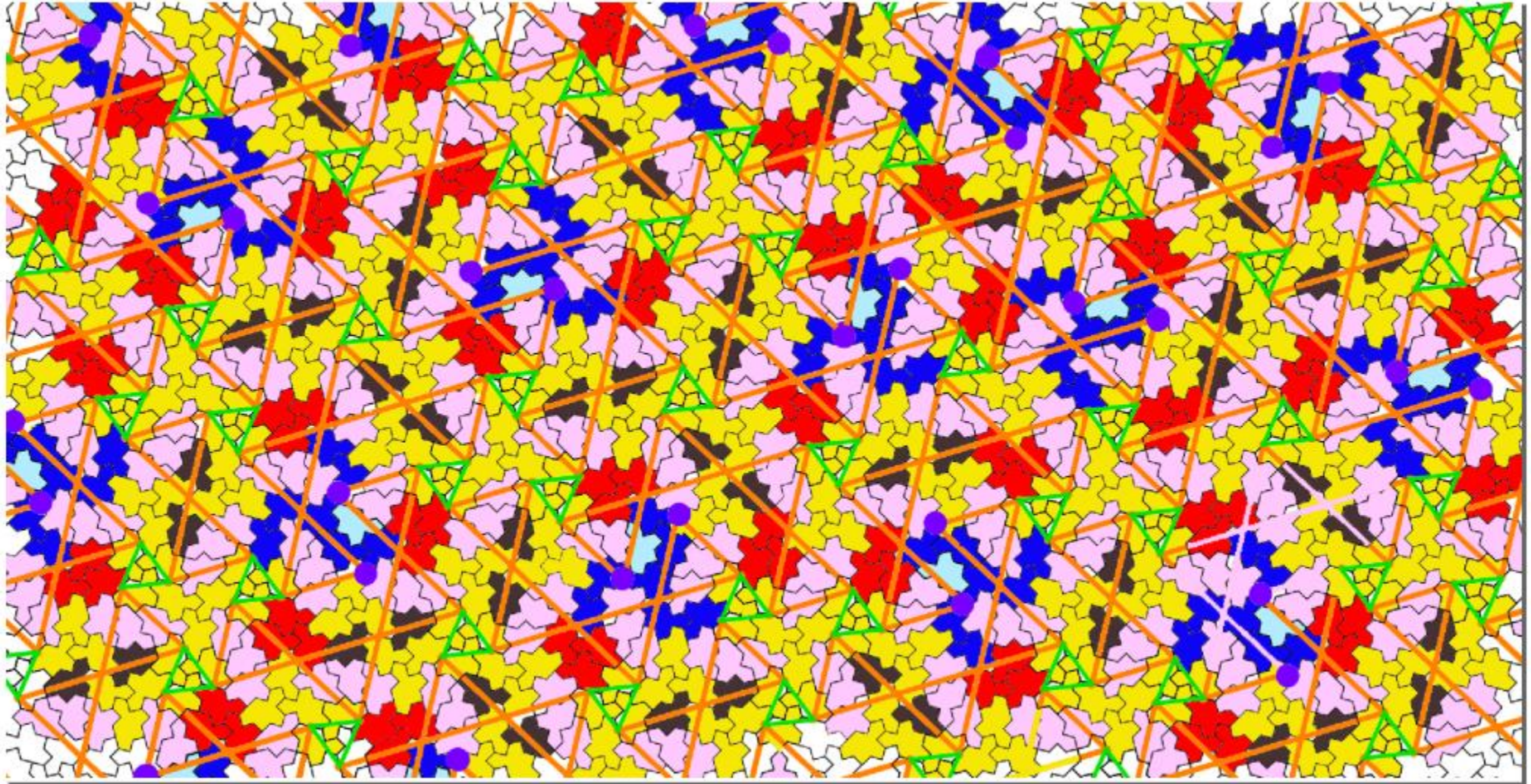
- $\backslash\theta$ and $\backslash\lambda$ always in couple $\rightarrow$ part of a propeller
- Large stars correspond to all $\backslash\Pi$ and $\backslash\Psi$ tiles
- $\backslash\phi$ tiles give all small stars plus isolated ones. Look at the position of the isolated ones (called here $\backslash\phi_{bis}$) ?

Another version with less colors ...

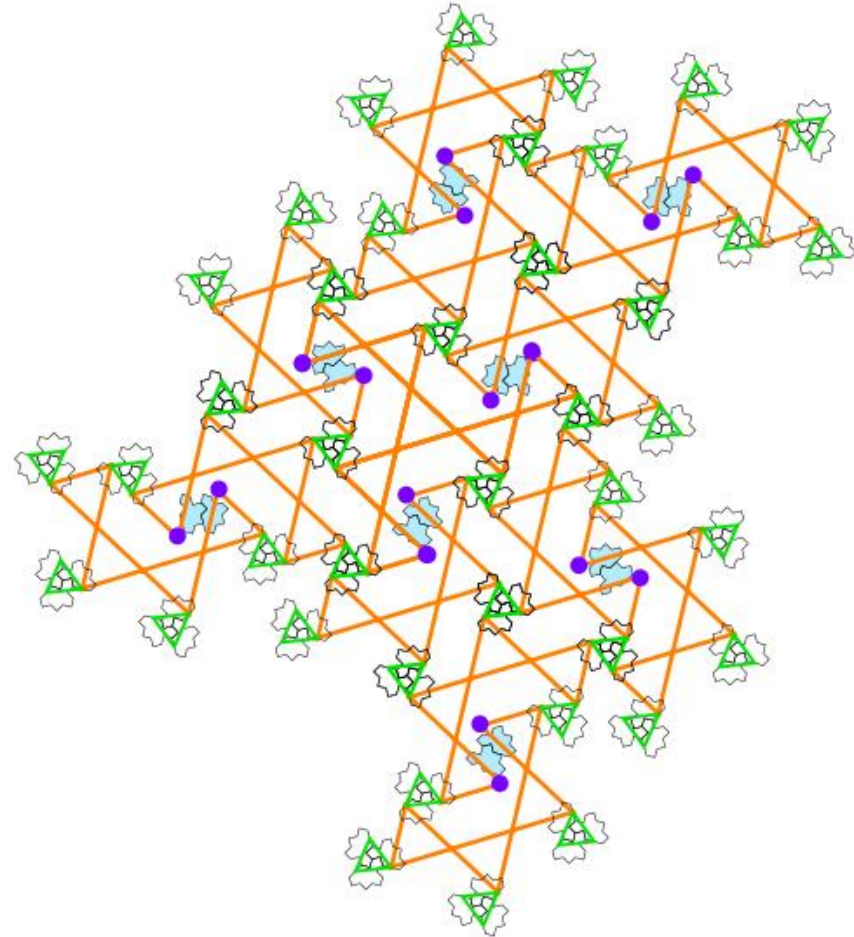
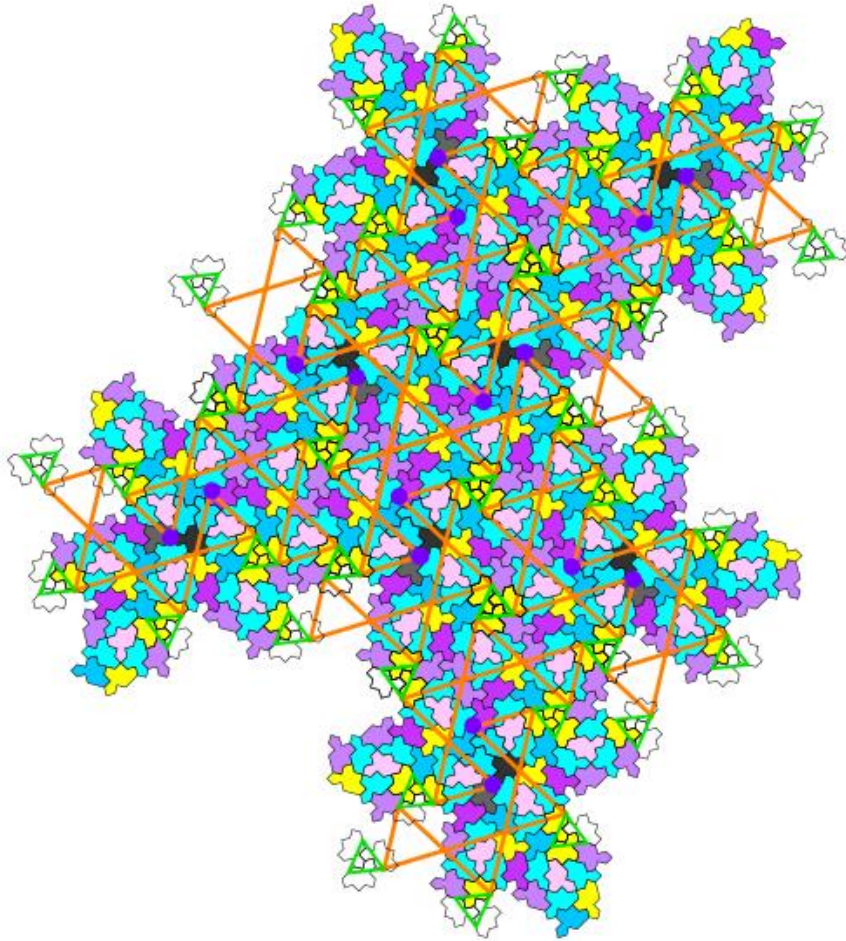


- θ and λ always in couple $\rightarrow$ part of a propeller
- Large stars correspond to all Π and Ψ tiles
- ϕ tiles give all small stars plus isolated ones. Look at the position of the isolated ones (called here ϕ_{bis}) ?

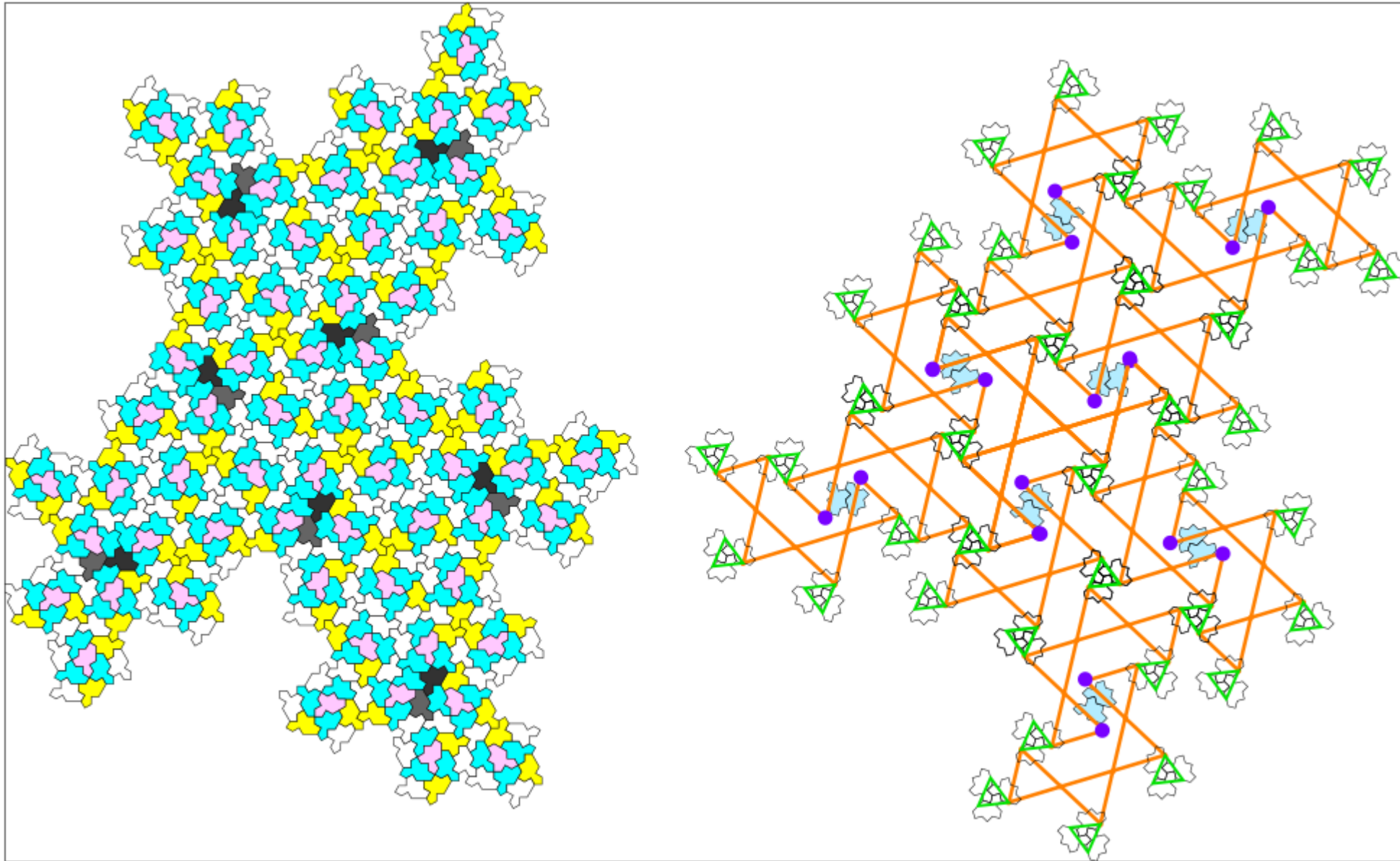
Link with anatomy by Boris Horvat ?



Third inflation: identification of the 2-arms, 3-arms and diabolos

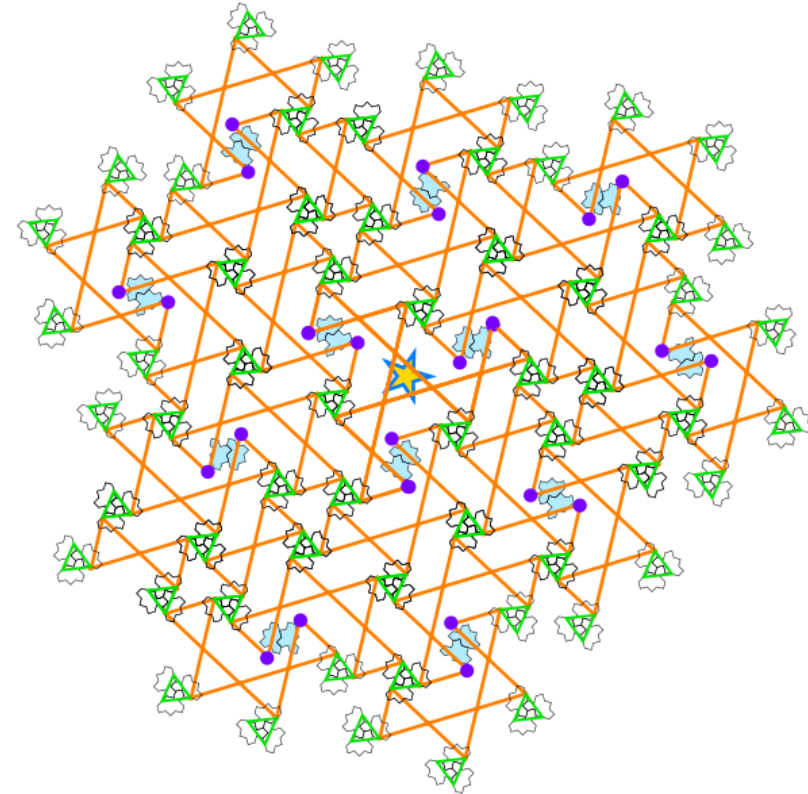
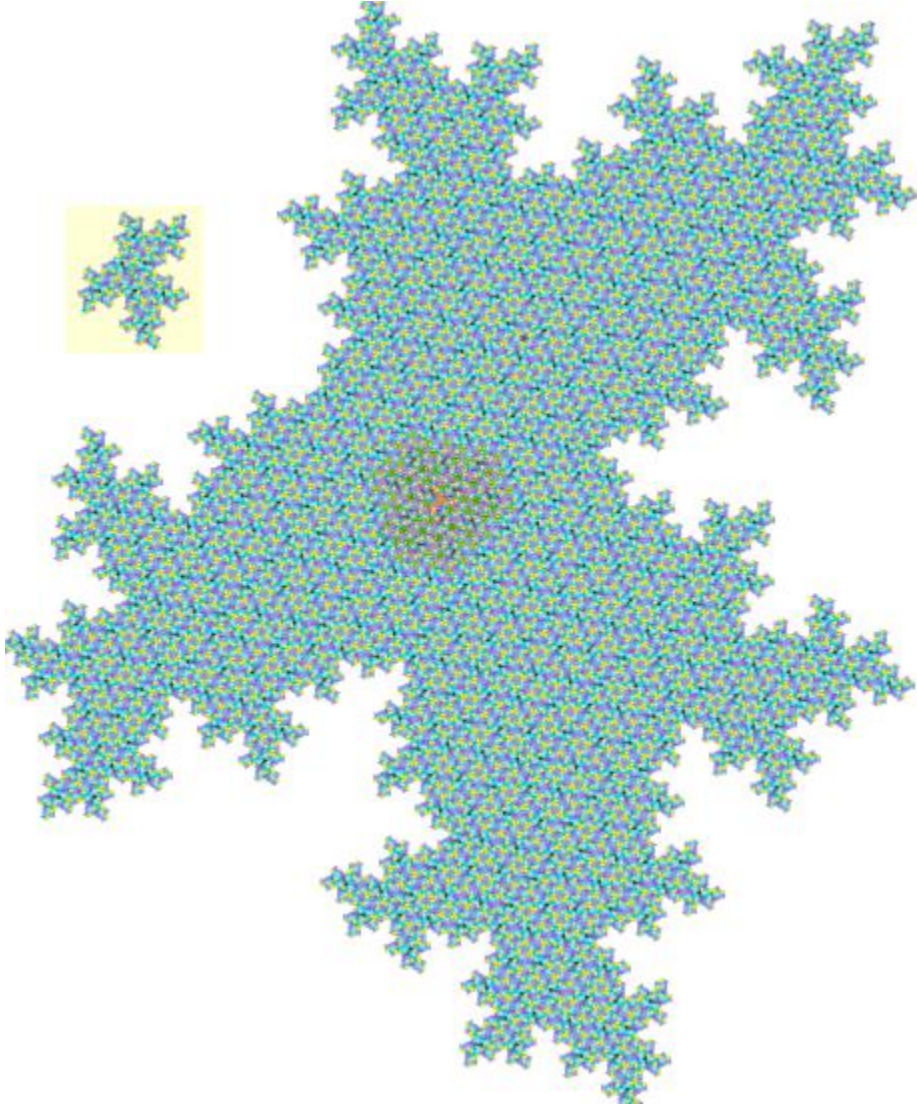


Another version with less colors ...



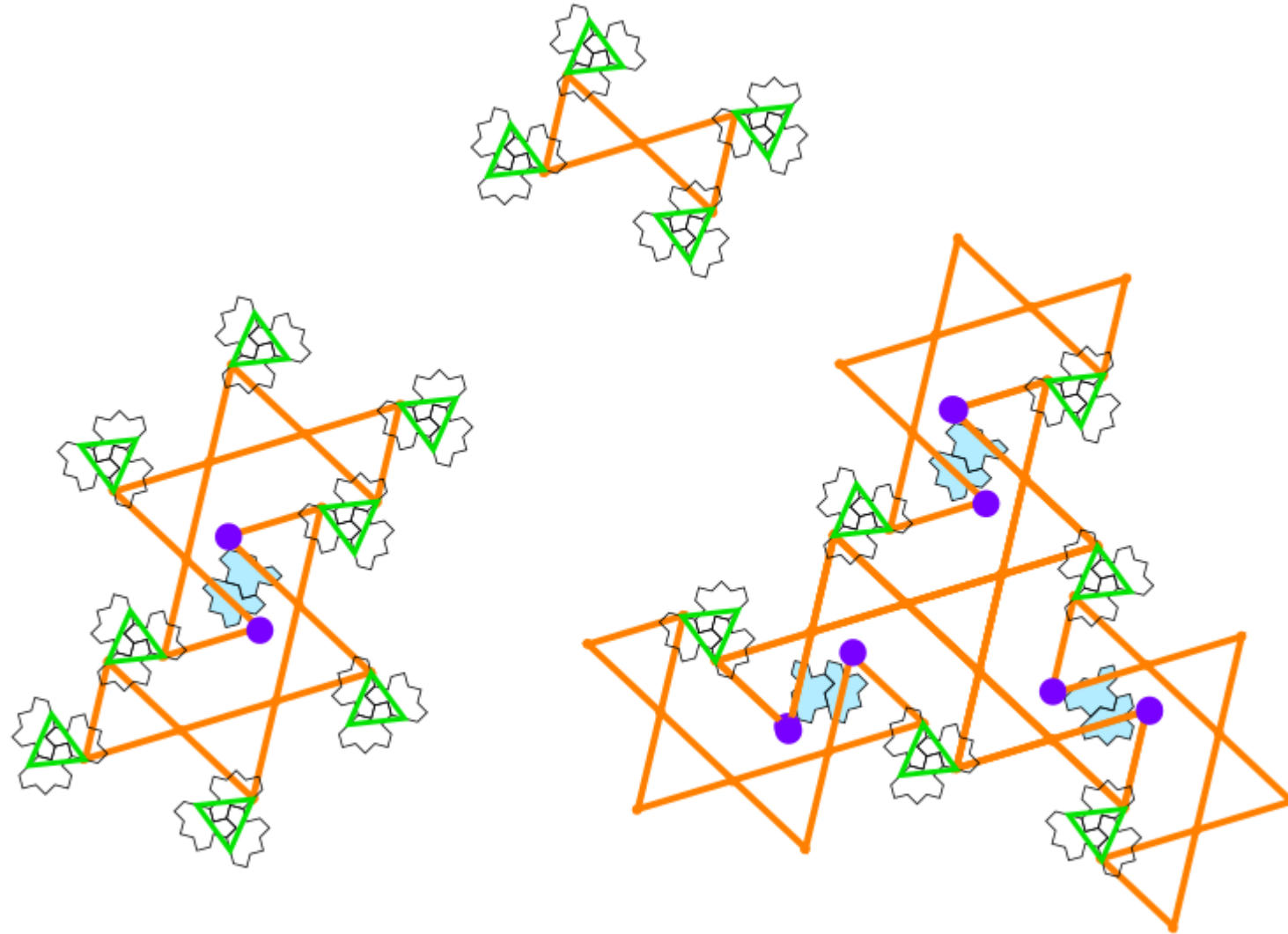
■ $\backslash\theta$ and $\backslash\lambda$ couple
-> part of a propeller ->
highlight a
specific
configuration
of two full
clusters !

Third to fifth inflation: identification of the 2-arms, 3-arms and diabolos ...



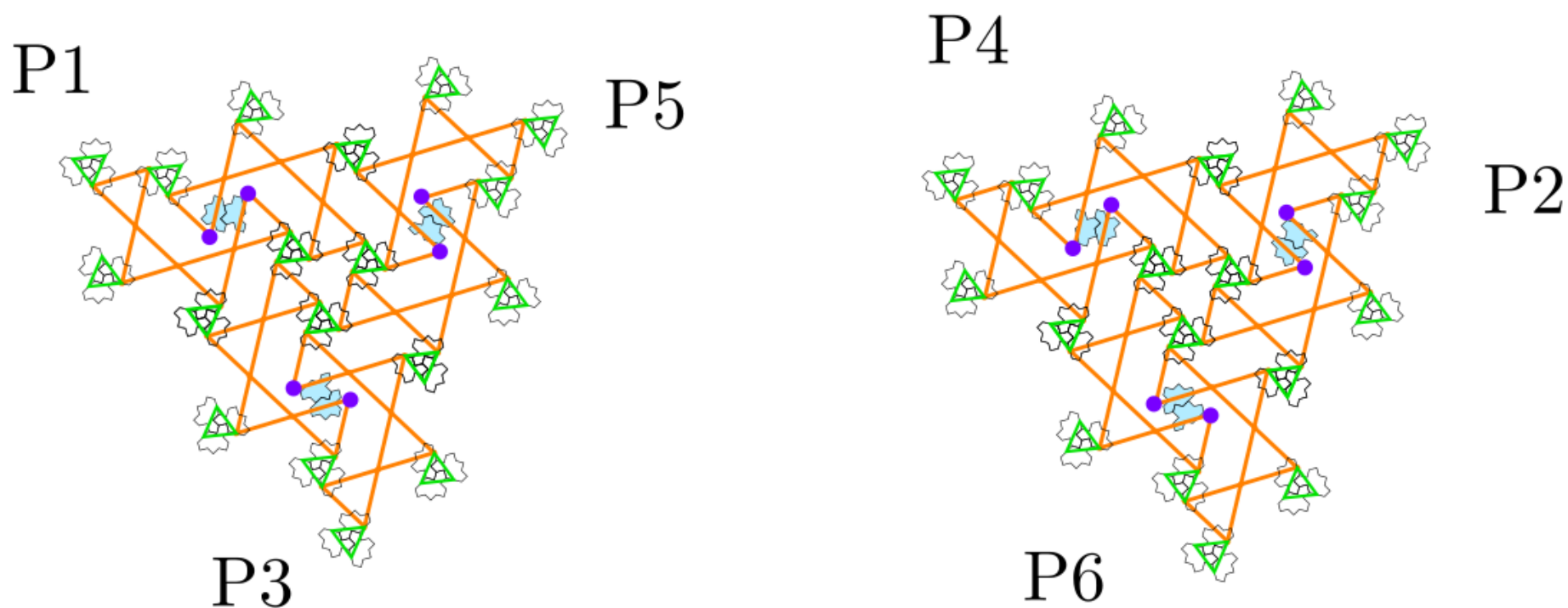
- Very long to do by hand !!!

New puzzle



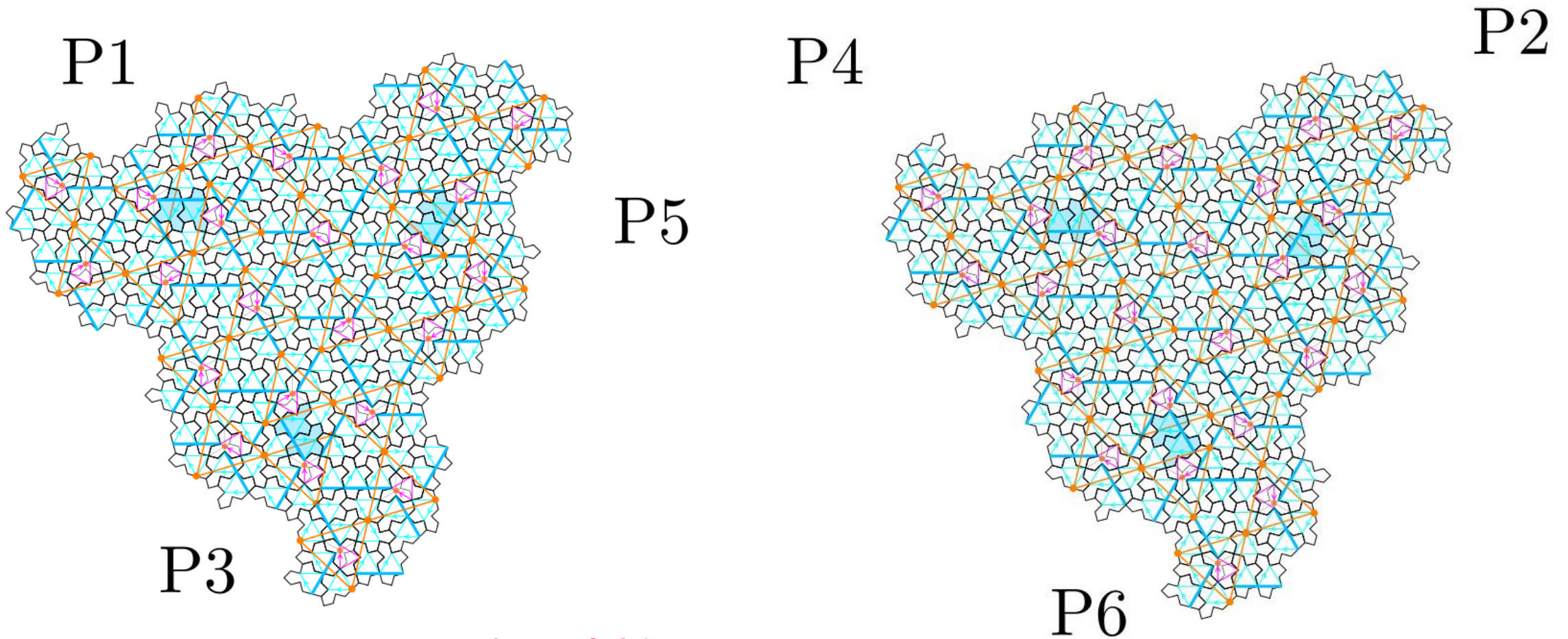
Possible candidate for a seed ?

- Three 2-arms can be combined in two ways:



Possible candidate for 3-fold symmetry center

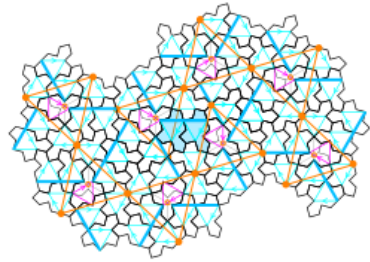
- Compare with center of giant star cluster by Boris Horvat ???



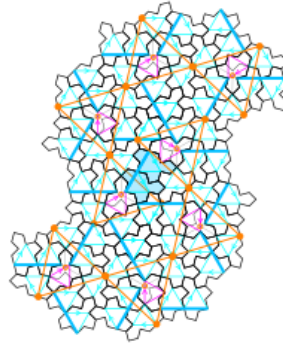
- True three fold symmetry point at center ???

Six orientations of the 2-arms

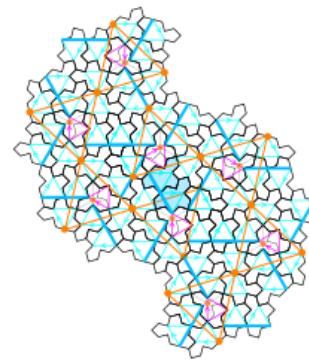
P1



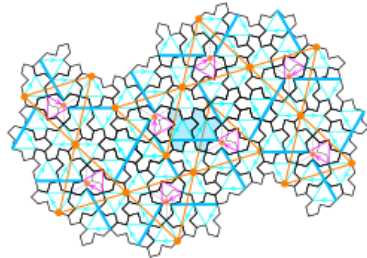
P2



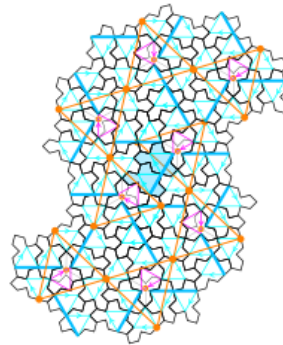
P3



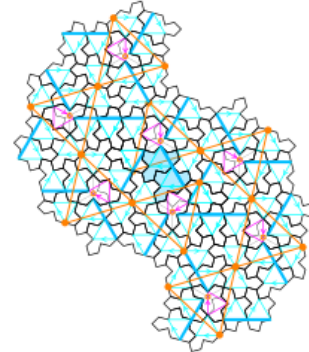
P4



P5

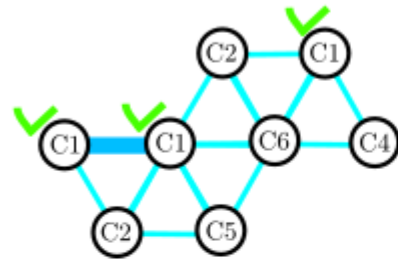


P6

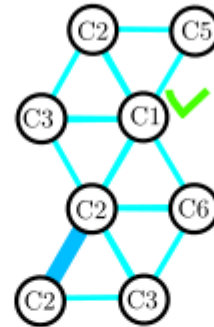


Six orientations of the 2-arms

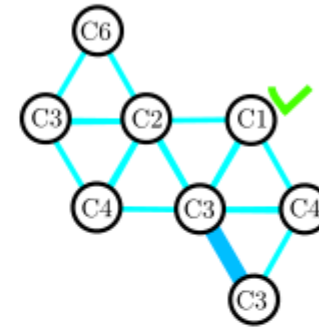
P1



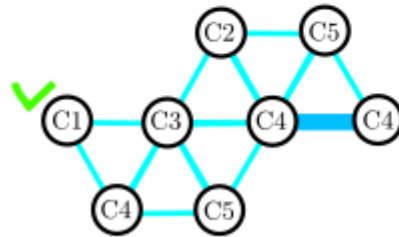
P2



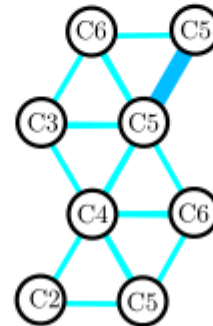
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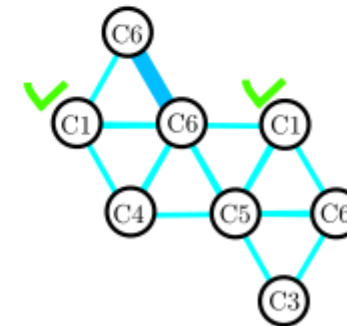
P4



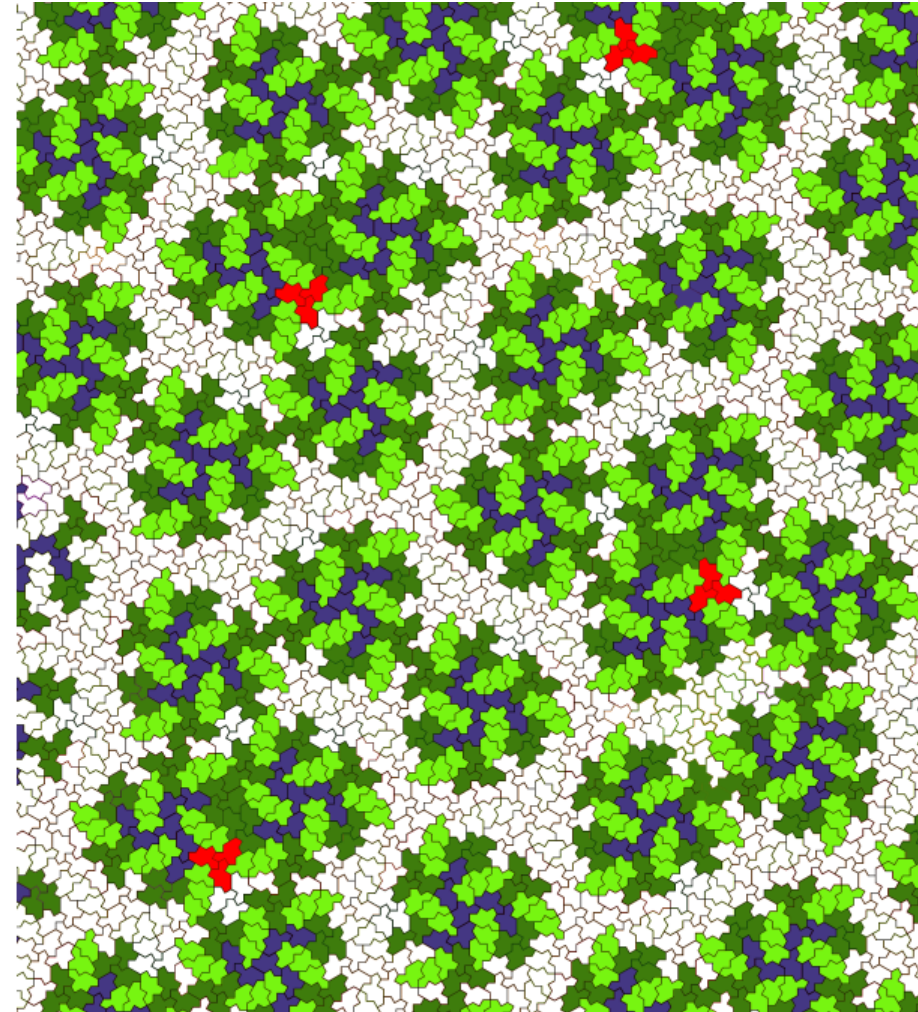
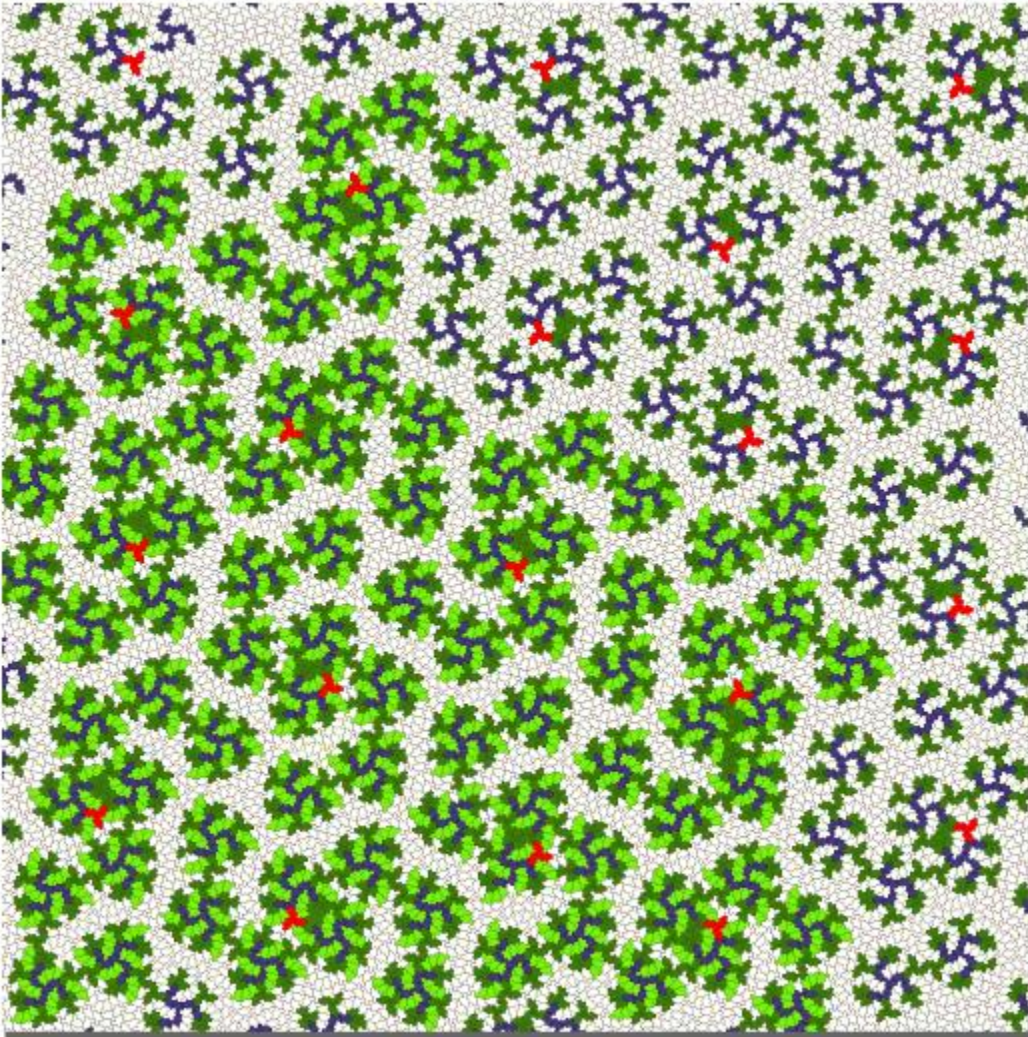
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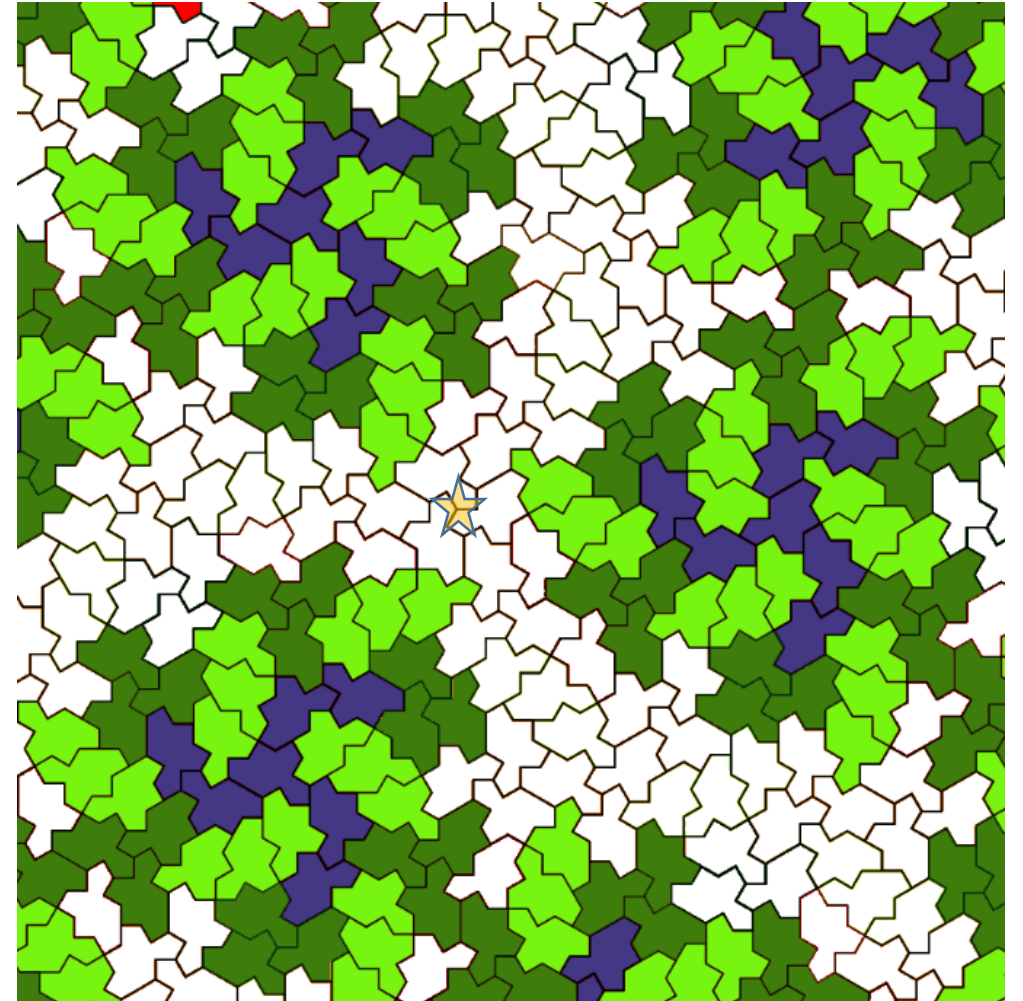
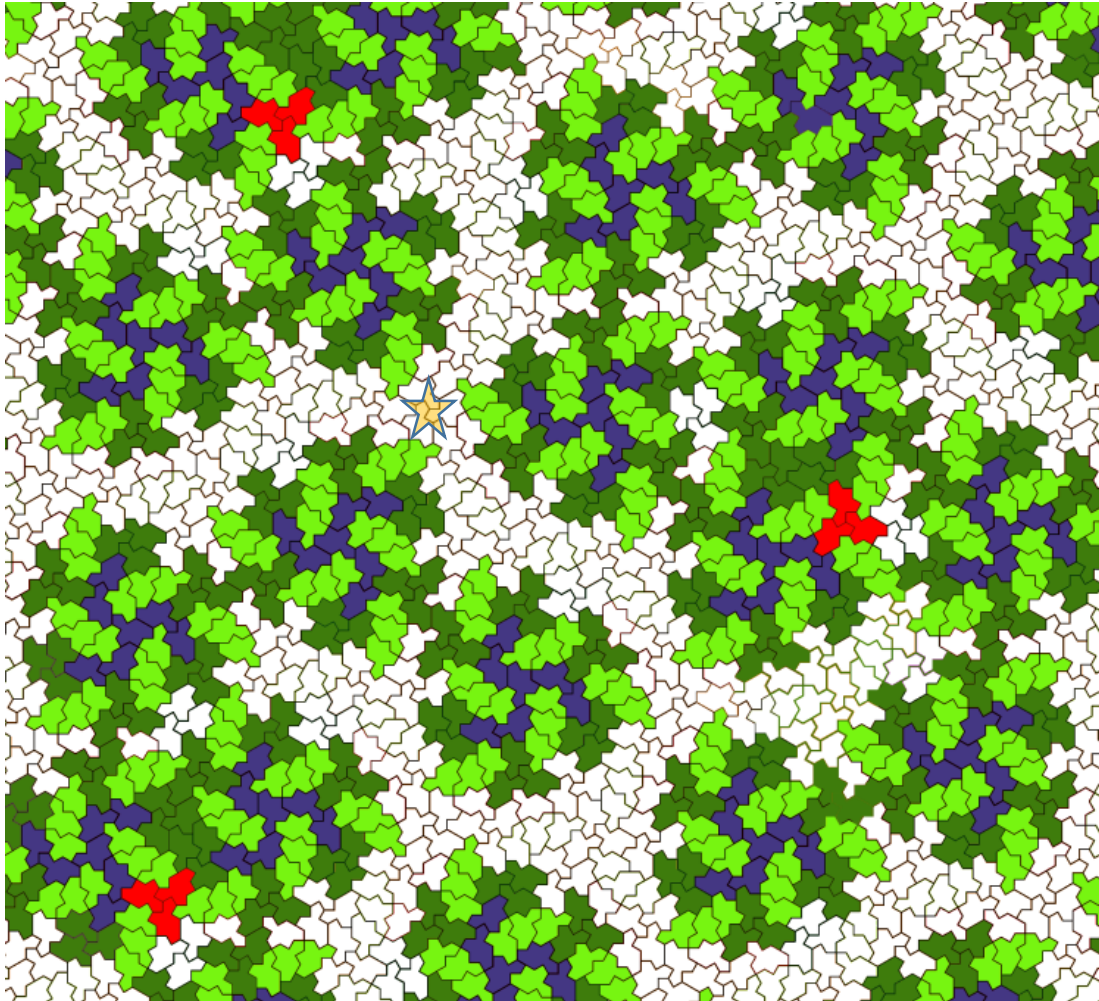
P6



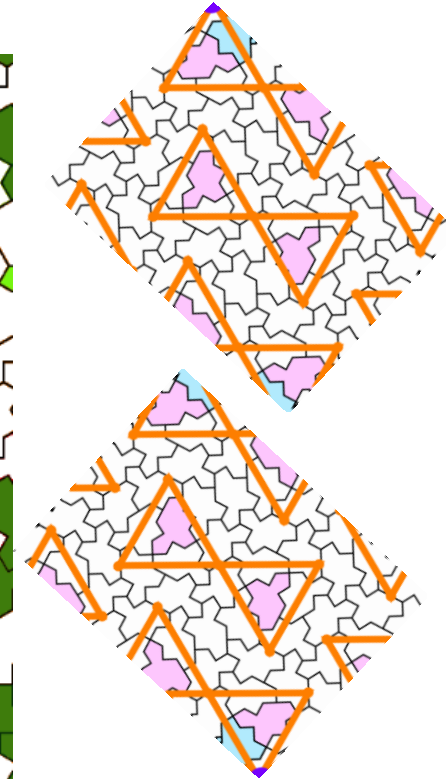
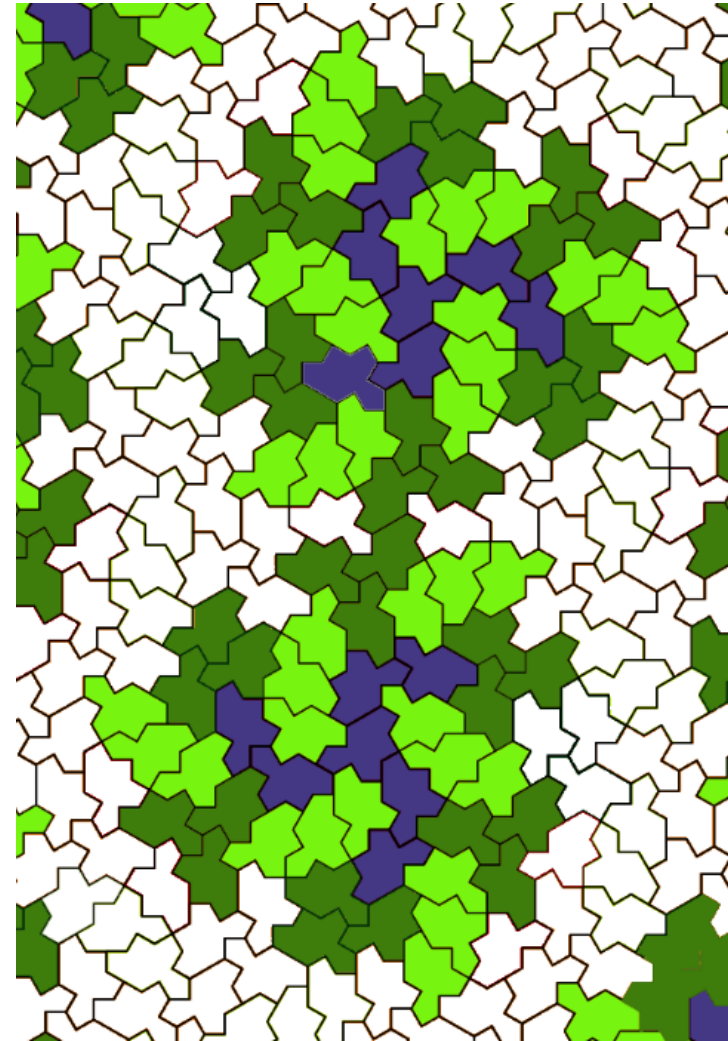
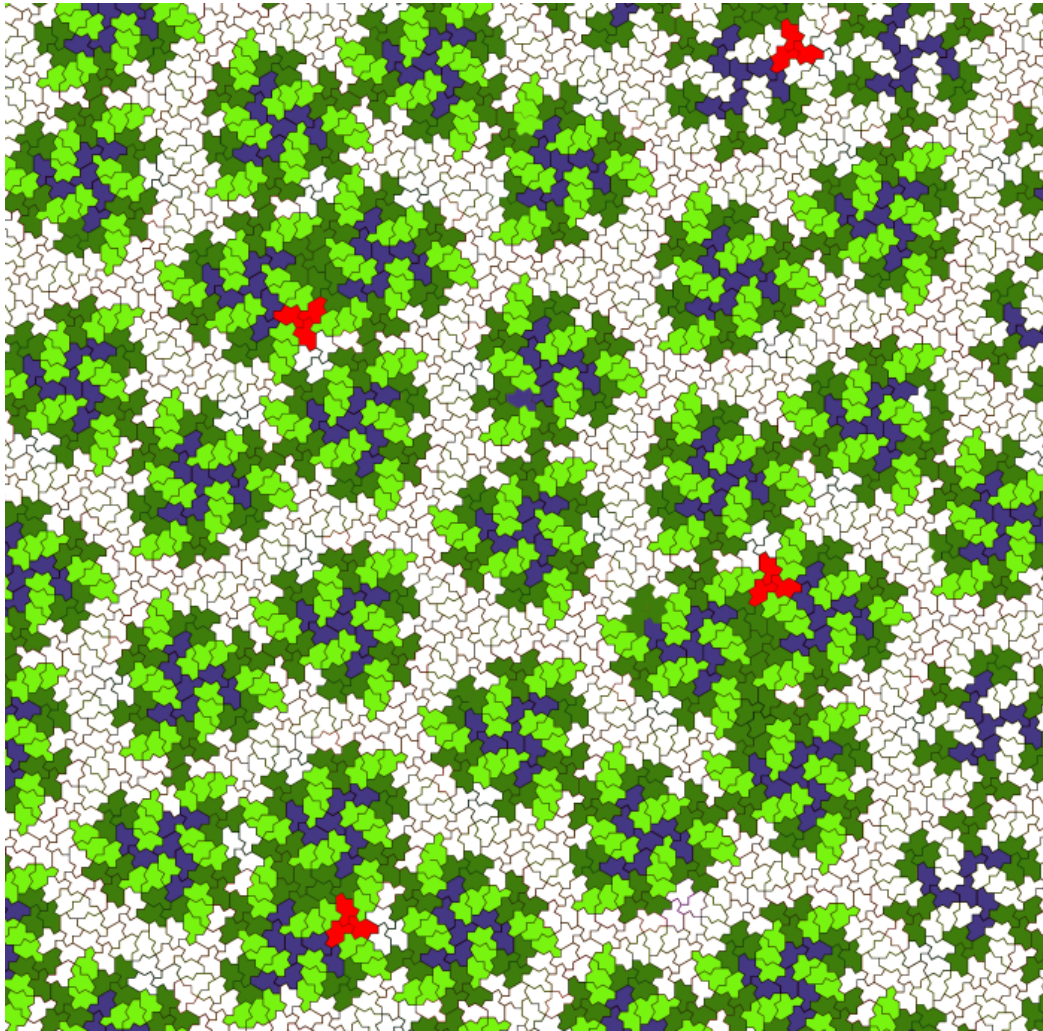
Diabolo/2-arms/3-arms at level 2 by Boris Horvat



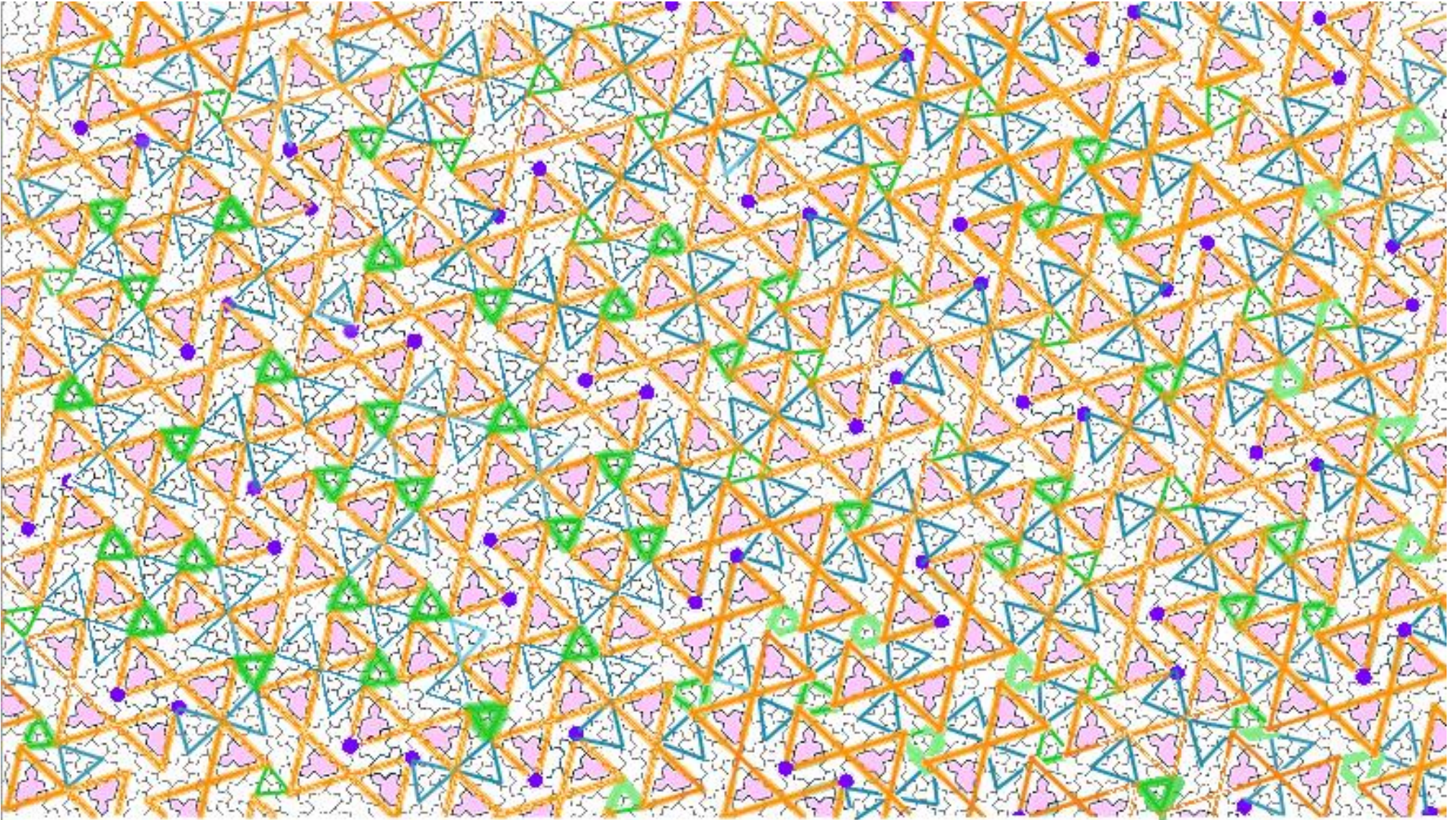
Diabolo/2-arms/3-arms at level 2 by Boris Horvat

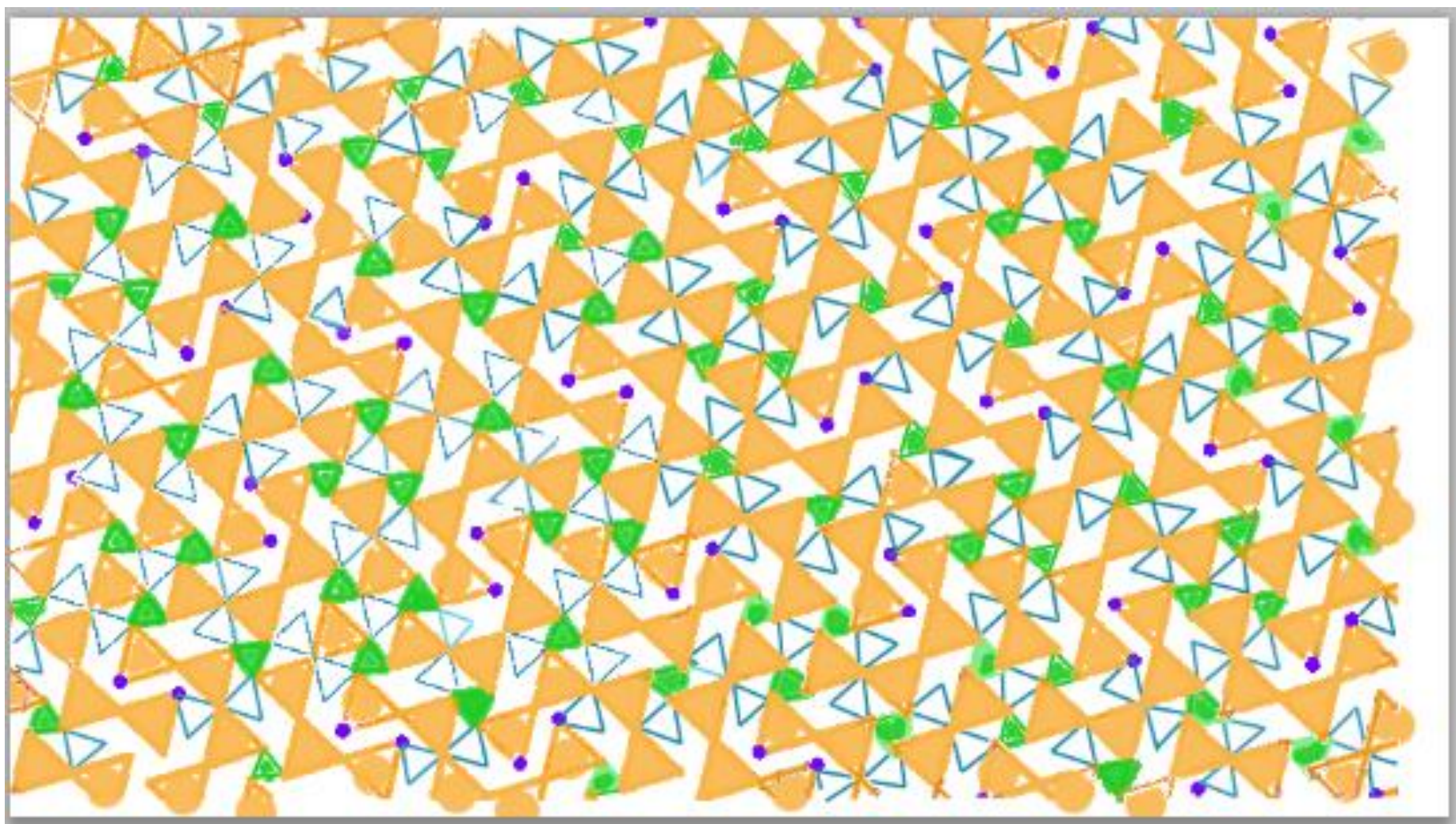


Diabolo/2-arms/3-arms at level 2 by Boris Horvat



SMALL AND LARGE STARS





Espaces entre triangles oranges

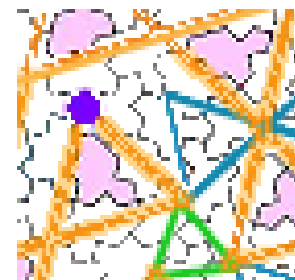
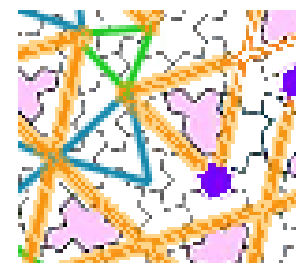
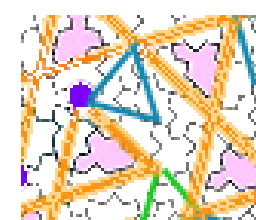
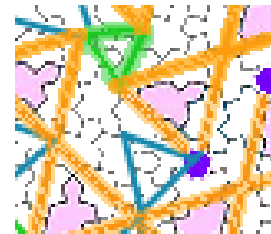
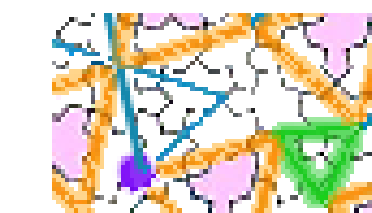
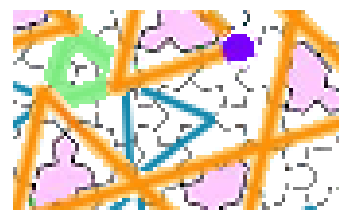
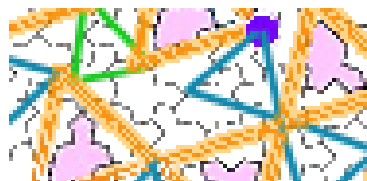
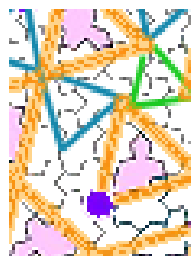
Parallelogrammes



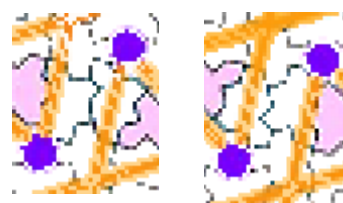
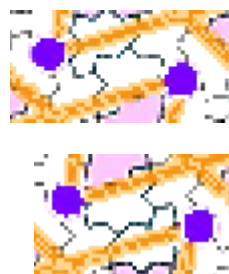
Triangles

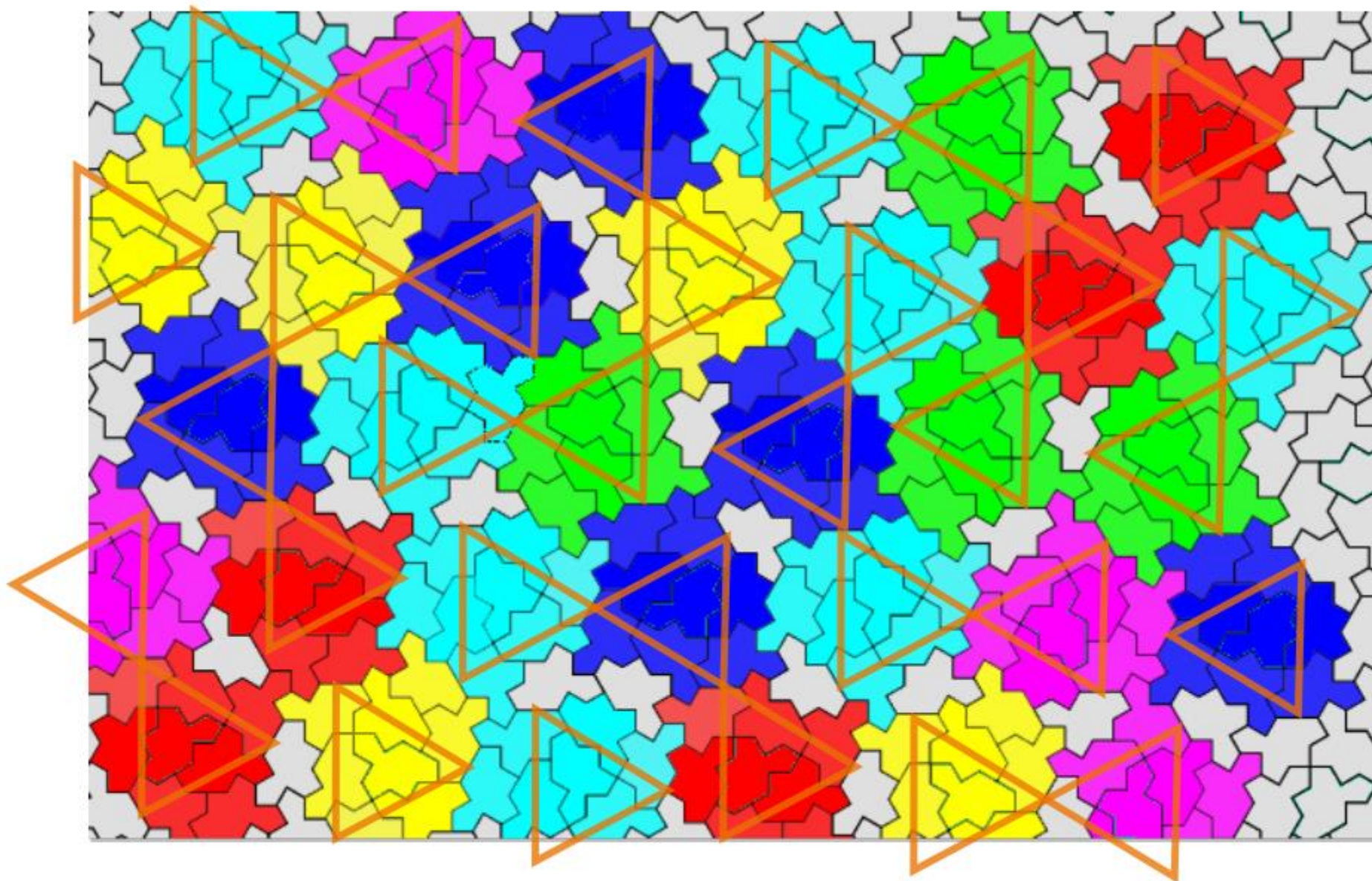


Bateaux

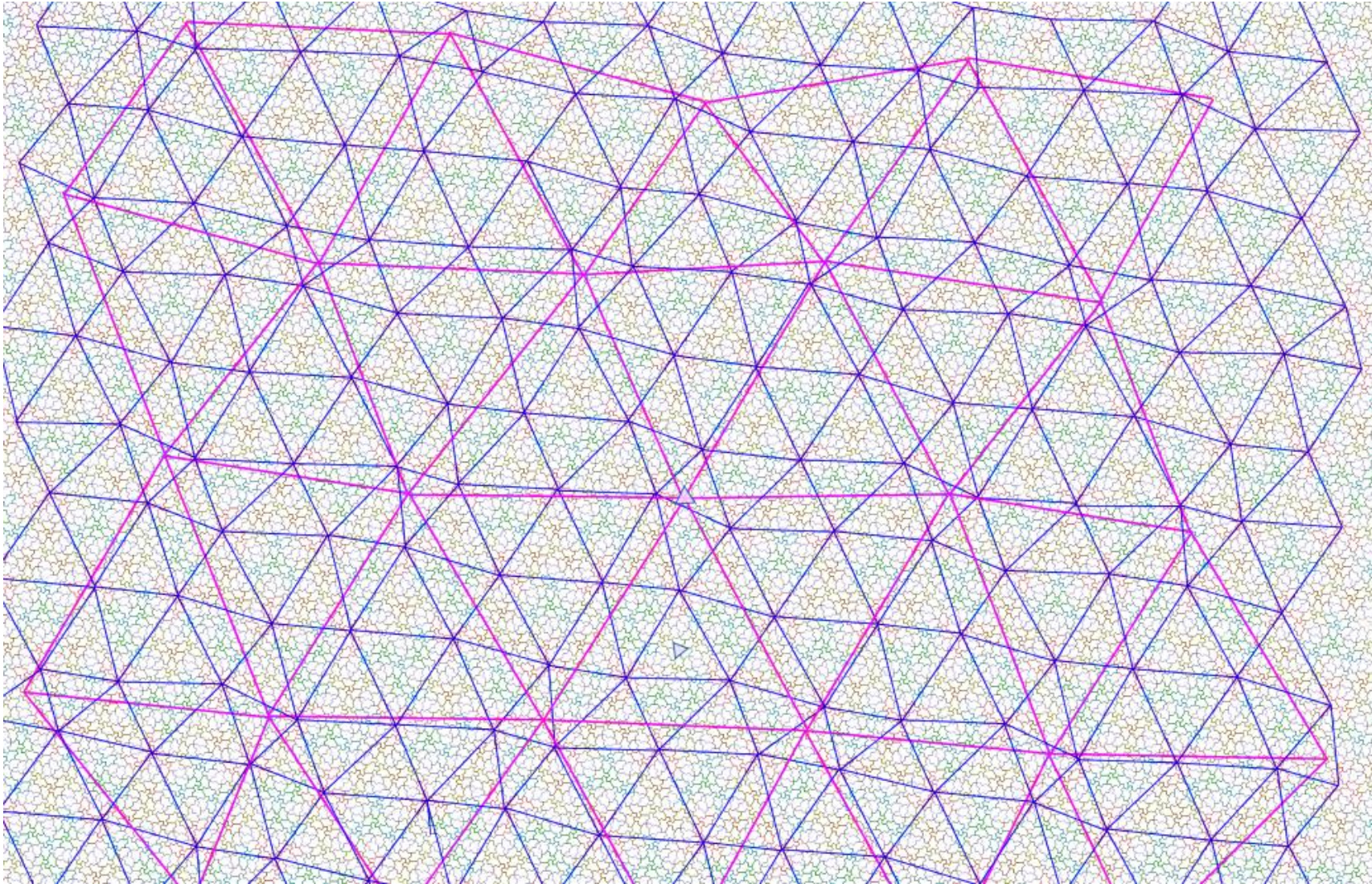


Petits parallelogrammes
2 orientations de tuiles



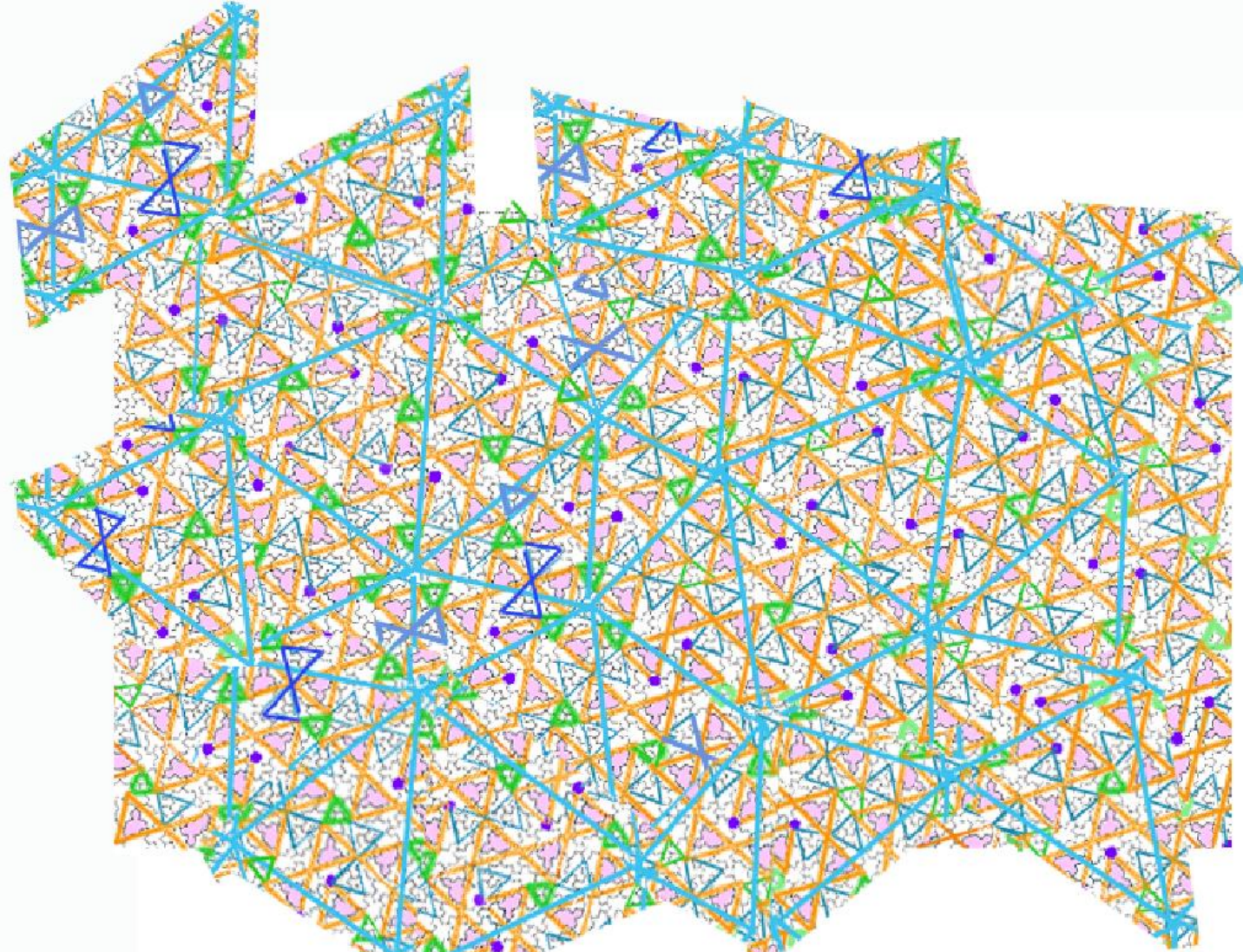


Triangular networks by Boris Horvat



Magenta color: triangular network between the “lone stars”

Triangular networks by Jean-François



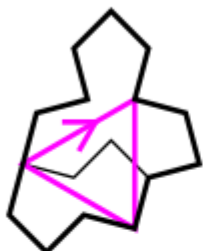
Same network between the “lone stars”

ODD and EVEN orientations for Tile(1,1)

$M_1 = M_{O_1}$



$M_3 = M_{O_2}$



$M_5 = M_{O_3}$



$M_7 = M_{O_4}$



$M_9 = M_{O_5}$



$M_{11} = M_{O_6}$



$M_2 = M_{E_1}$



$M_4 = M_{E_2}$



$M_6 = M_{E_3}$



$M_8 = M_{E_4}$

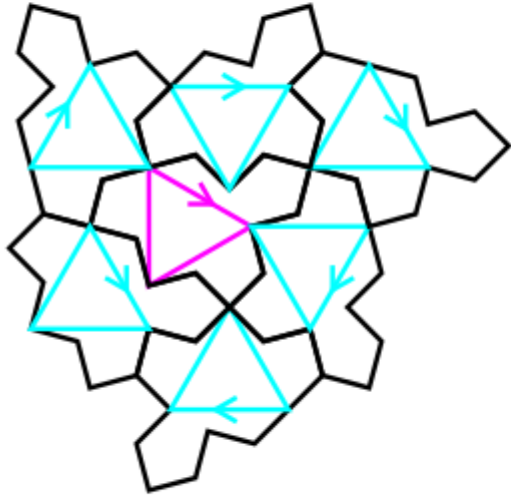


$M_{10} = M_{E_5}$

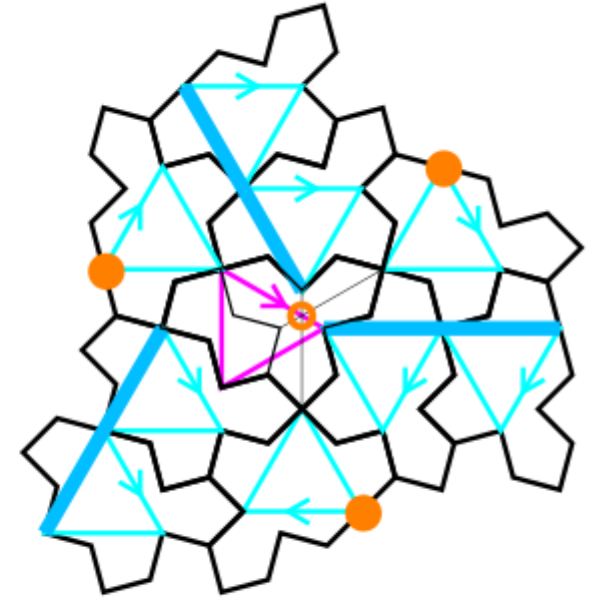
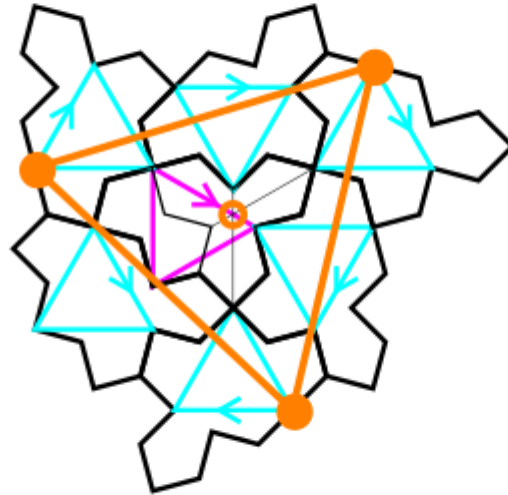


$M_{12} = M_{E_6}$

- Tile(1,1)



full cluster

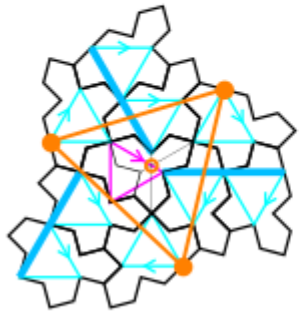


- C cluster: one ODD tile at the center + 6 EVEN tiles around
- Three glue points (orange color): regular triangle centered on the trifle shape of the ODD tile
- Full cluster: 3 more EVEN tiles obtained by translations (three dark blue lines)

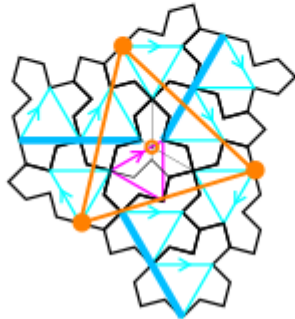
full cluster with its six rotations: $n(2\pi/6)$

- Tile(1,1)

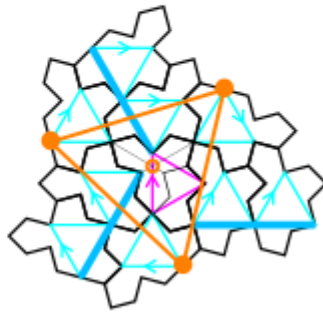
C1



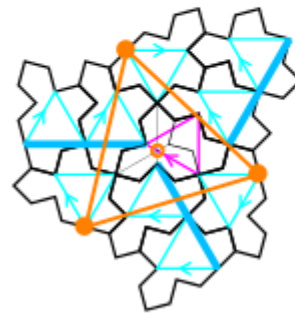
C2



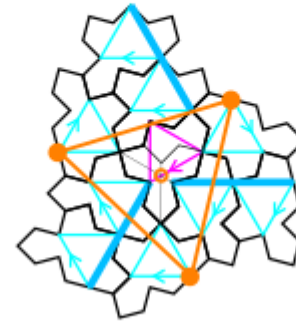
C3



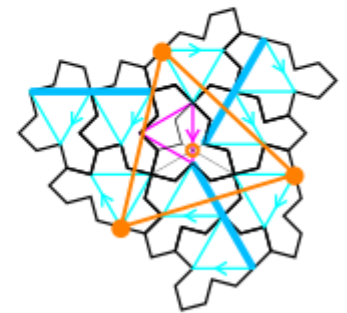
C4



C5

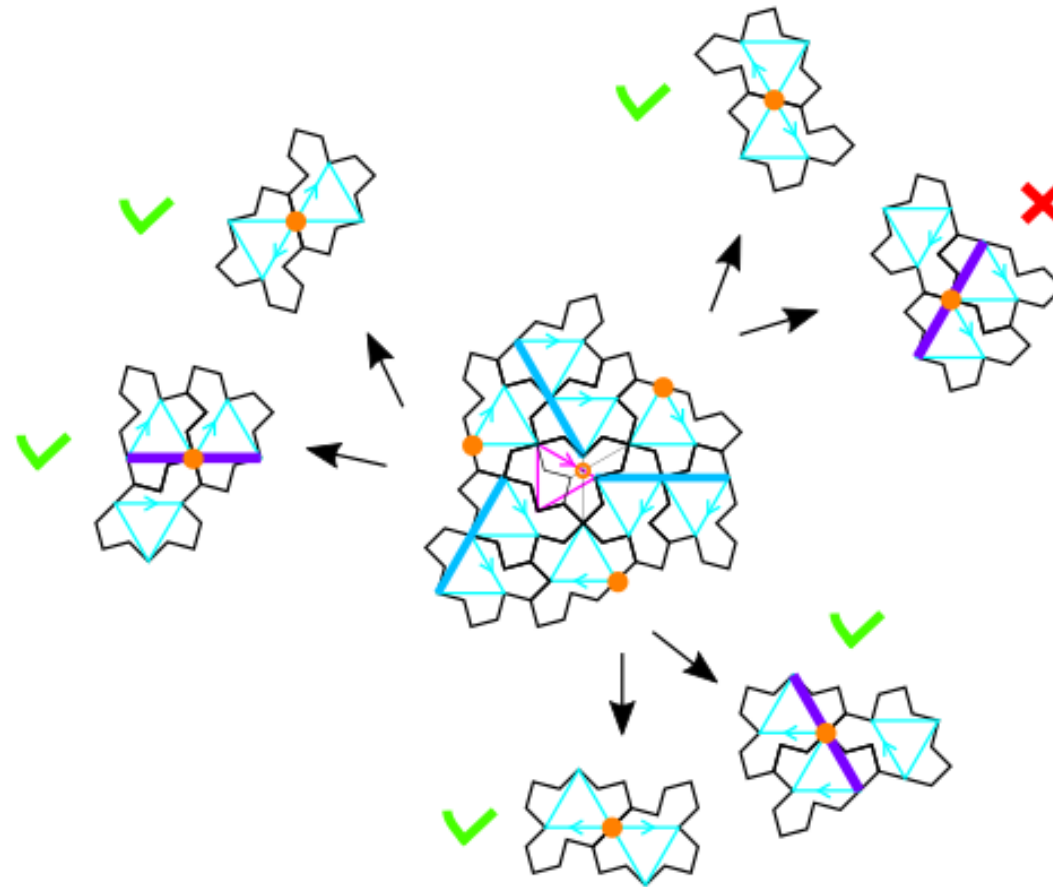


C6



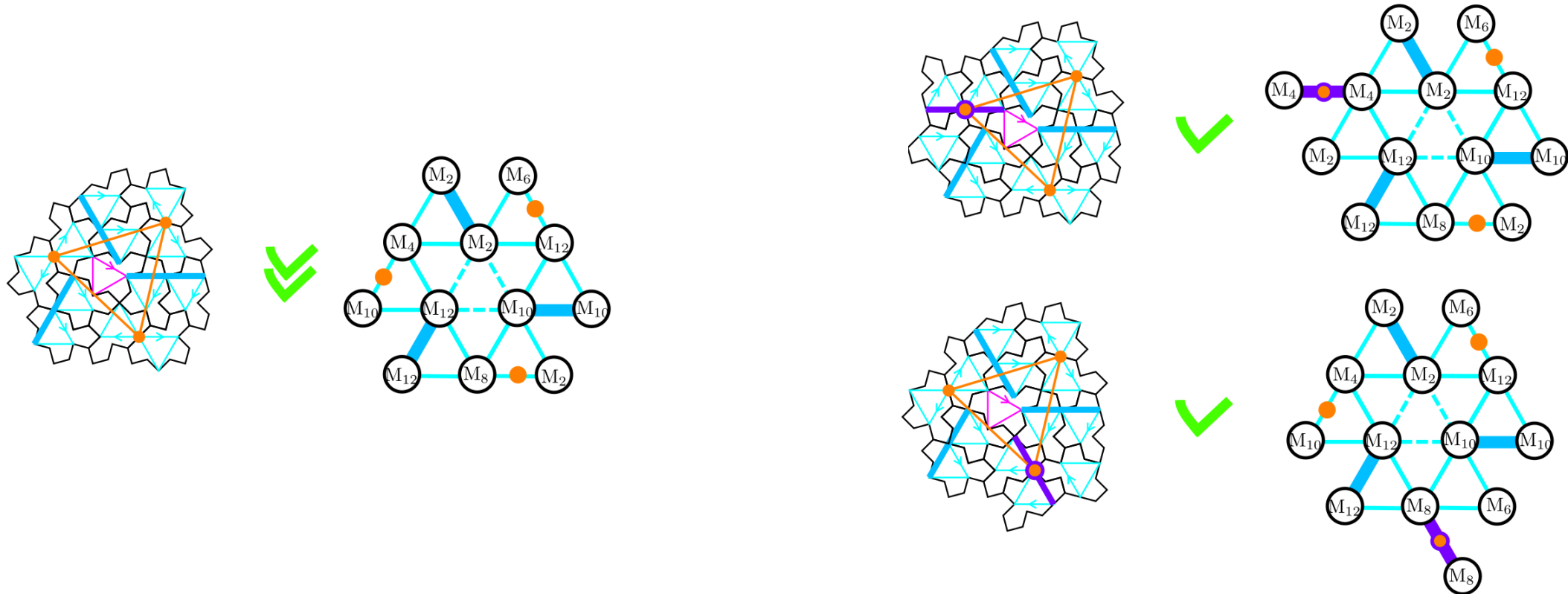
- 13 tiles in a full cluster: one ODD tile at the center + 9 EVEN tiles around
- Blue triangles (EVEN) have one horizontal edge when pink (ODD) have a vertical one

Possible configurations at the glue points ?



- Half-turned or translated monotile (violet line)

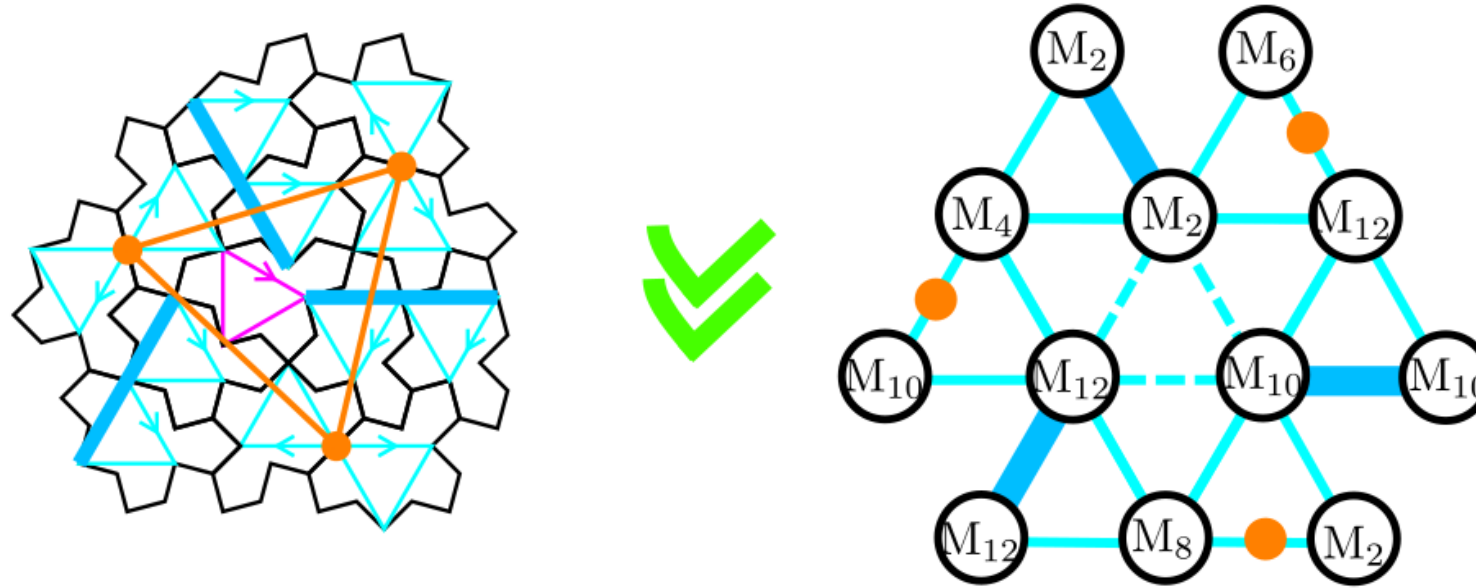
Three possible configurations at the glue points



- configuration with three half-turned tiles is the most common
- The other two are always coupled together in the tiling:
- One violet line in a 2-arms and three violet lines in a 3-arms

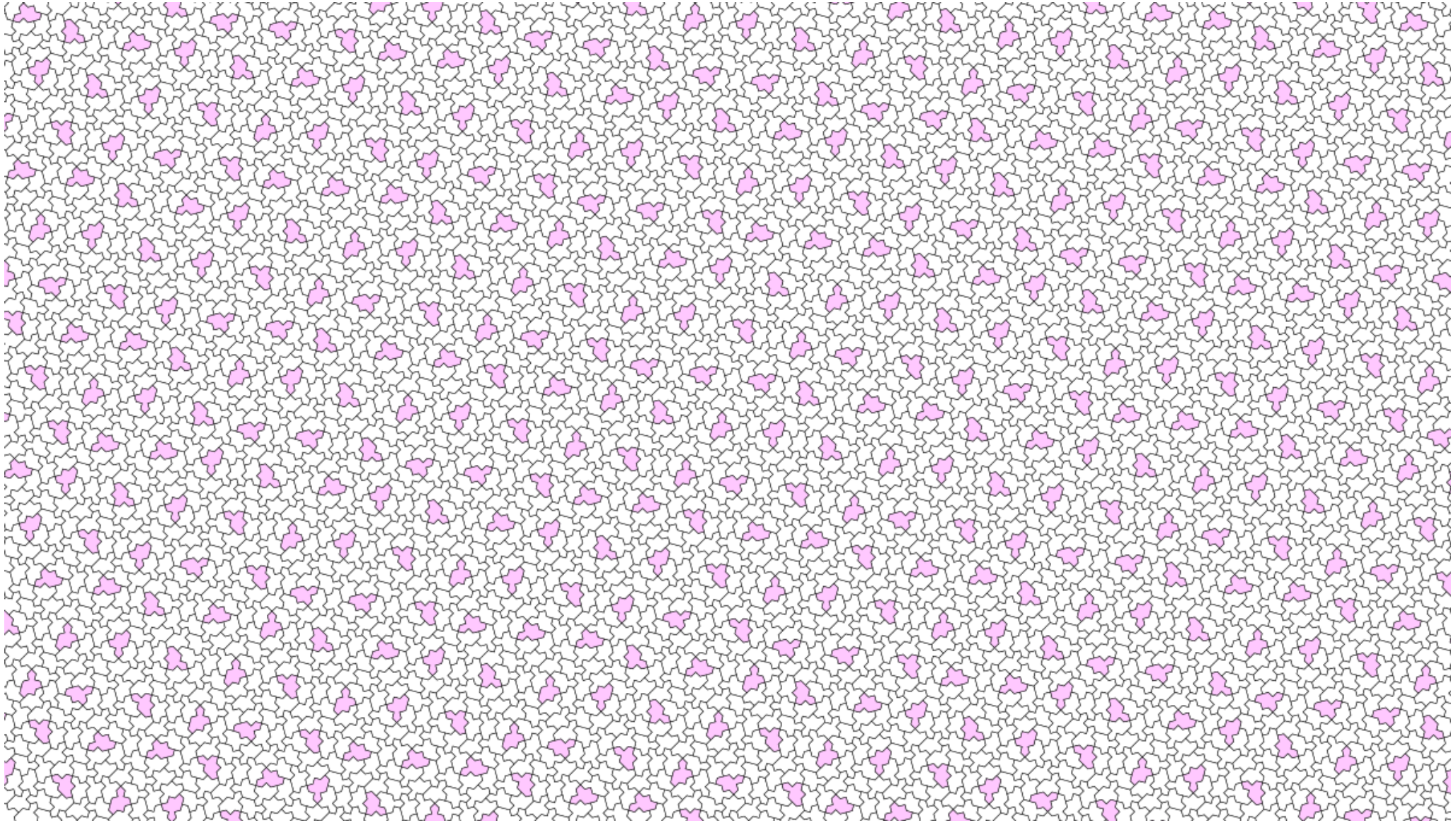
Orientation of the EVEN tiles in the most frequent configuration

- Tile(1,1)



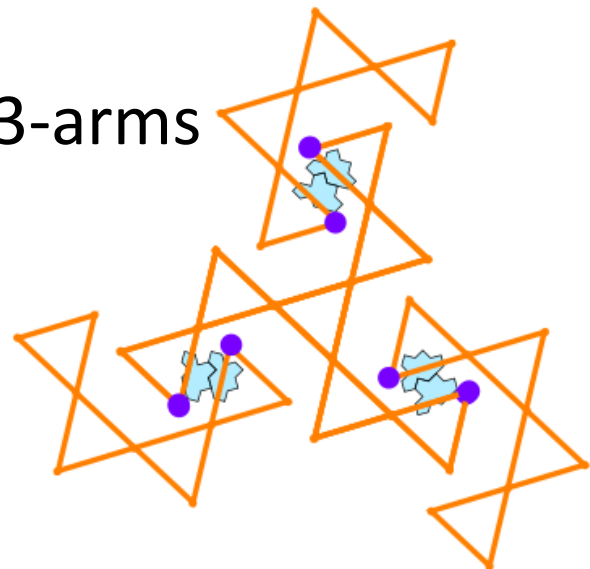
- 6 tiles in the C-cluster (corona made of M12, M4, M2, M12, M10, M8)
- 3 translated (M2, M10, M12) (along blue lines, the edges of blue triangles)
- 3 half-turned (M4 → M10, M12 → M6, M8 → M2)

Construction for the whole tiling ???

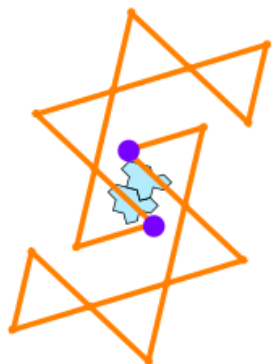


Three patterns in the whole tiling 😊

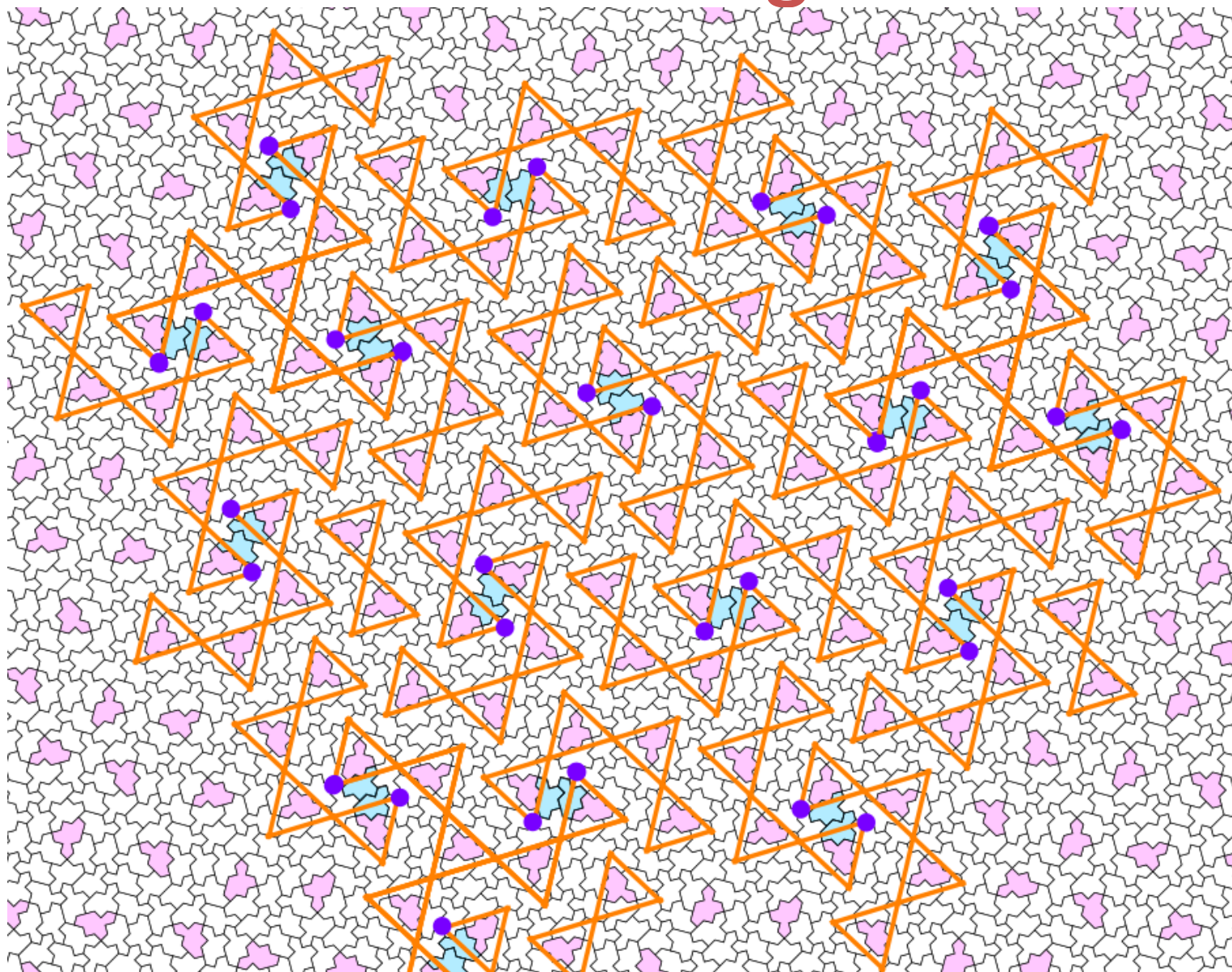
■ 3-arms



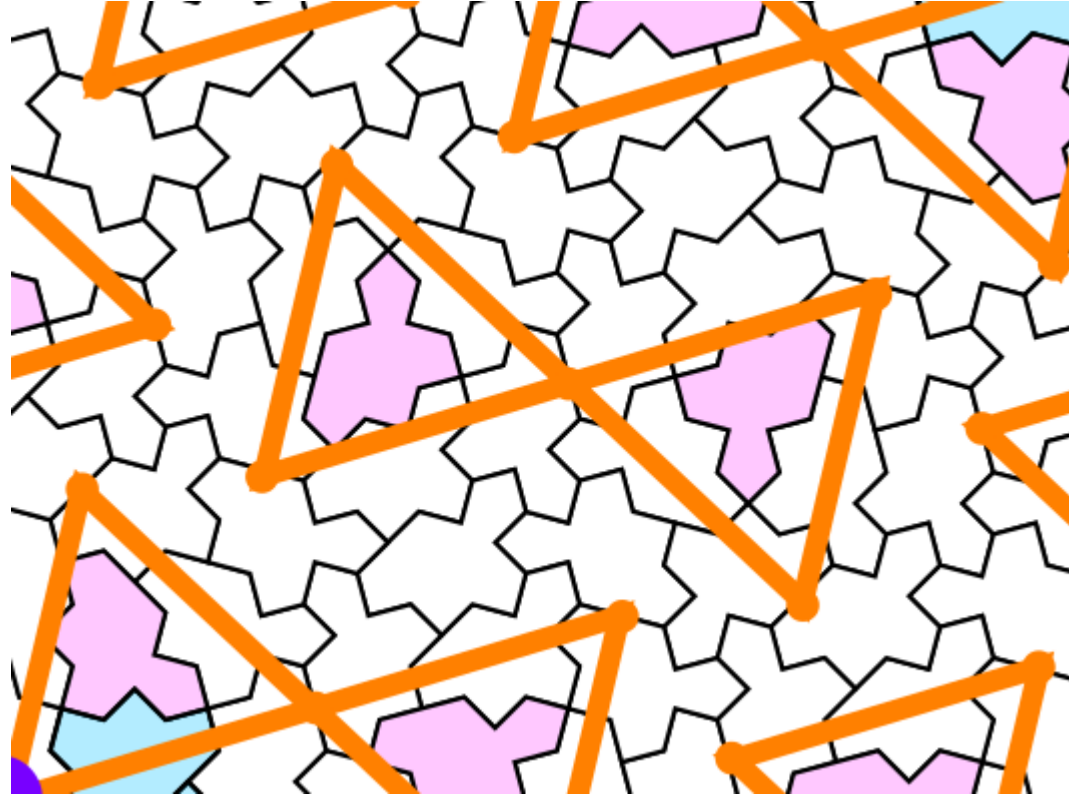
■ 2-arms



■ diabolos

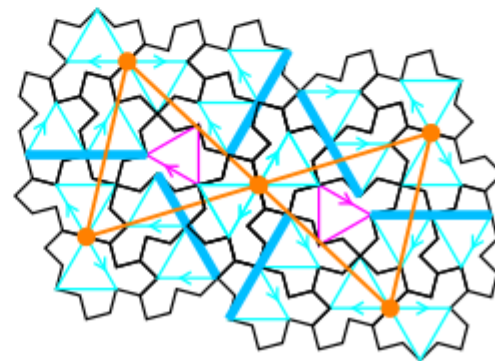
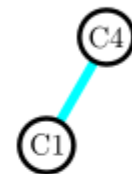
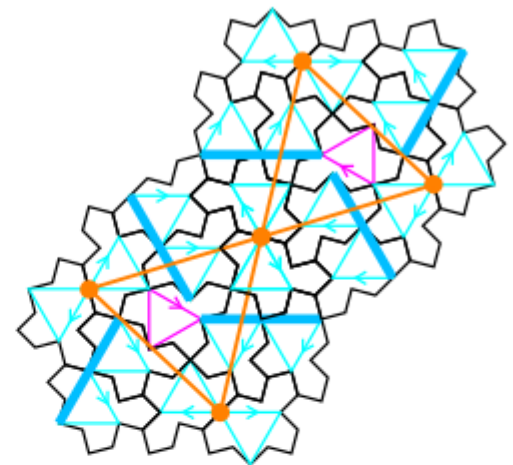
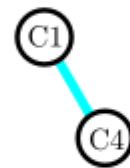
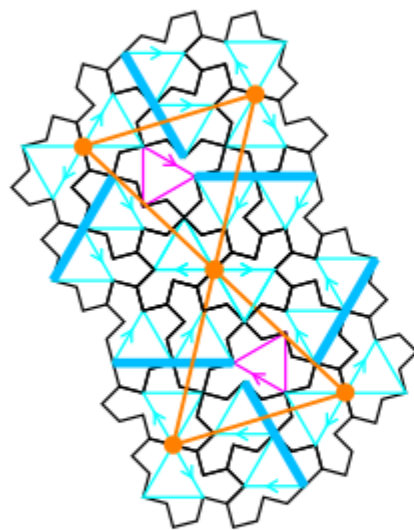
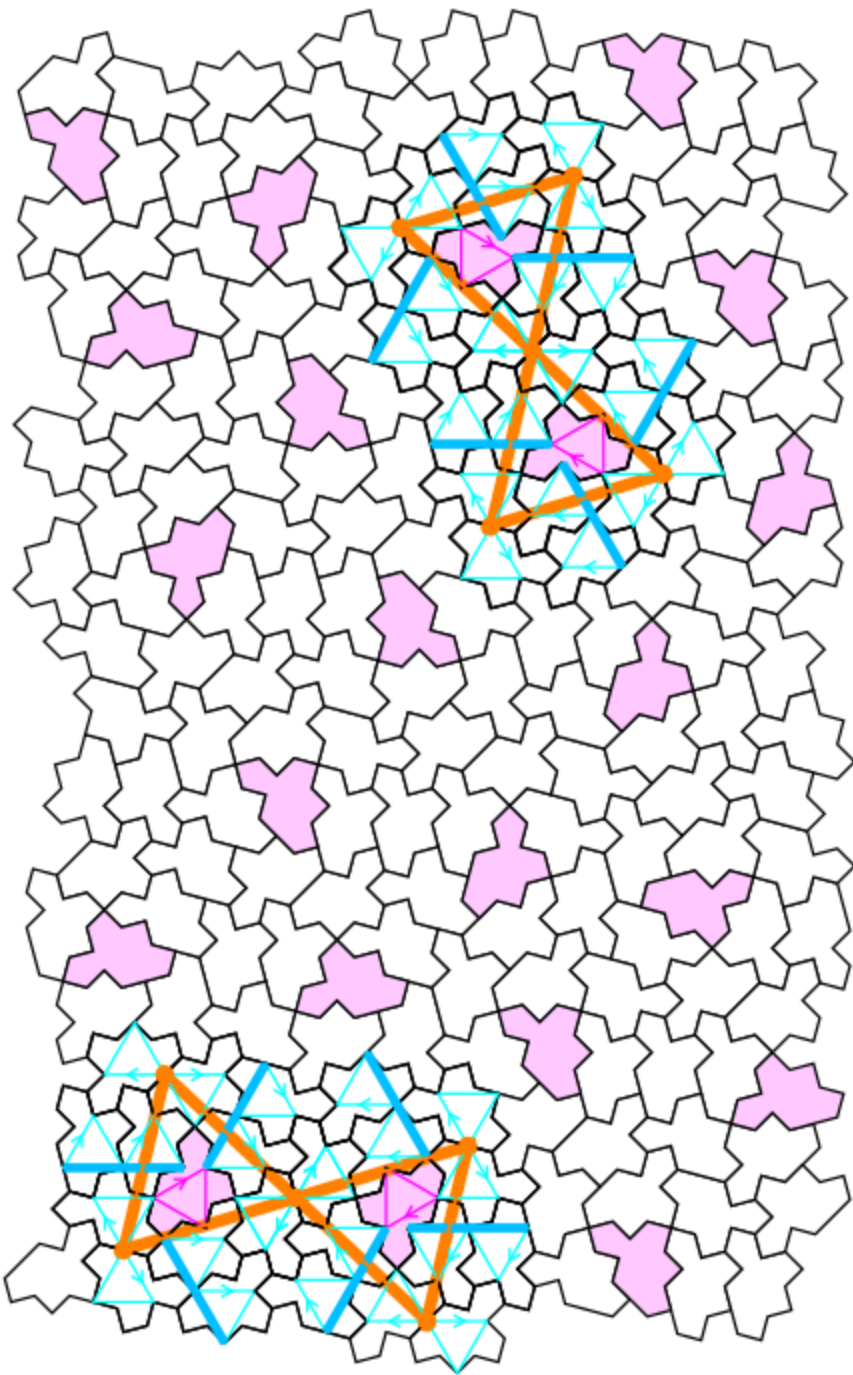


Diabolo pattern

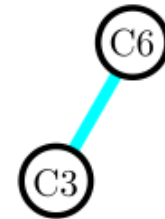
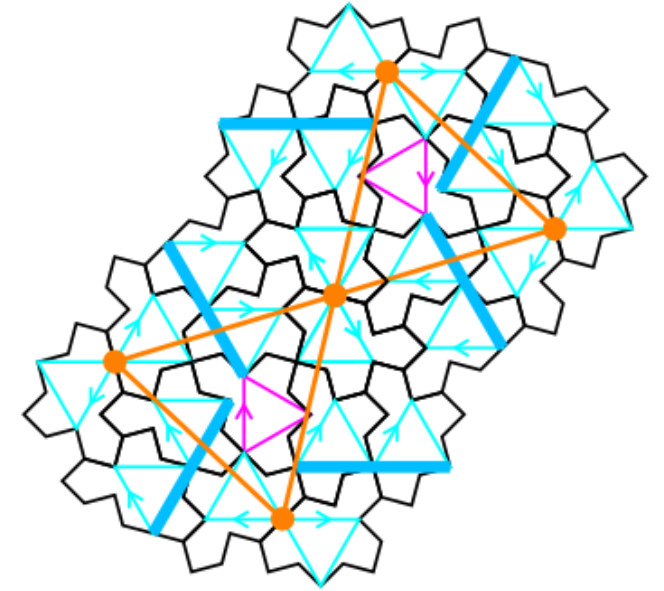
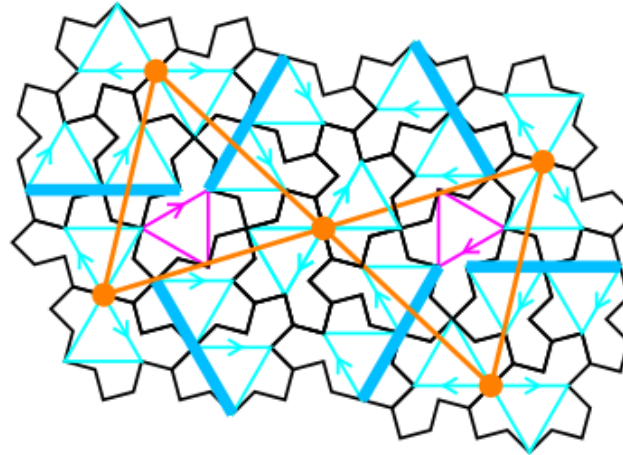
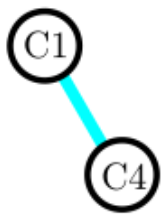
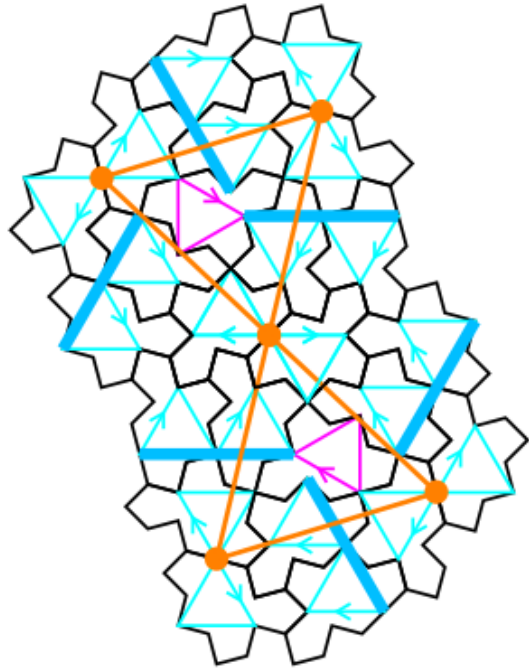


- Local 2-fold axis

Diabolo pattern

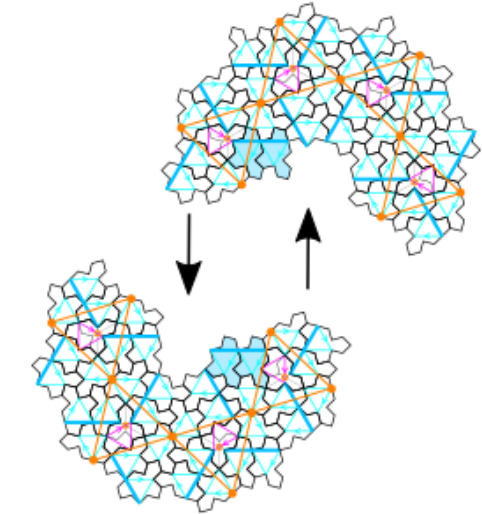
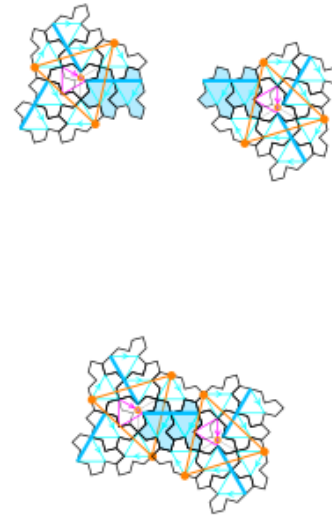
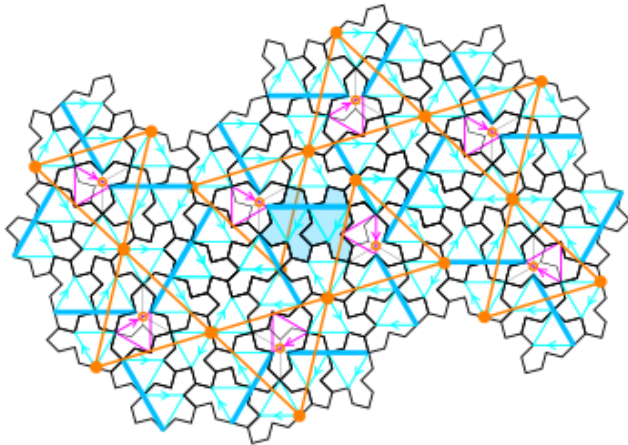
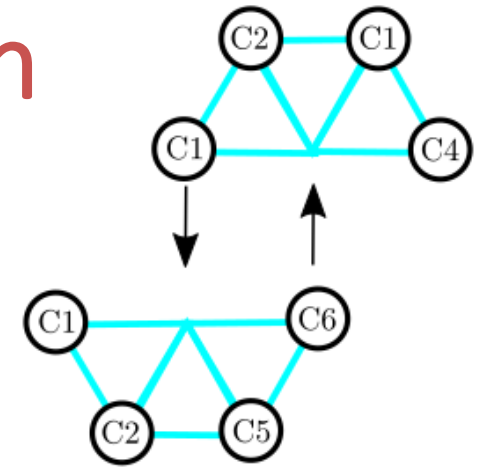
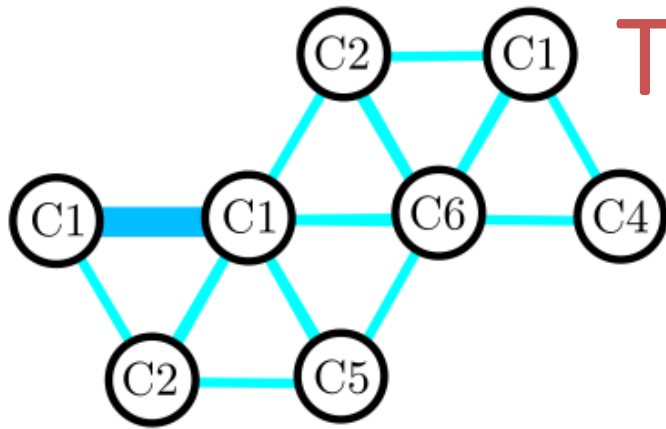


Diabolo pattern



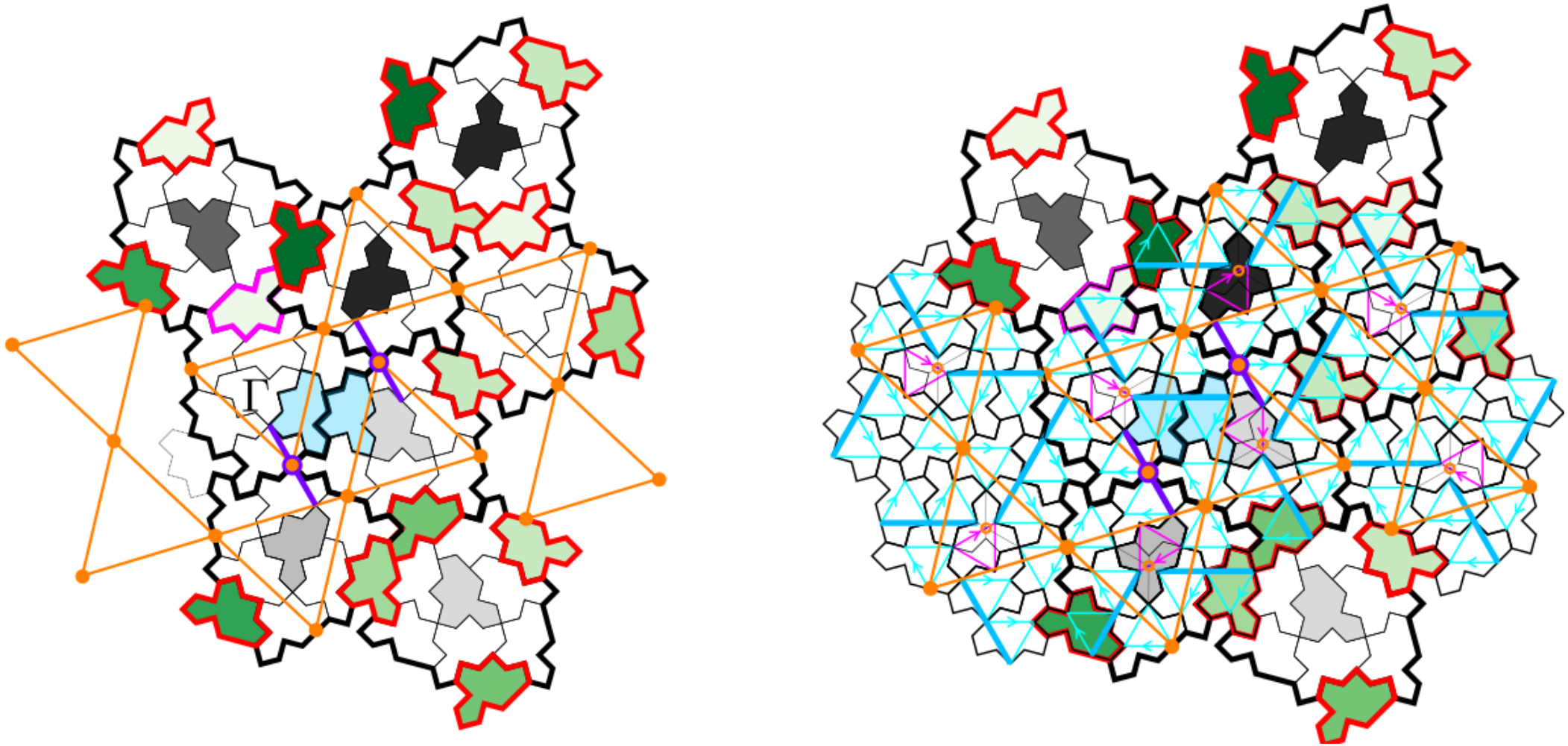
- Three orientations of the same diabolo pattern

Two-arms chiral pattern



- Two 'croissants' of 4 clusters
- Two translated monotiles in the middle + superposition of glue points
- Easy to code ! 😊

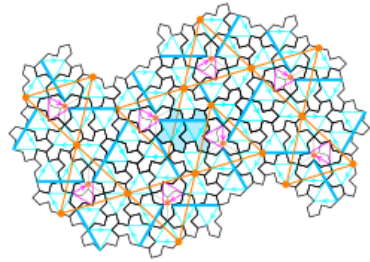
Link with triangular pattern



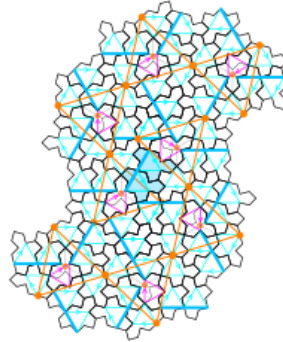
- The two other environnements with a violet lines are always coupled together

Six orientations of the 2-arms

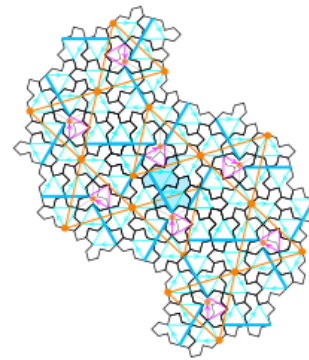
P1



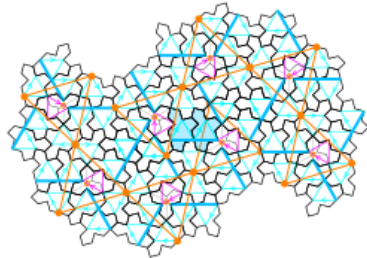
P2



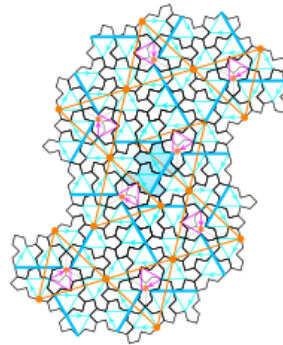
P3



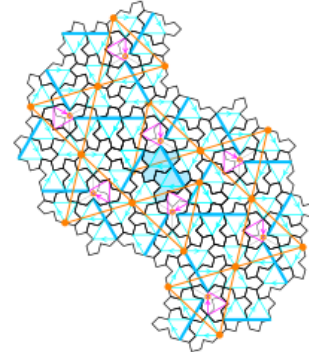
P4



P5

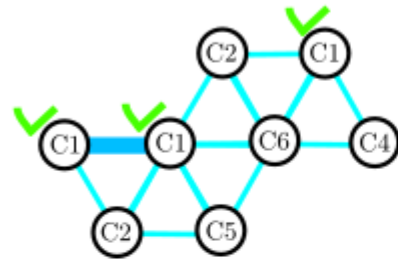


P6

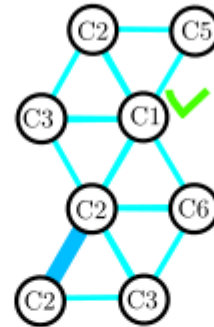


Six orientations of the 2-arms

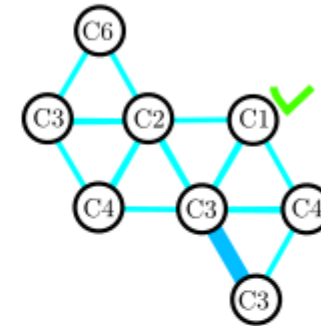
P1



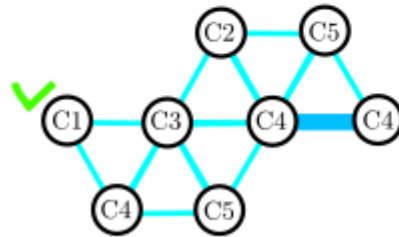
P2



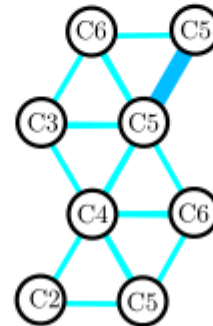
P3



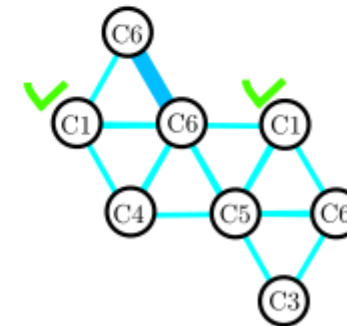
P4



P5

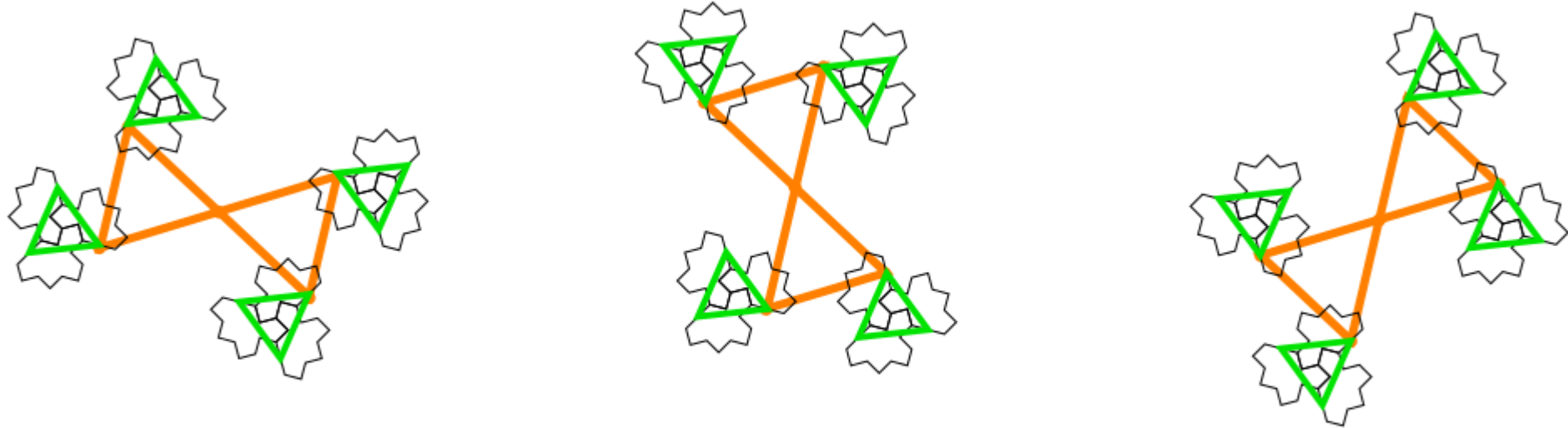


P6

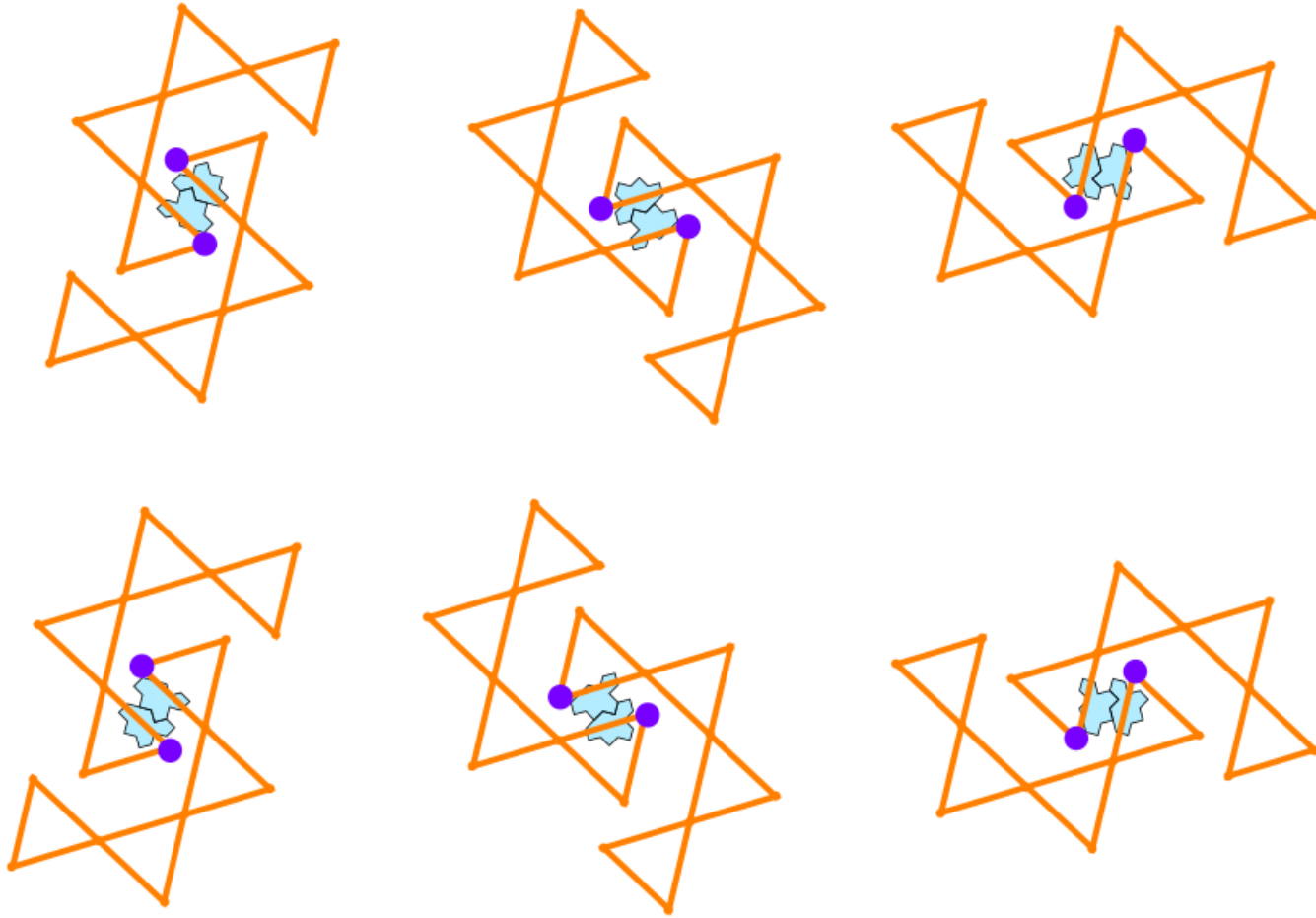


- 8 combinations for a given cluster orientation (green crosses)

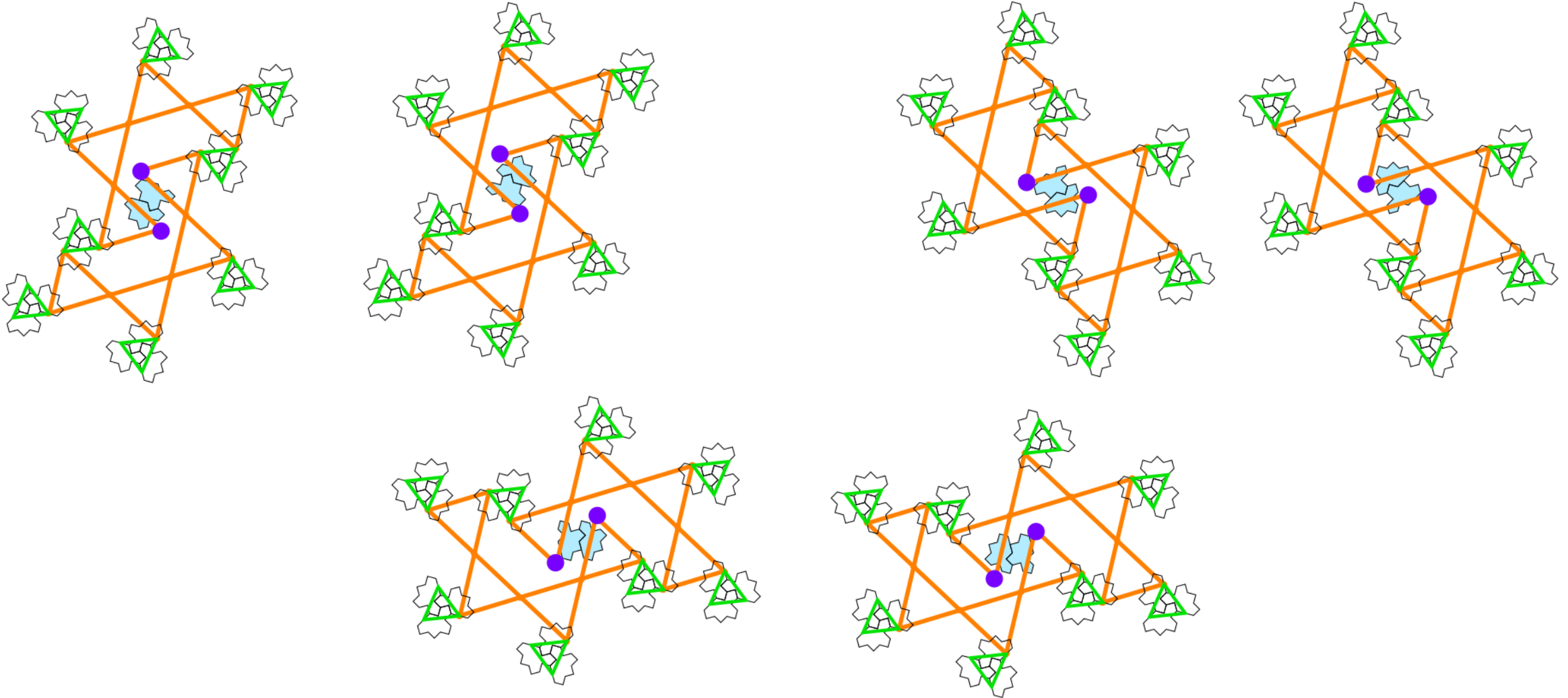
three orientations for diabolos



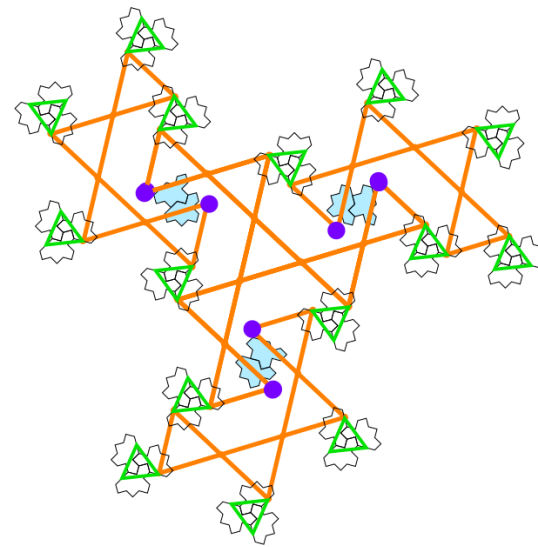
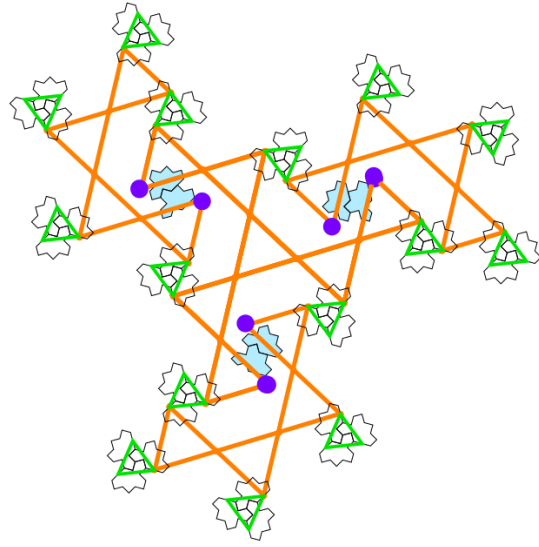
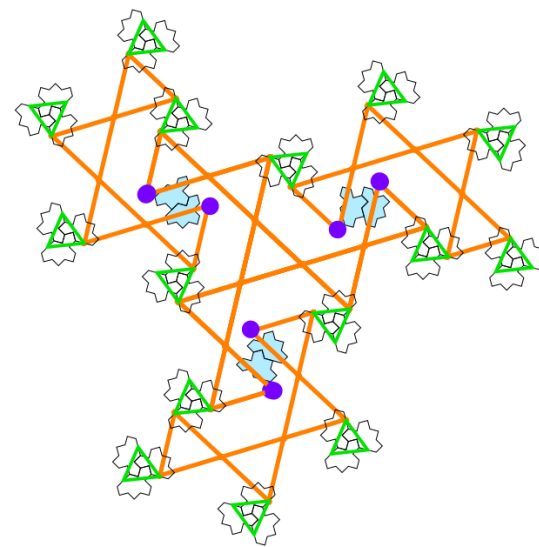
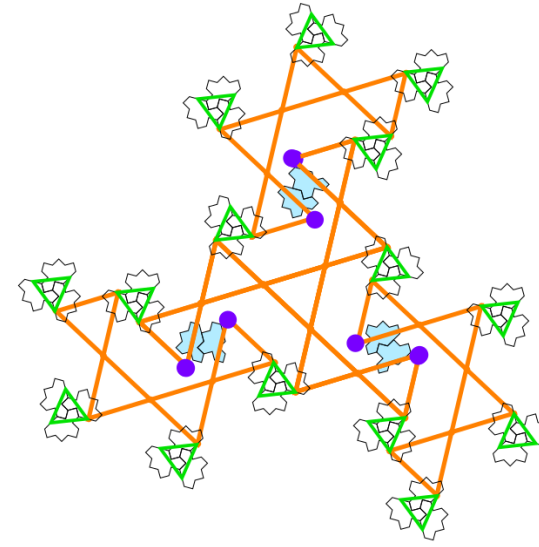
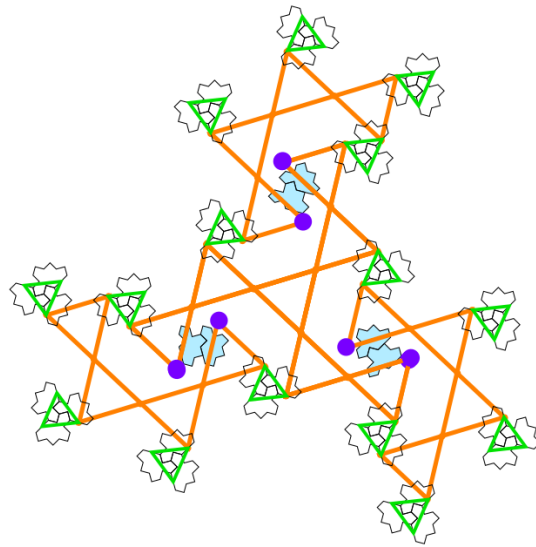
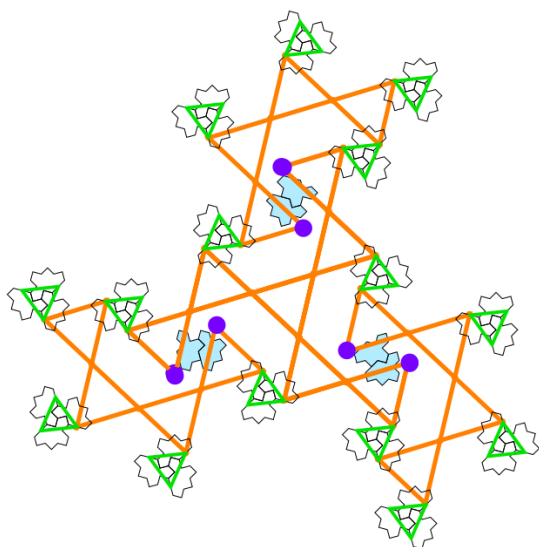
Six orientations of the 2-arms



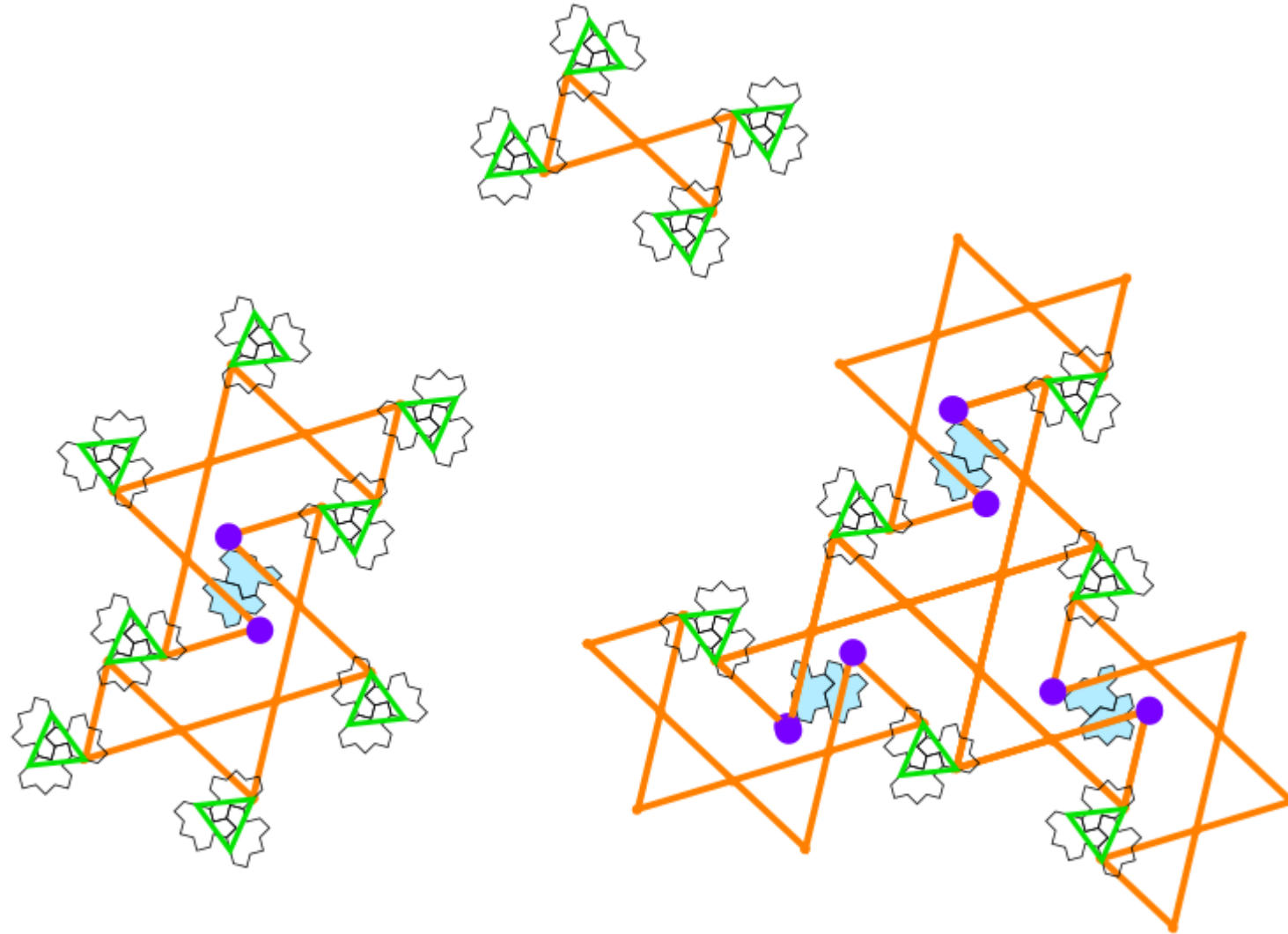
Six orientations of the 2-arms



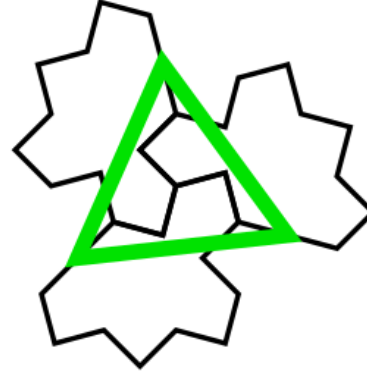
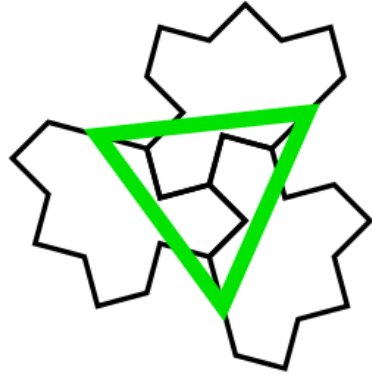
six orientations of the 3-arms



New puzzle

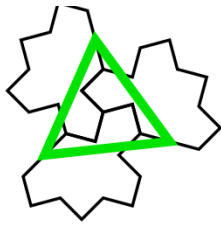
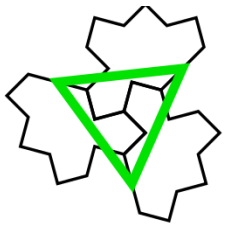
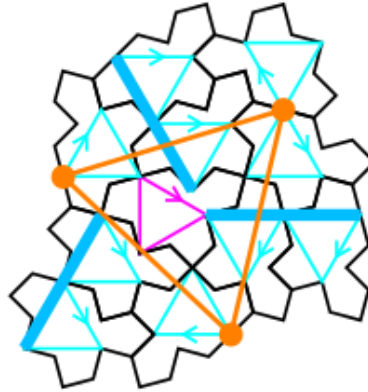
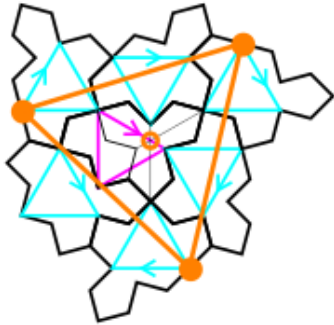
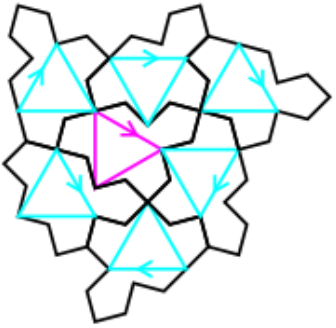


stars



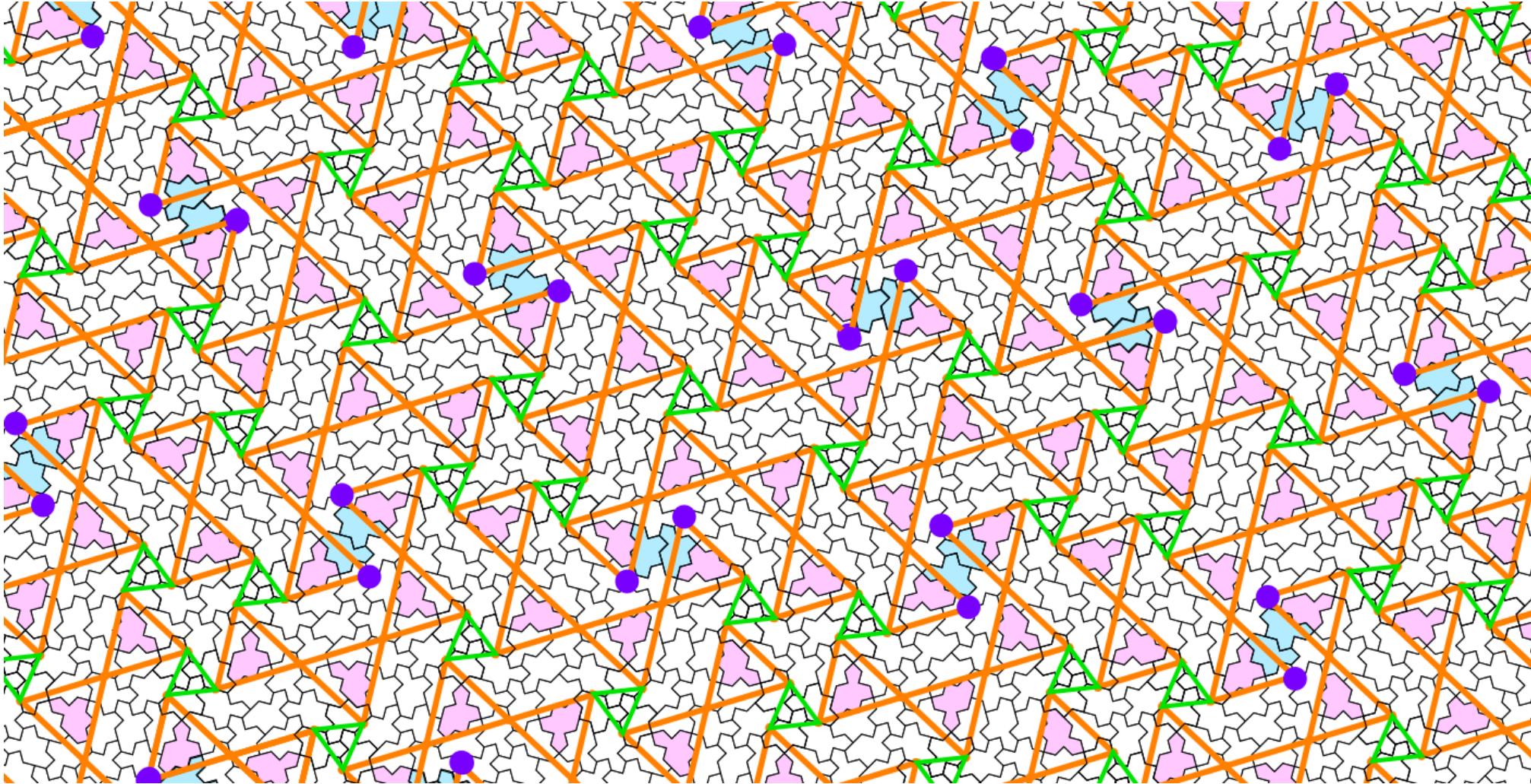
- Small and large stars with two possible orientations

Edge lengths of several triangles: pink, orange, green and blue



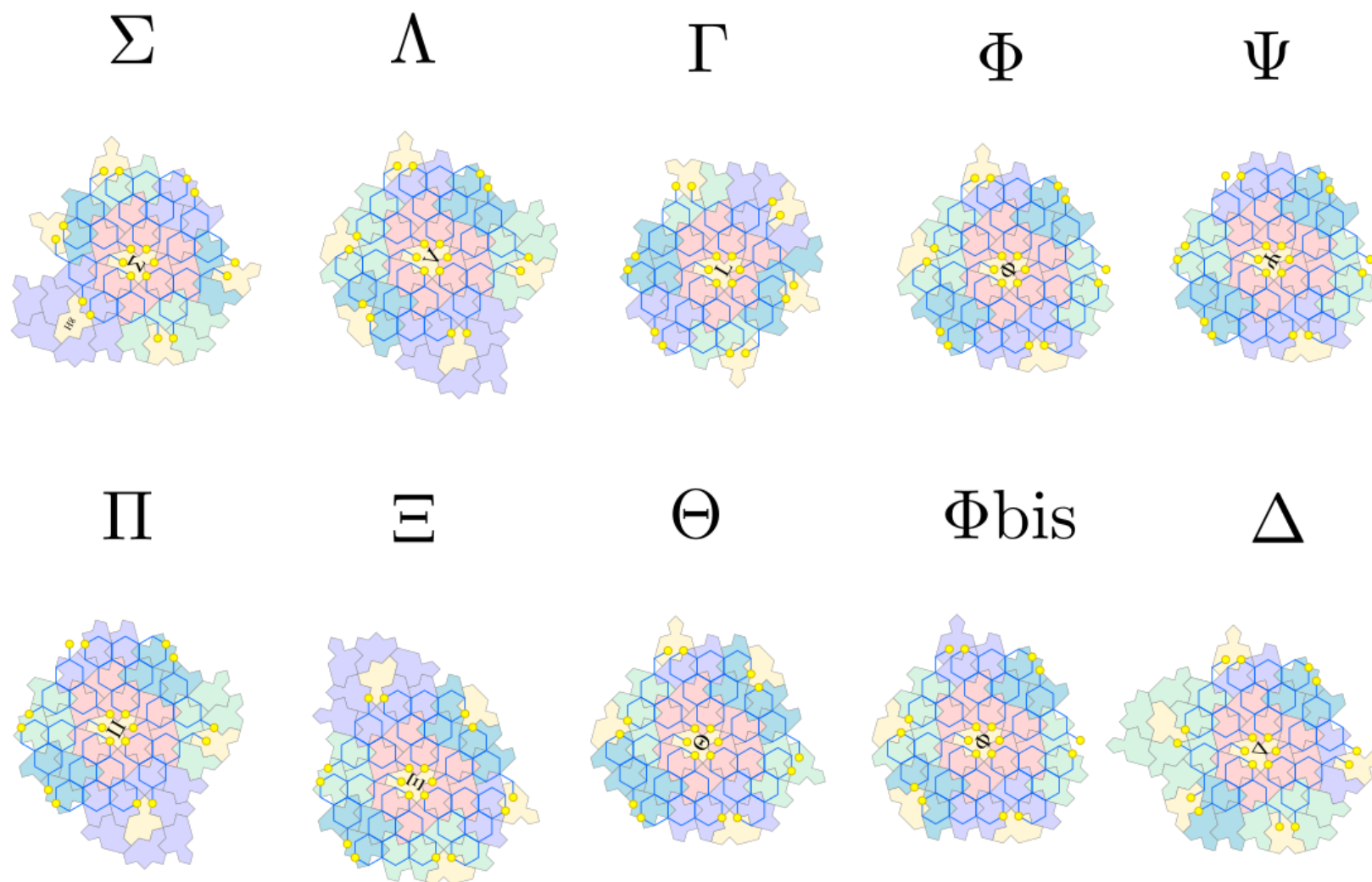
- TO DO: calculate the edge length for all these different regular triangles (take 1 for the edge length in the monotile)
- And do it after when applying the $T(a,b)$ transformation (all remain regular !)

Small stars connect the orange triangles



- Possibility to obtain the relative positions between diablo, 2-arms and 3-arms

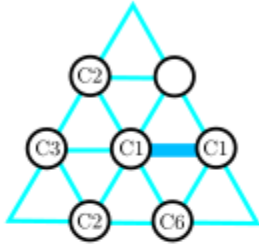
10 environments (C1 orientation)



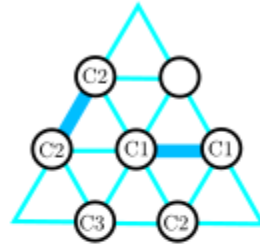
- Left handed version of the figure in Arnaud's paper.

10 environments for C1

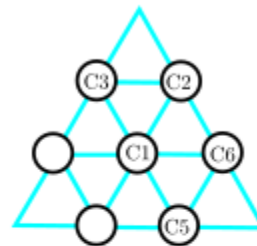
Σ



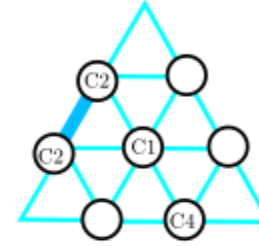
Λ



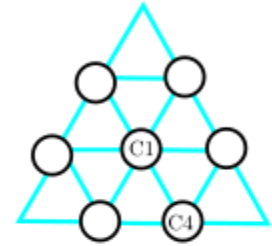
Γ



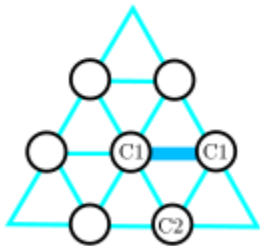
Φ



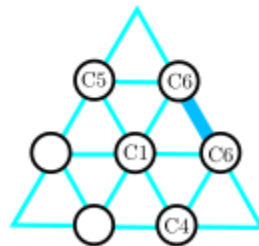
Ψ



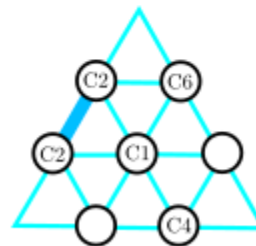
Π



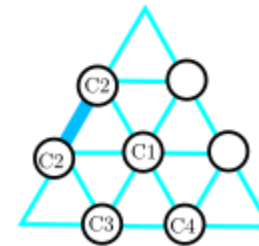
Ξ



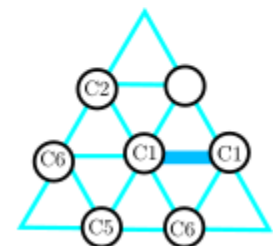
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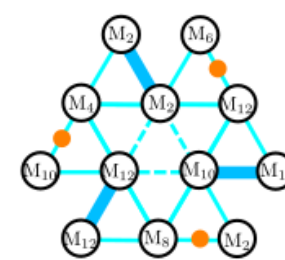
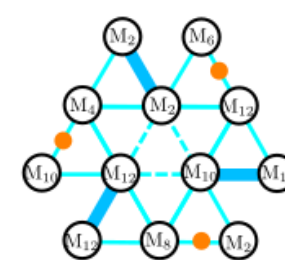
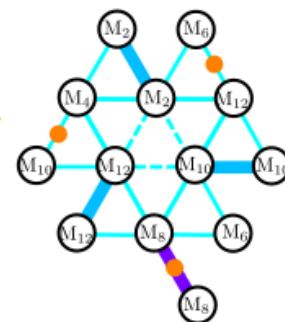
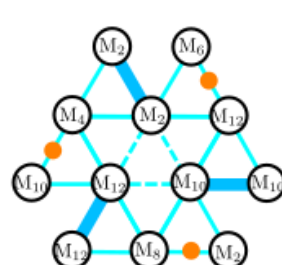
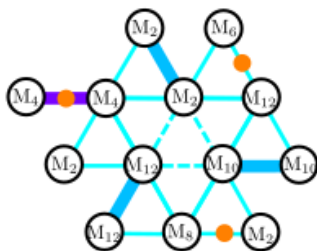
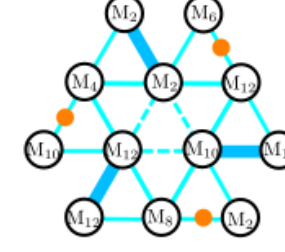
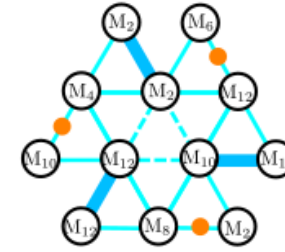
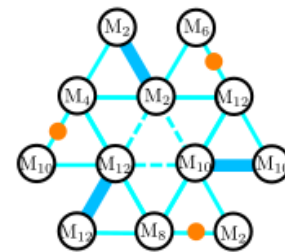
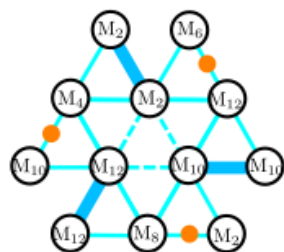
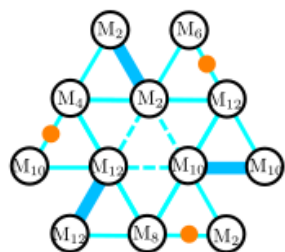
Φ bis



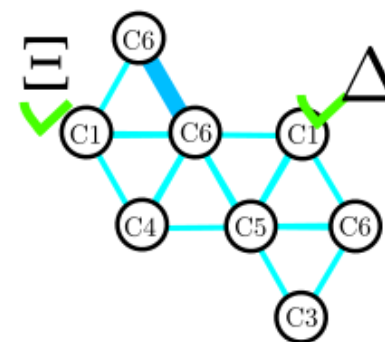
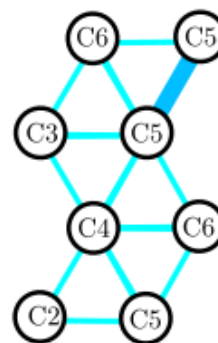
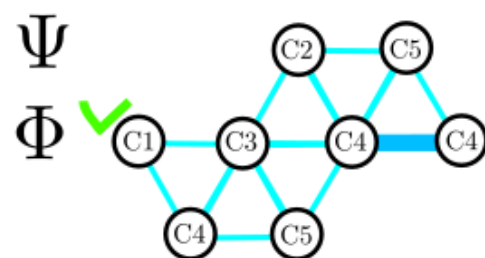
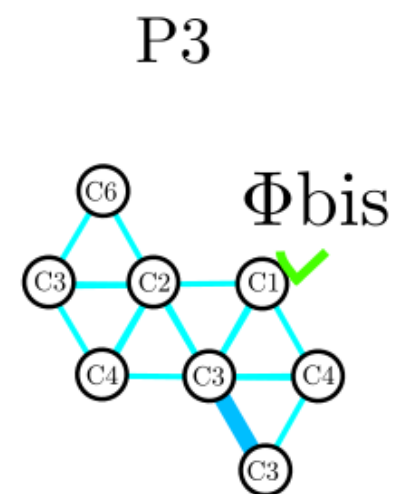
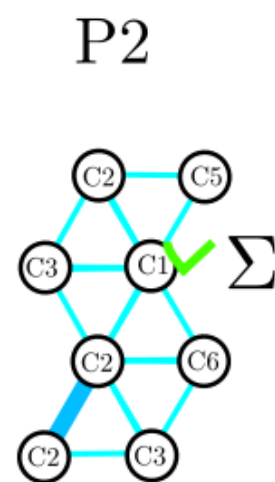
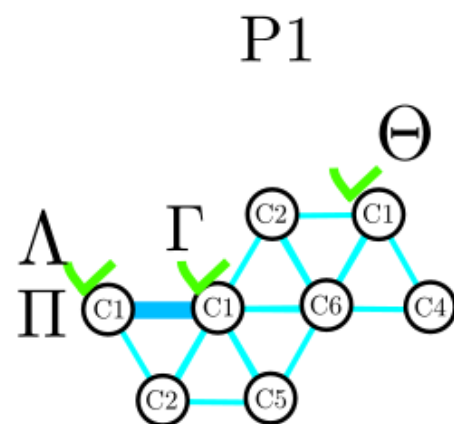
Δ



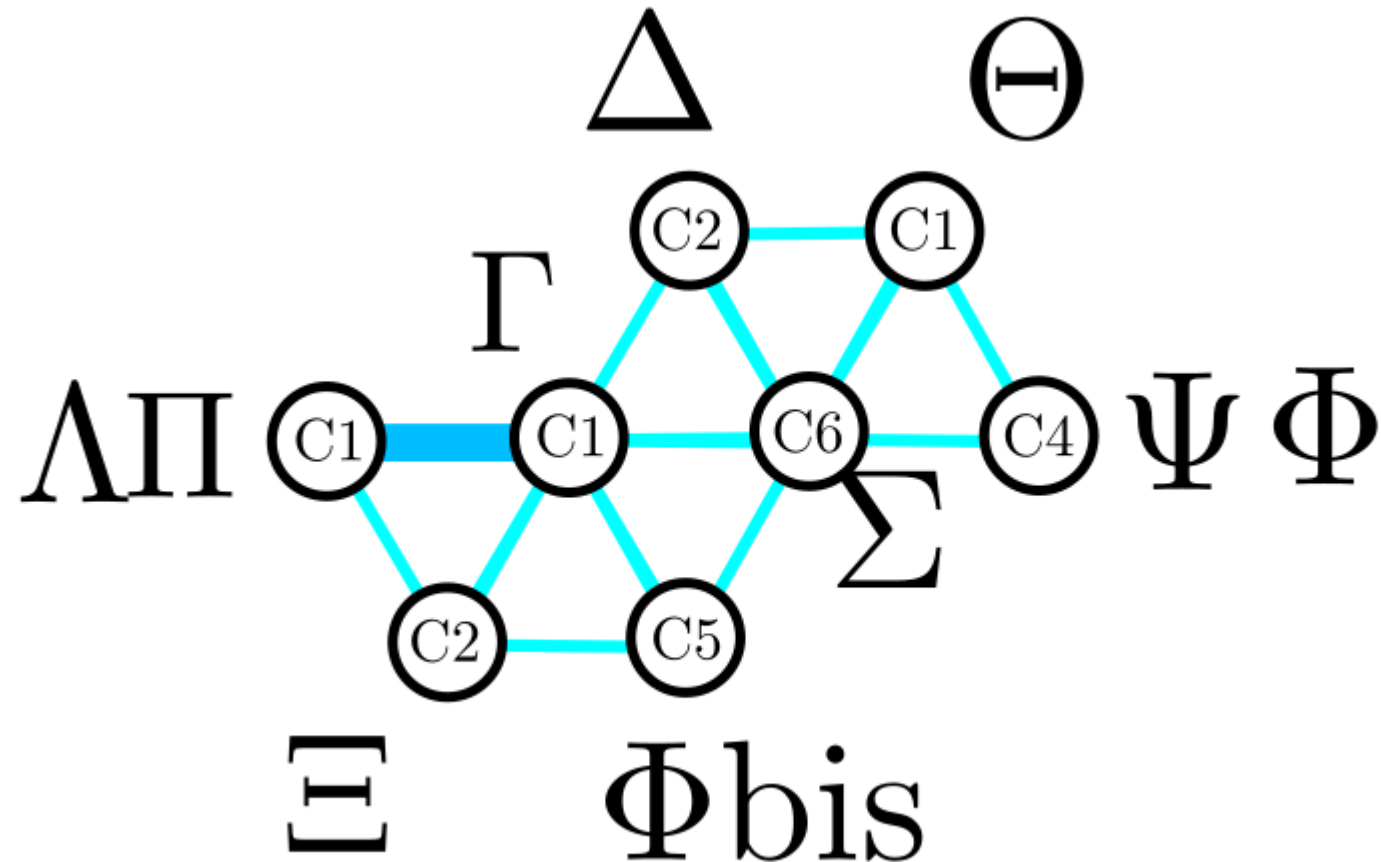
10 environments for C1

 Σ
 Λ
 Γ
 Φ
 Ψ

 Π
 Ξ
 Θ
 Φ_{bis}
 Δ


8 different locations in the chiral pattern

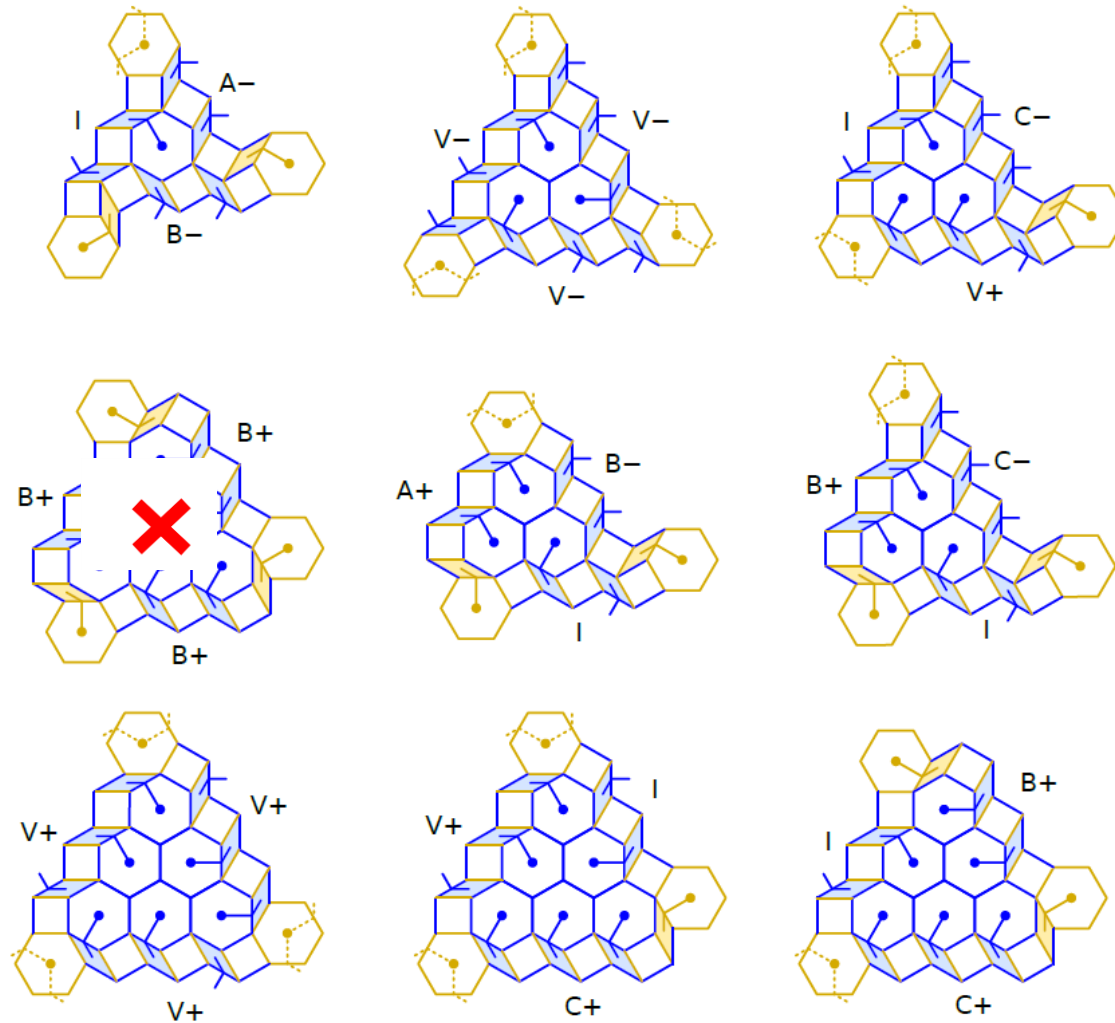


10 environments (capital greek letters)



- Need to reduce to 8 environments ?

8 types of hexagonal clusters



To do:
Locate the 8 hexagonal
clusters in the chiral pattern

Homochiral python code on github

- Under construction !
- <https://github.com/marimperorclerc/homochiral>

python code for drawing clusters with Tile(a,b) in svg graphics.

References

Code is based on the drawsvg package:

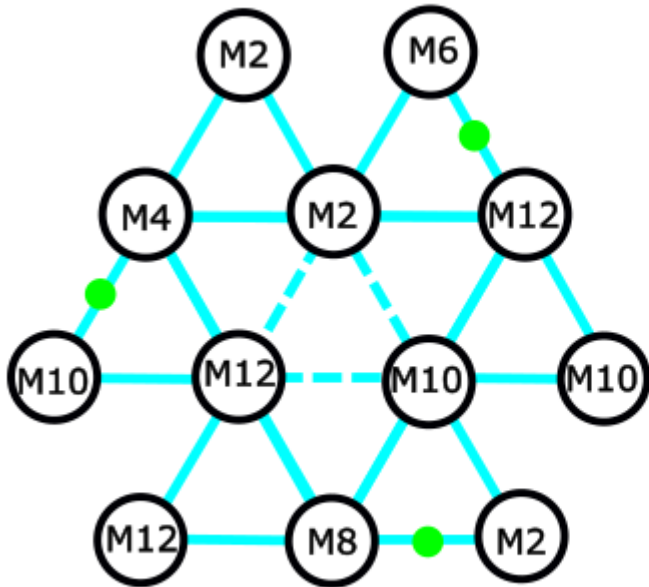
<https://pypi.org/project/drawsvg/>

And the python code with alternate chirality for inflation for spectre tiling:

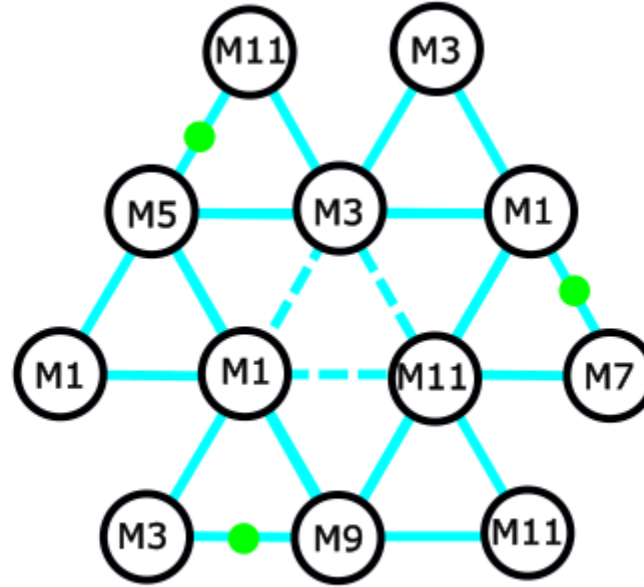
<https://github.com/shrx/spectre/blob/master/spectre.py>

Inflation step 1: substitution rule for C1

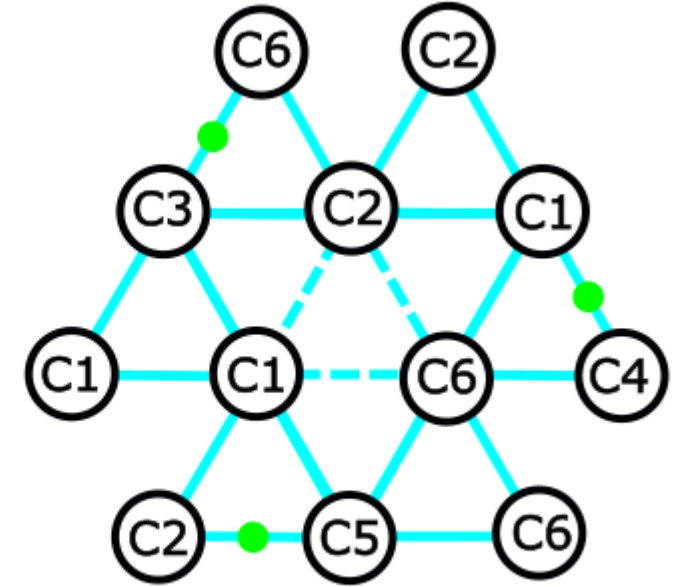
EVEN tiles in C1



ODD tiles for step 1

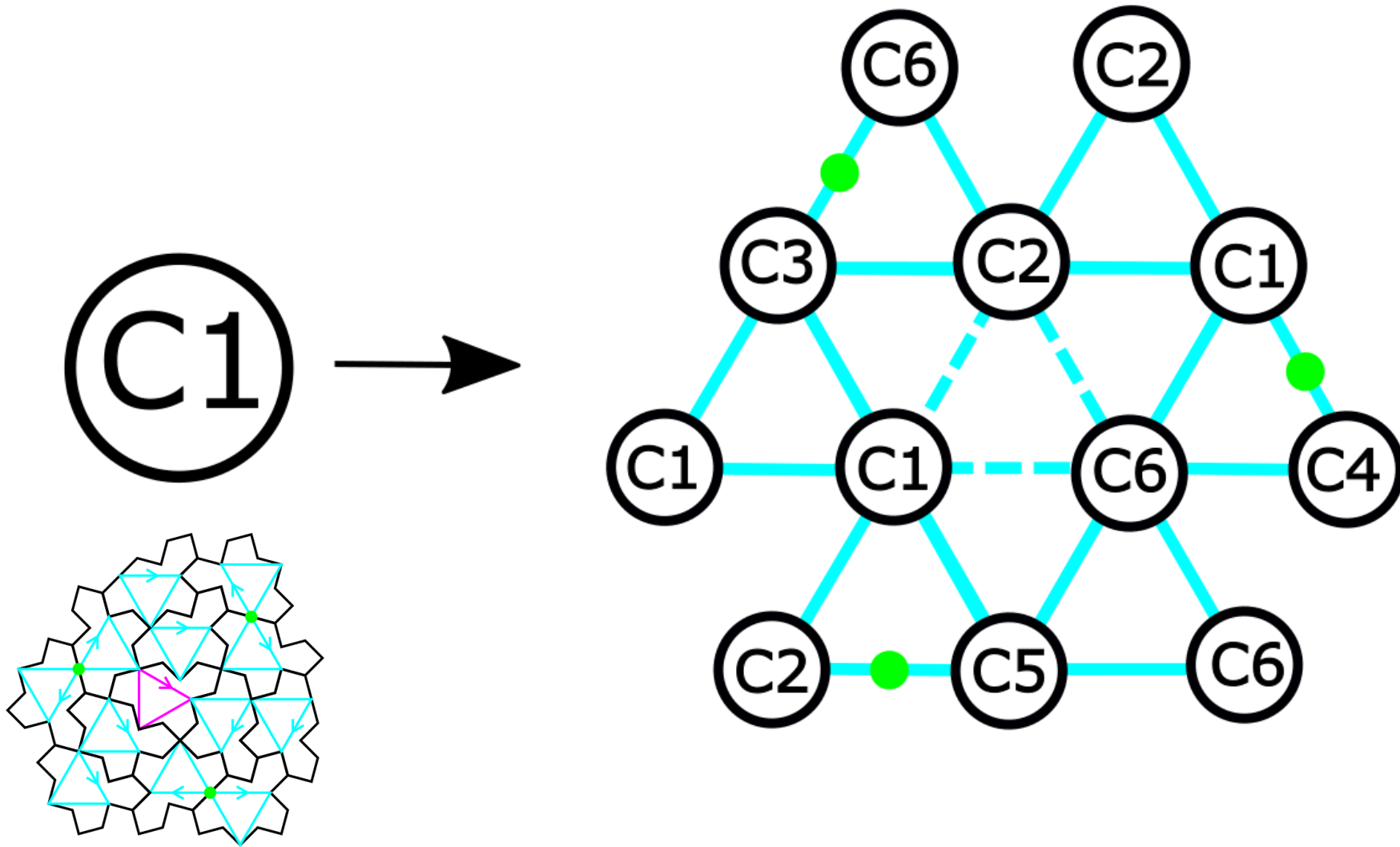


same as full clusters

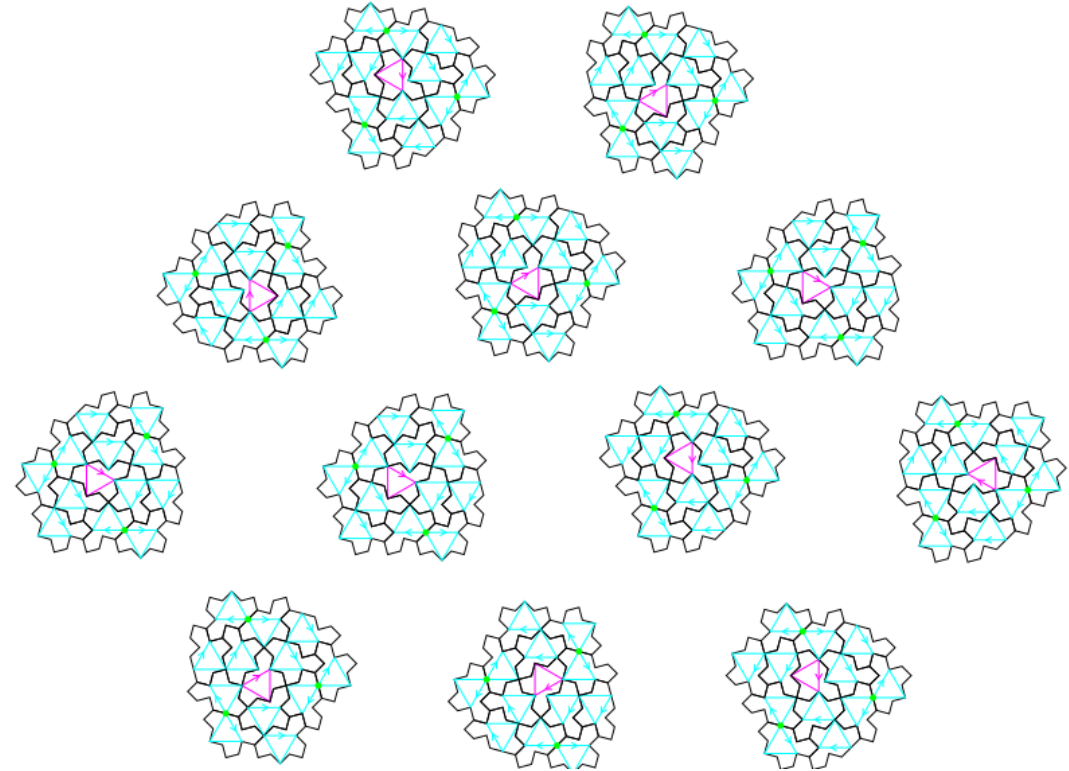
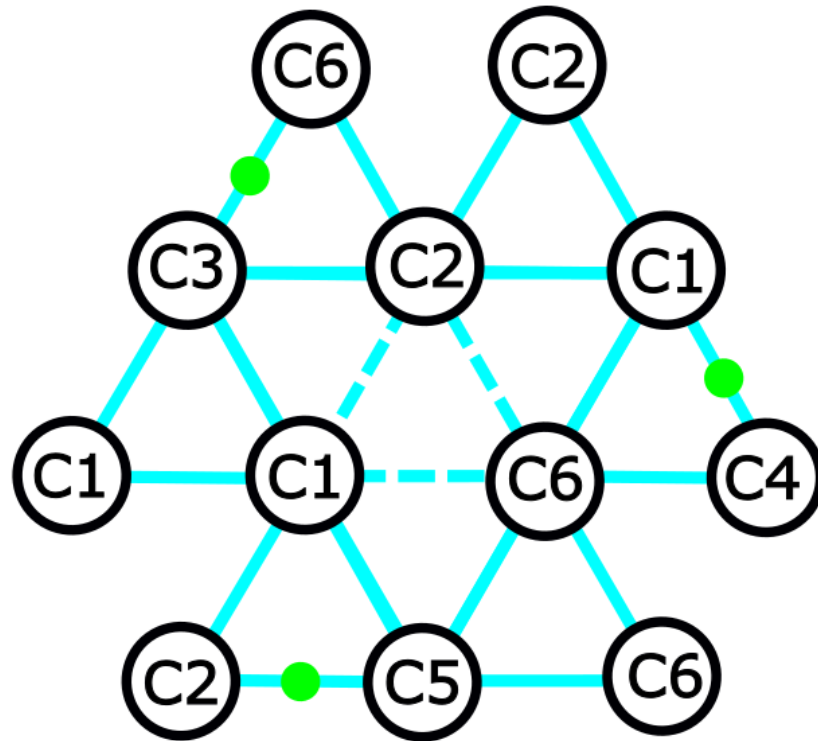


- ODD tiles for step 1 are obtained by adding 1 (mod 12) to all EVEN tiles in the C cluster (6 tiles)
- Directions of translated and half-turned tiles are exchanged (**see the change of position for green points**). Then three tiles are translated (M1, M3, M11) and three are half-turned (M5 $\rightarrow$ M11, M1 $\rightarrow$ M7, M9 $\rightarrow$ M3)
- ODD tile index is equivalent to the orientation of a full C cluster 😊

Inflation step 1

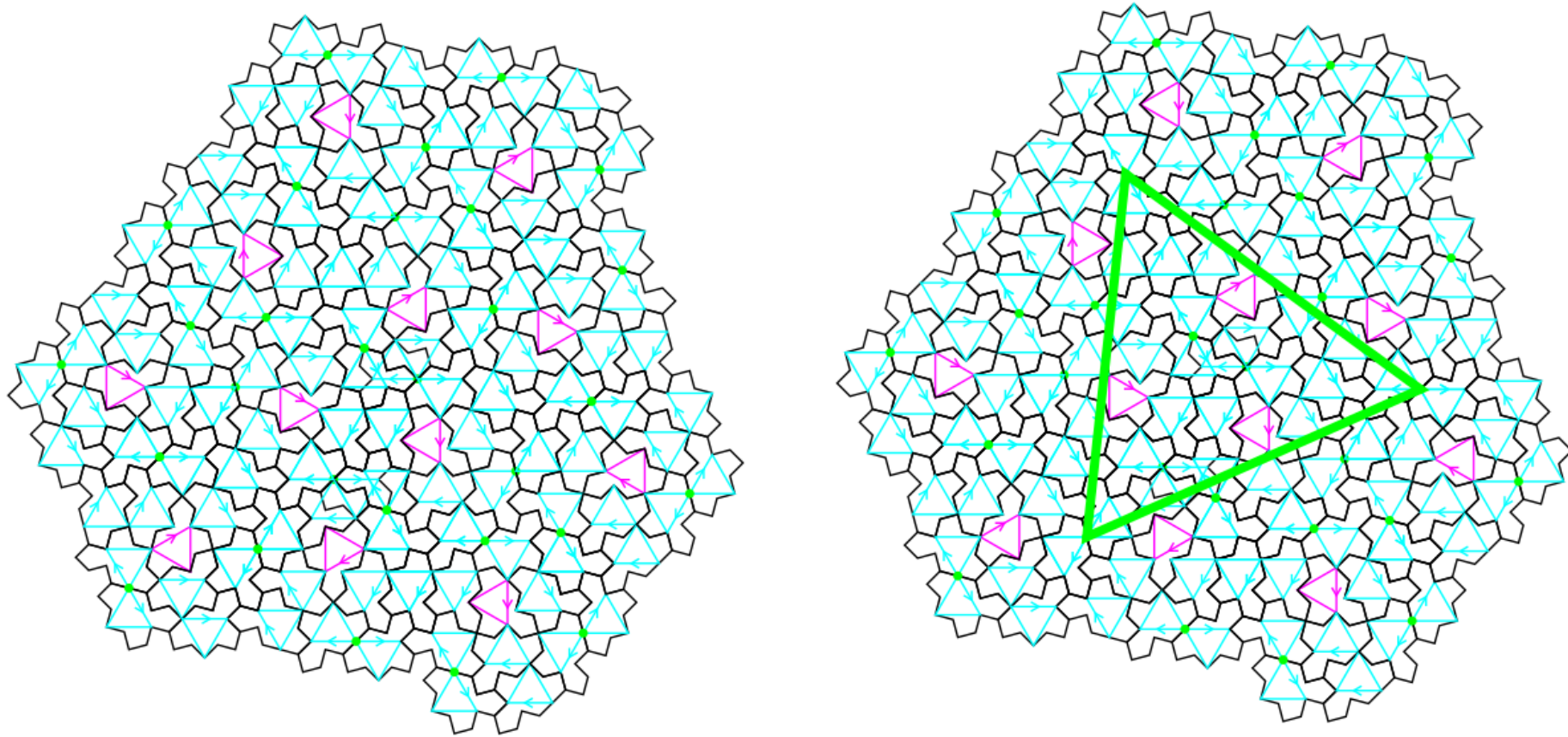


Inflation step 1



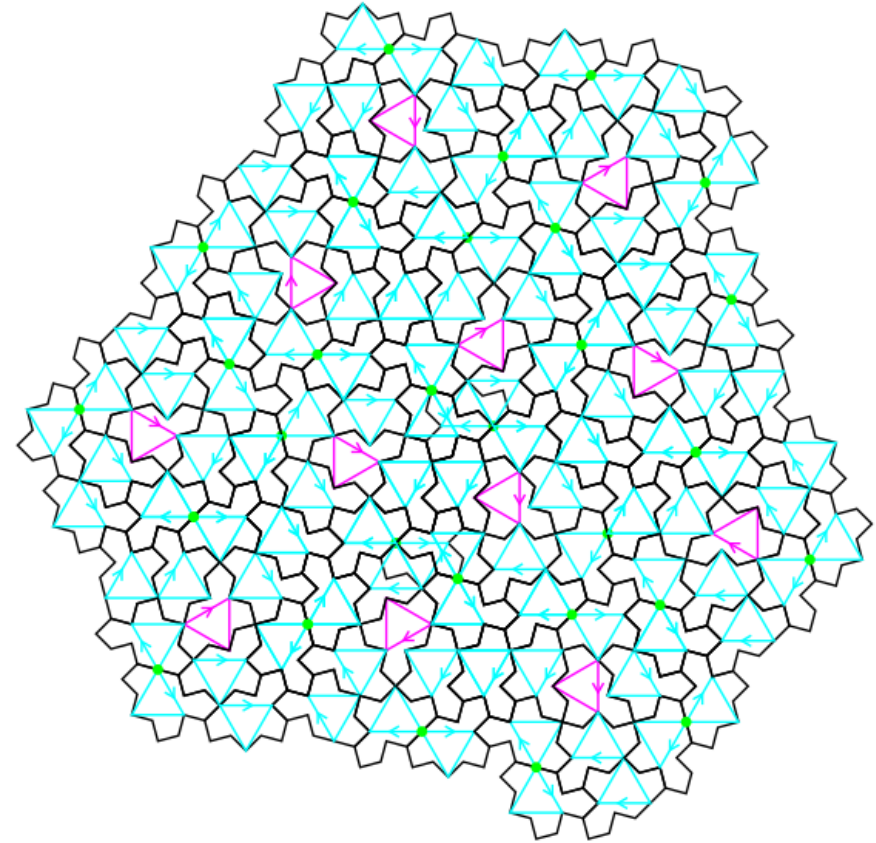
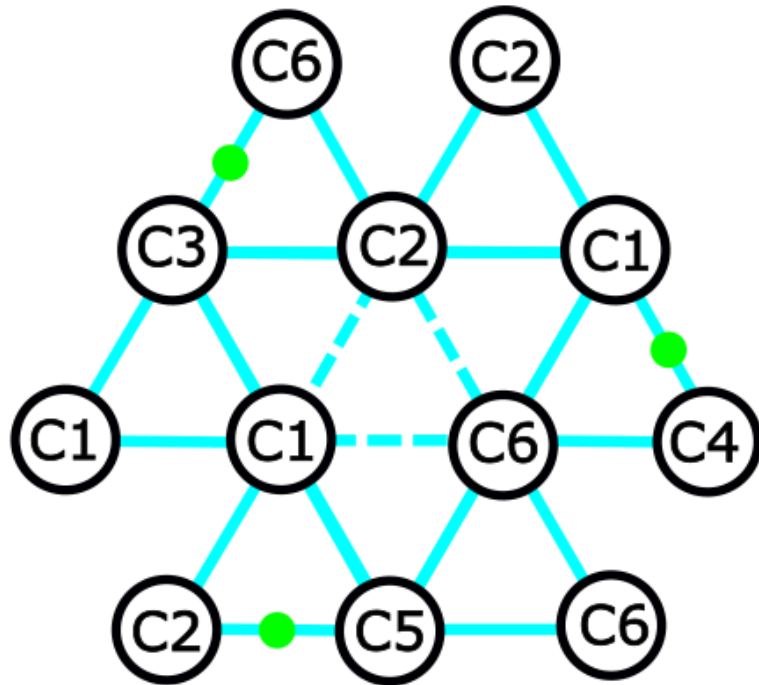
- Directions along the blue lines are the edge directions of the blue triangles !!!

Inflation step 1 for full cluster C1



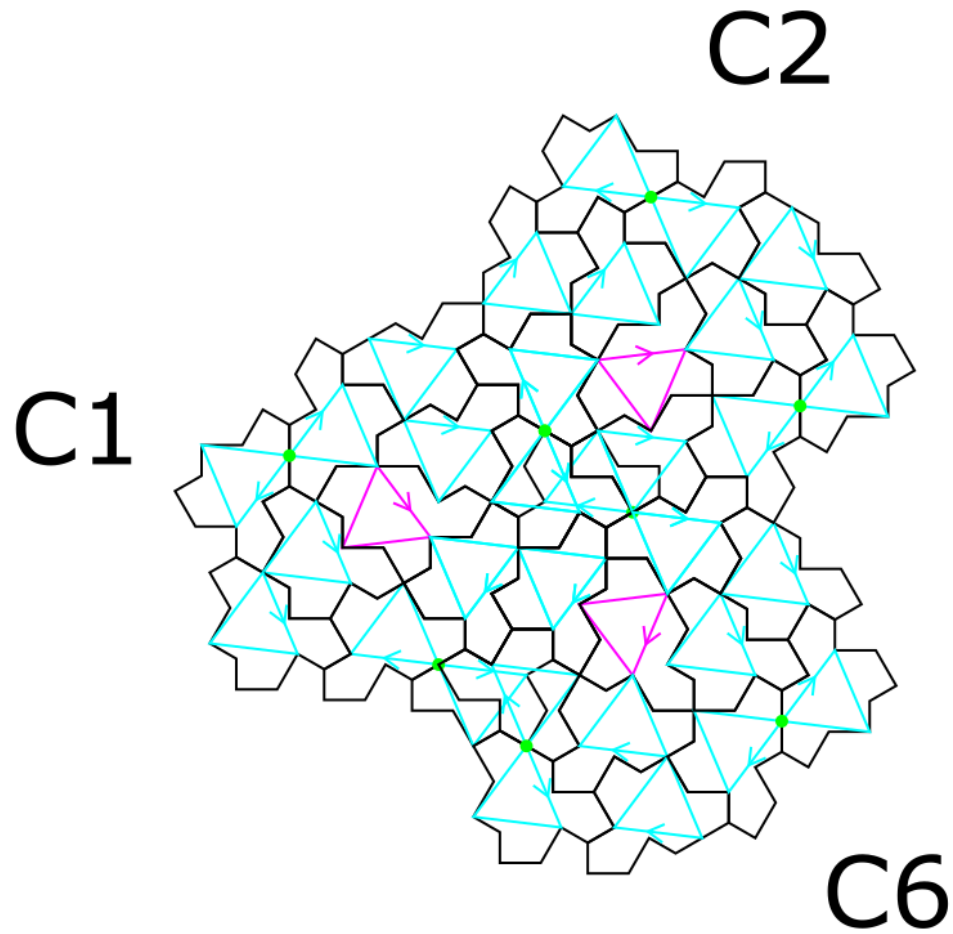
- Assembly of the 12 full clusters works fine !!! (done manually so far for the relative positions of clusters)
- Some tiles are superimposed
- Some half-turned tiles need to be removed with a simple rule: **translation wins over half-turn !**
- A bigger green triangle is identified, centered on a trifle at the center of the figure ☺

Inflation step 1 for full cluster C1

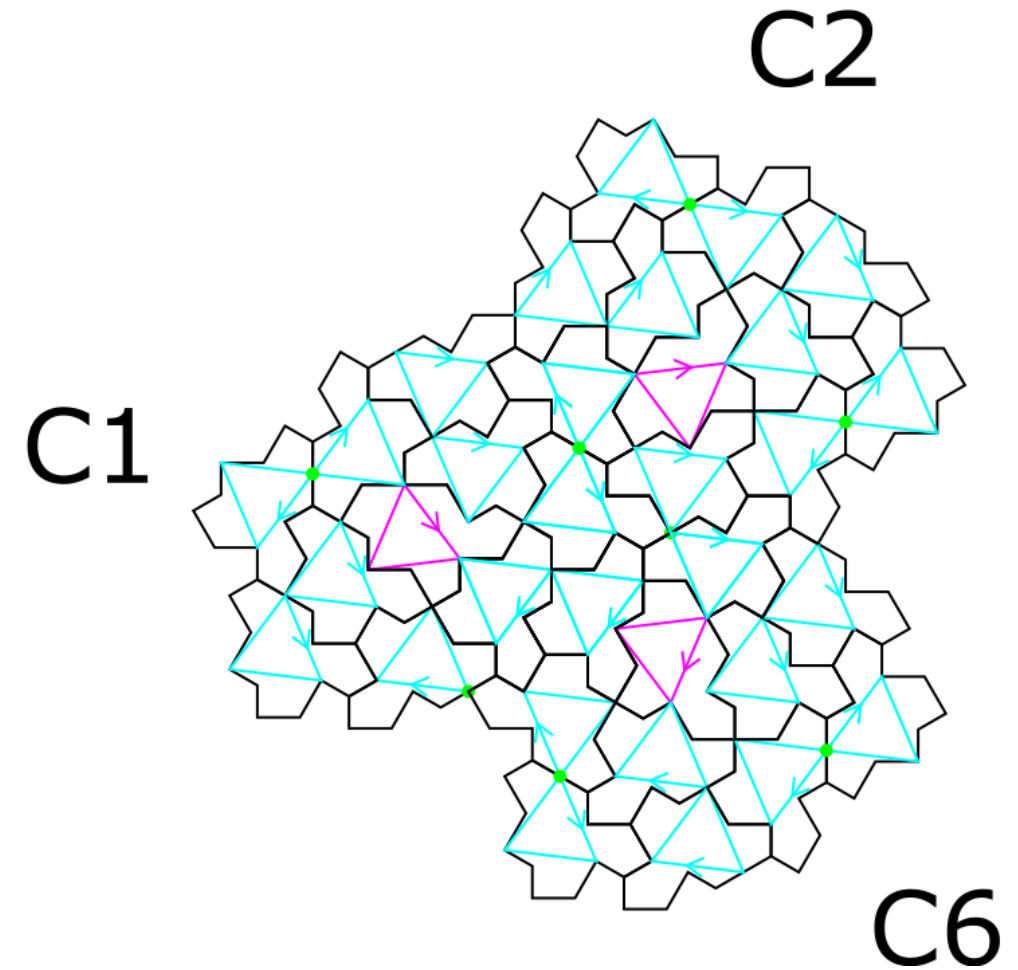


- TO DO: Find a rule for the relative positions between clusters in the assembly
- Clusters are translated from each other along the blue lines
- Use superposition of green points from two adjacent clusters ???

Defect healing

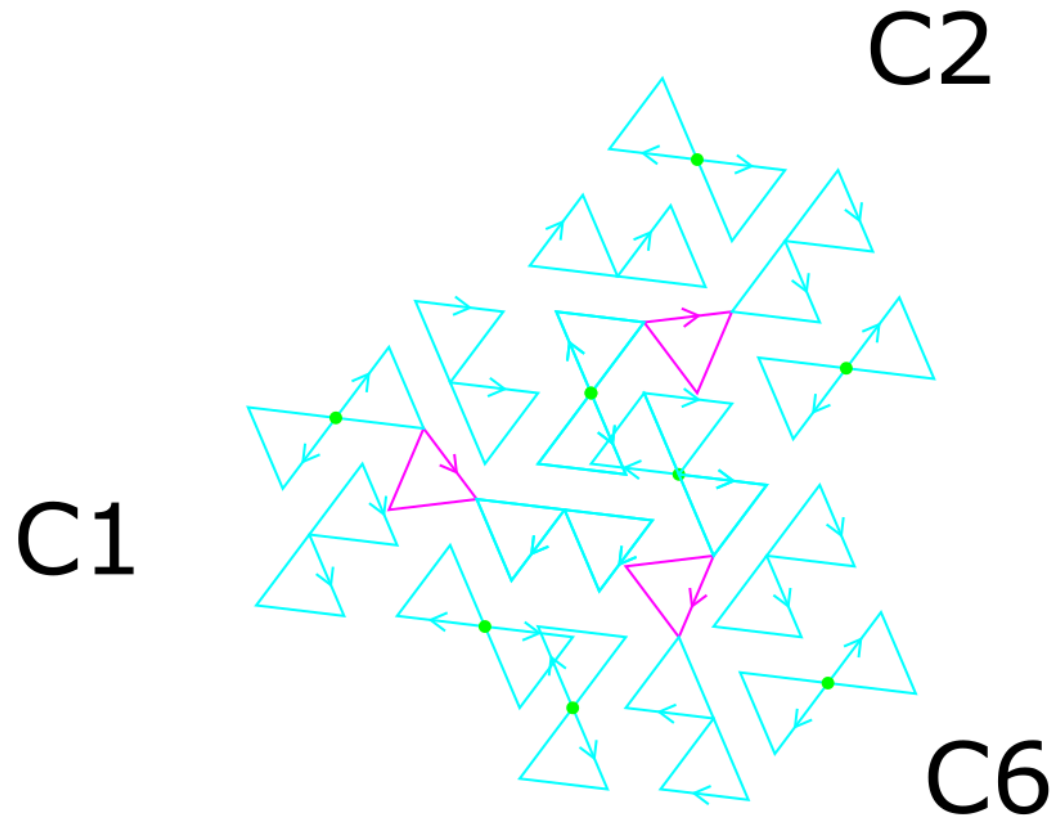


- Starting point for worms ☺

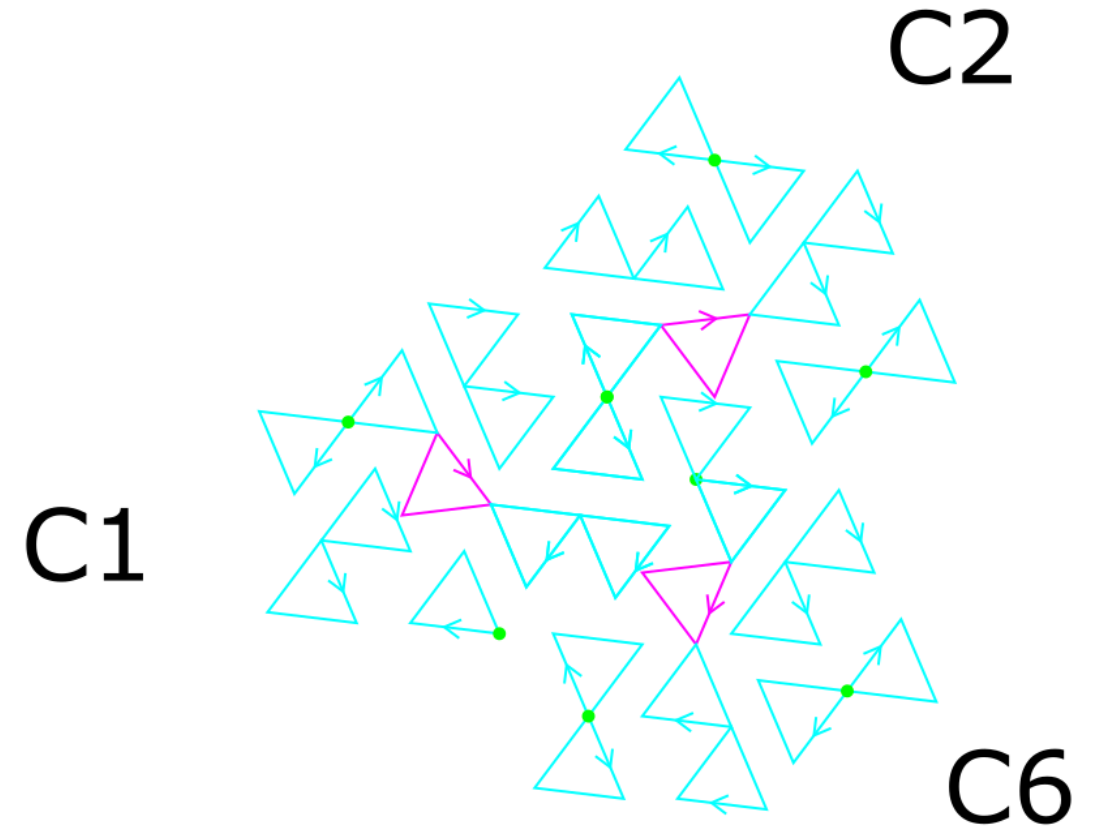


- After removing two half-turned tiles

Defect healing

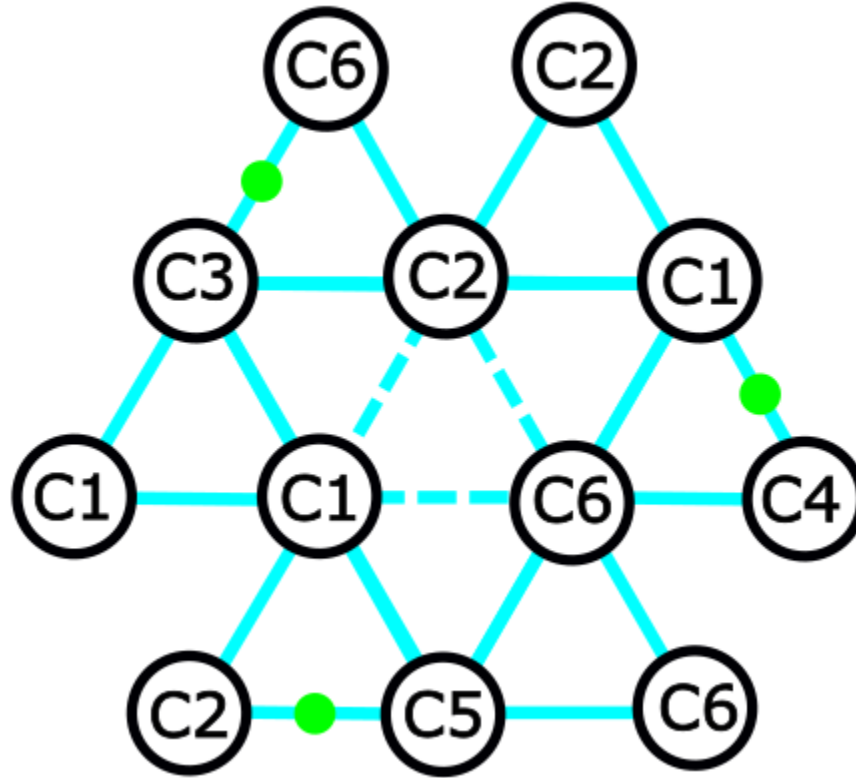


- Starting point for worms 😊



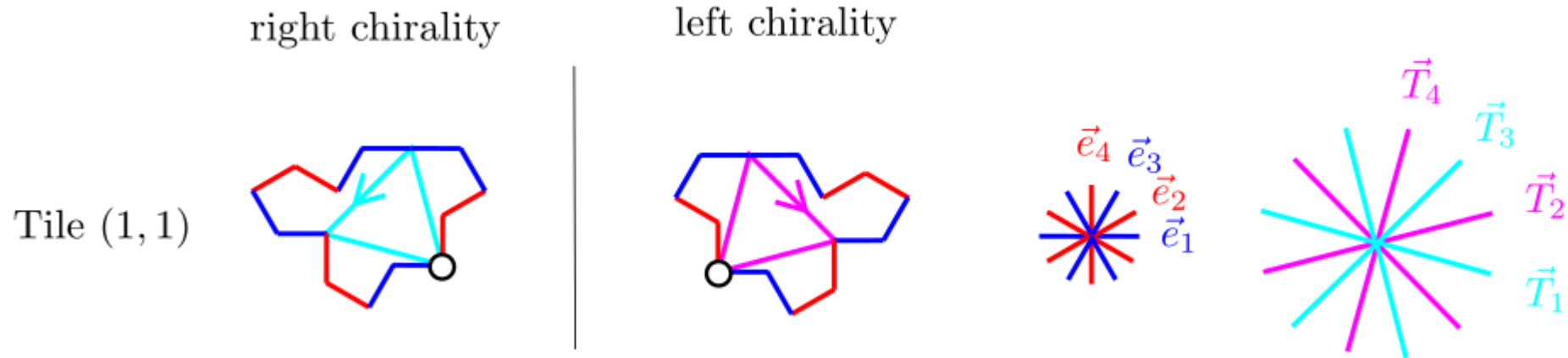
- After removing two half-turned tiles

TO DO: Step 2



- Replace all the 12 full clusters in this pattern with the same rule used previously for C1 (just a rotation of the pattern for C2, C3, C4, C5 and C6)

New reference triangle in Tile(a,b)

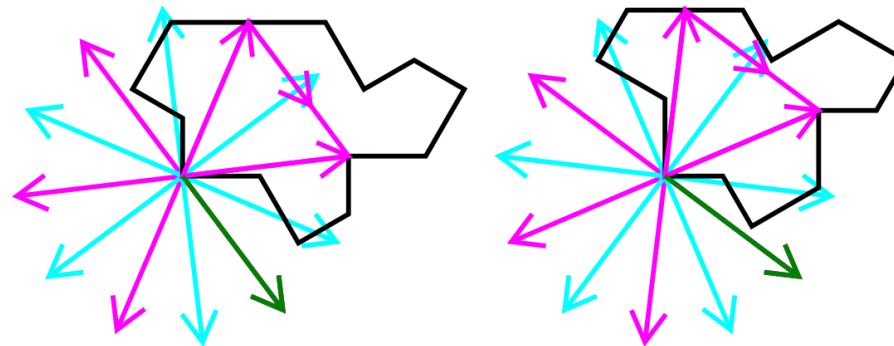


Robust for any T(a,b) tile ☺

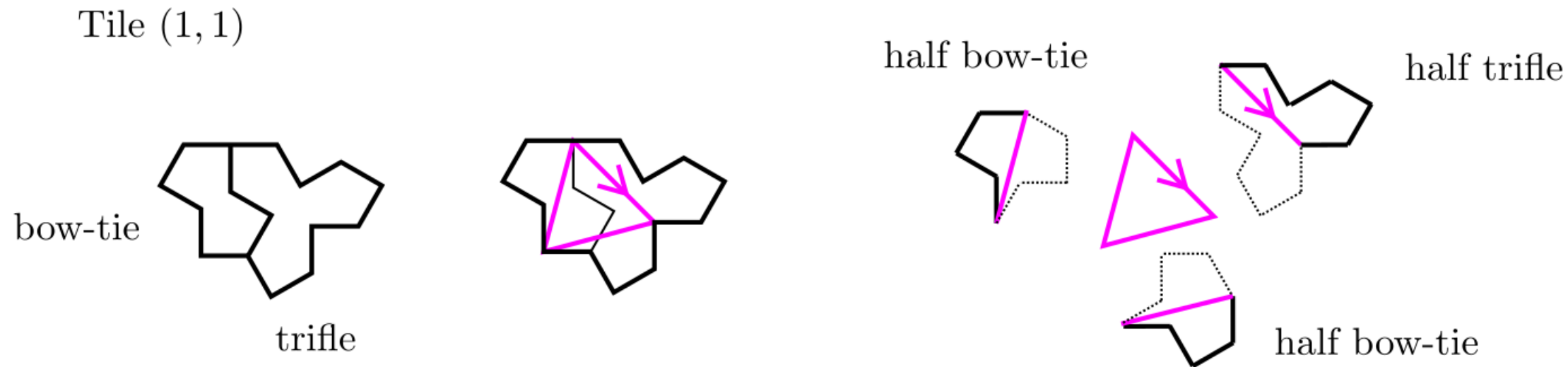
Tile(1.6,1.2)

Tile(1.2,1.6)

$a=1.6$
 $b=1.2$

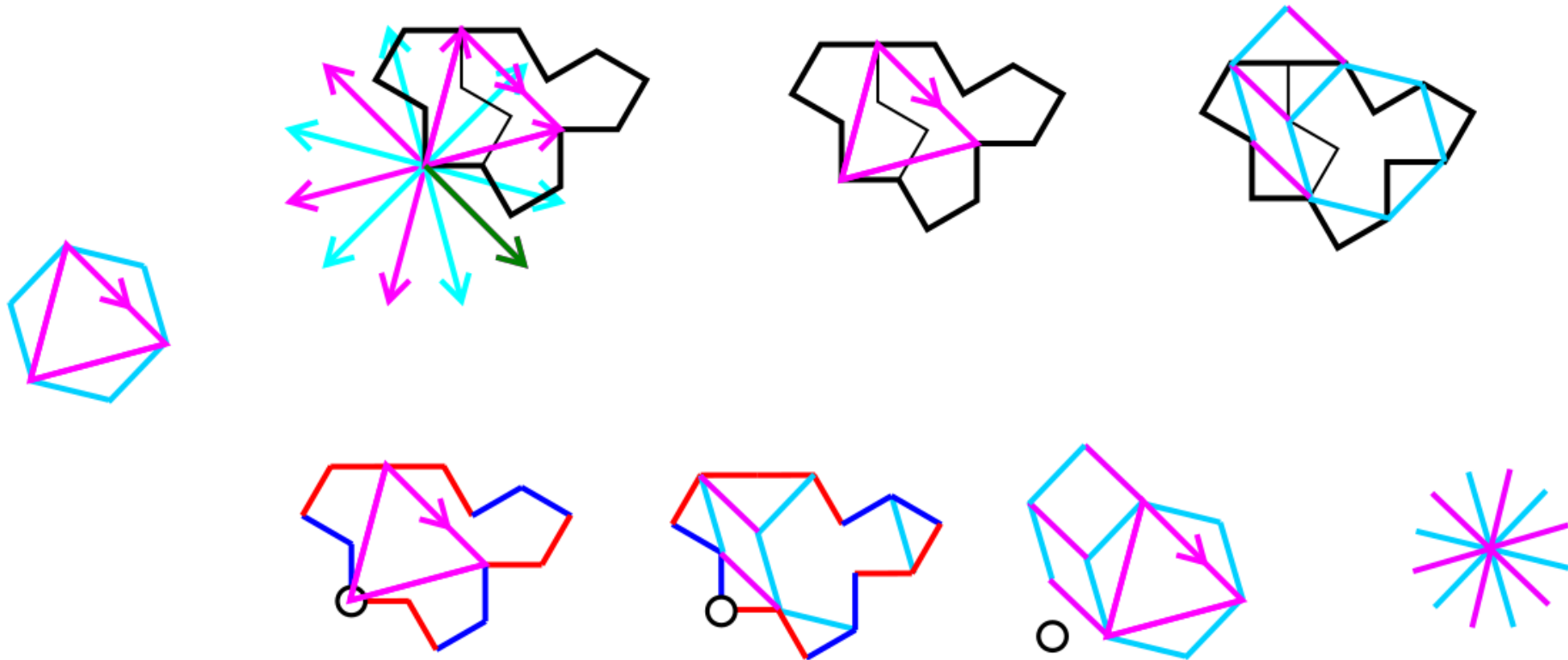


Link with trifle and bow-tie in $\text{Tile}(1,1)$



Comparison with hexagon/square/rhombus decomposition

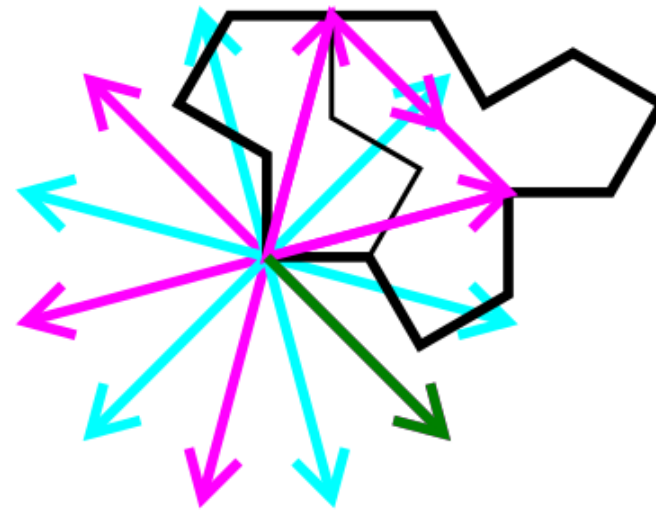
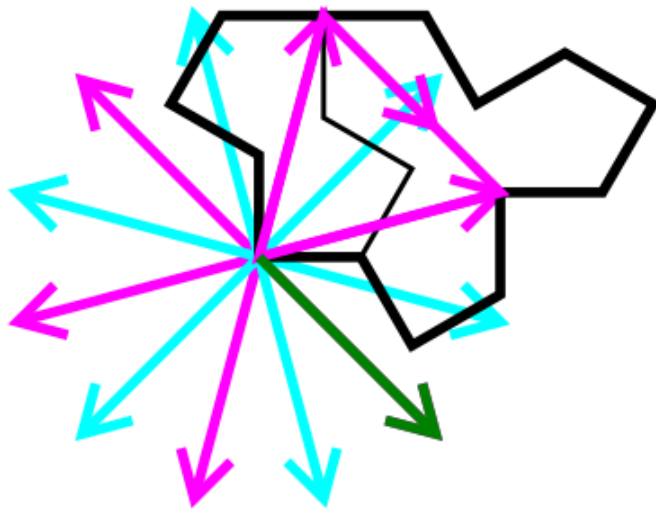
Tile(1.0,1.0)



Tile(1,1) : spectre tile

Tile(1.0,1.0)

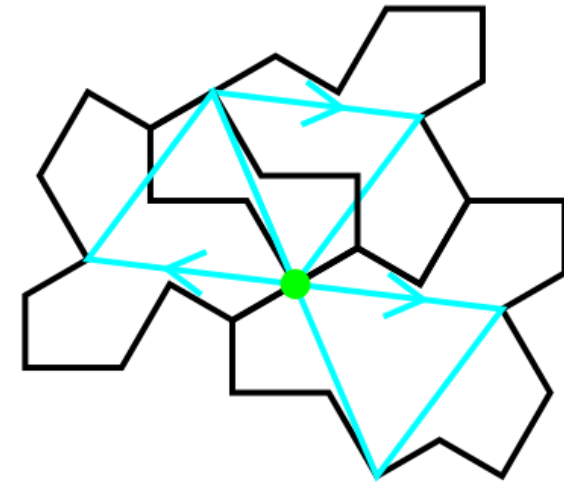
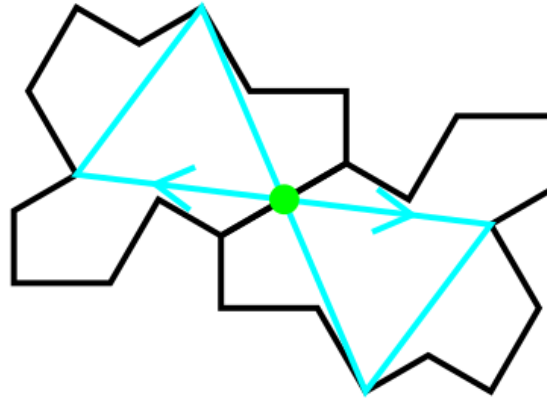
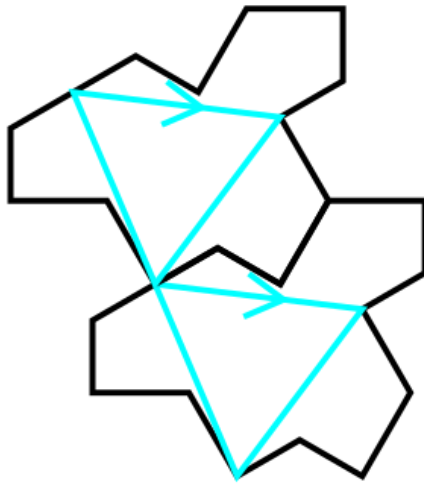
Tile(1.0,1.0)



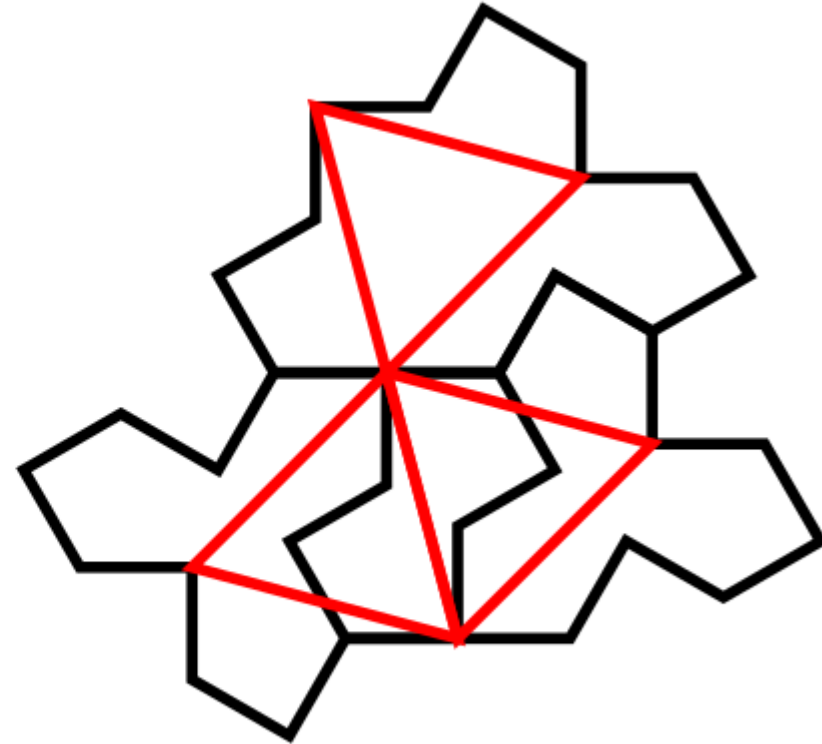
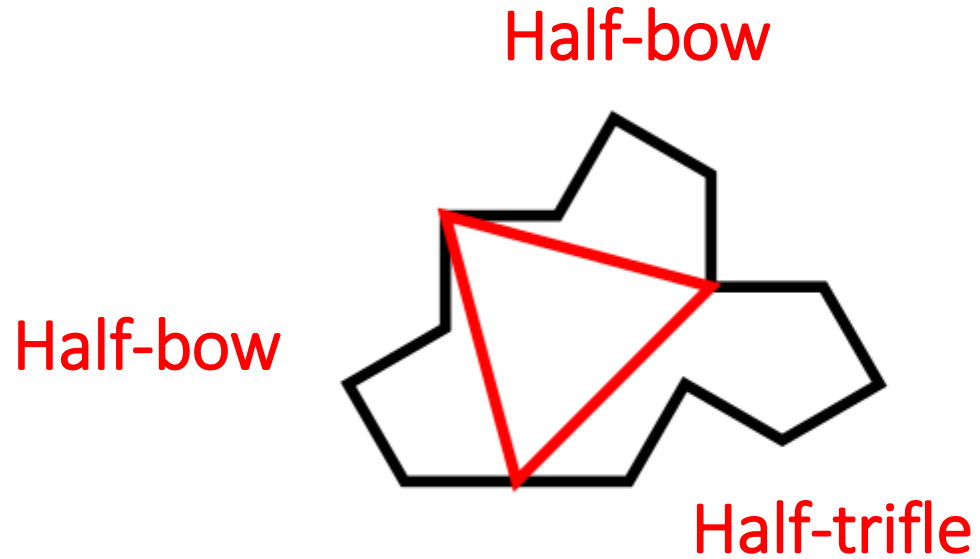
translation versus half-turn for any $\text{Tile}(a,b)$

$a=1.6$

$b=1.2$



Regular triangle inside Tile(1,1)

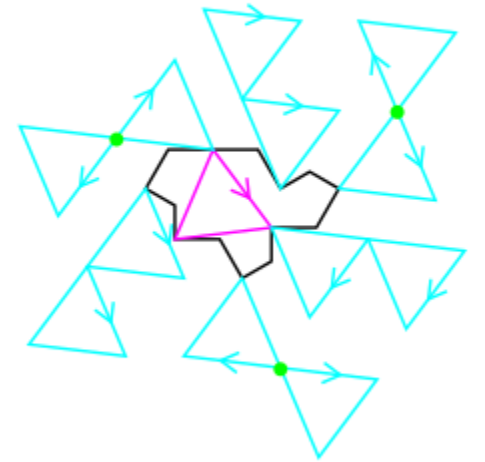
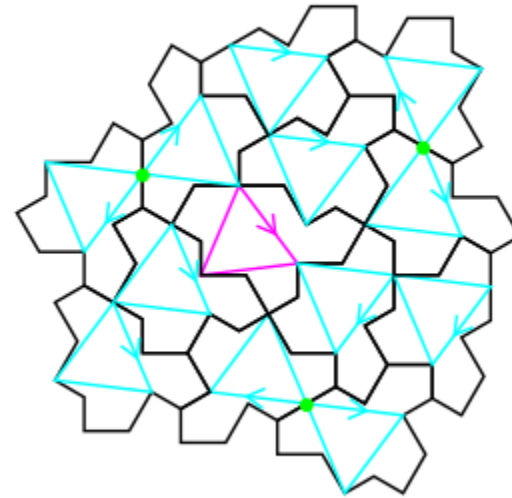
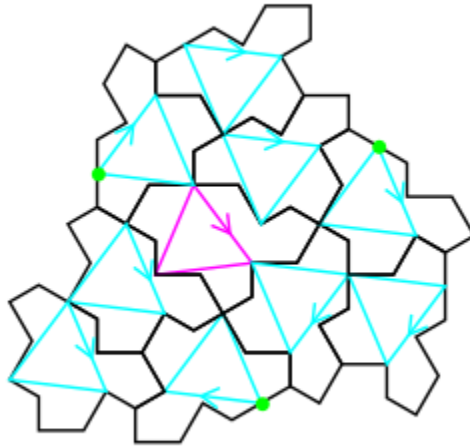
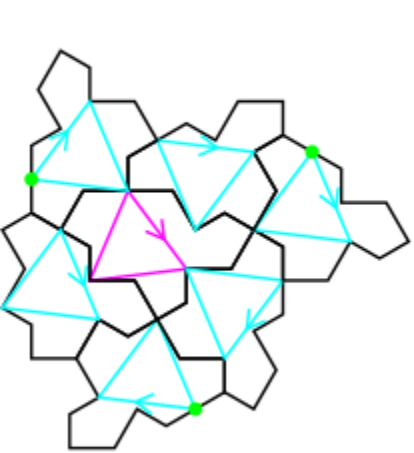


2 other shapes are left:
-> 2 half-bows
-> 1 half-trifle

Cluster and full cluster

$$a=1.6$$

$$b=1.2$$



C cluster
(corona around a
ODD tile)

The three
translated tiles
are all in the tiling

Three more possible tiles
with half-turn around the
green points